

ARTICLE

Double descent, and statistical learning/physics

Bui Gia Khanh,¹ Luong Van Tam,¹ and Dang Tri Trung²¹Department of Physics, Hanoi University of Science, Vietnam National University, Hanoi Vietnam²Monash University, Department of Computer Science, Melbourne, Australia

Author for correspondence: Bui Gia Khanh, Email: fujimiyaamane@outlook.com.

Abstract

Classical statistical learning theory, grounded in Computational Learning Theory (CoLT) Valiant 1984 and Statistical Learning Theory (SLT) Vapnik 1999, predicts a monotonic bias-variance tradeoff Geman, Bienenstock, and Doursat 1992 governing model selection: increasing model complexity reduces bias at the cost of increased variance, implying the existence of an optimal complexity. However, modern overparameterized models consistently violate this prediction, exhibiting the double descent phenomenon Belkin et al. 2019a; Nakkiran et al. 2019, where test error first increases then decreases beyond the interpolation threshold. In this paper, we review the theoretical progress on double descent, situating existing explanations from Random Matrix Theory and replica methods within their equilibrium assumptions. We then develop a Non-Equilibrium Statistical Physics (NESP) framework that explains double descent through the interplay between anisotropic SGD noise and loss landscape curvature, analytically demonstrating a curvature-noise coupling $\Sigma(W) \sim H(W)$ in a linear teacher-student model. This coupling drives a dynamical selection mechanism—“survival of the flattest”—providing a physical mechanism for the implicit regularization of stochastic gradient descent.

Keywords: Statistical learning theory, double descent, deep learning, computational complexity theory, complexity theory, statistical physics, spin-glass models, theoretical physics

The modern machine learning landscape is dominated and guaranteed by the foundation of theoretical machine learning, specifically computational learning theory (CoLT) and statistical learning theory (SLT). The theory of machine learning and its modern practice has been developed and researched, of a substantial portion by empirical and heuristic approach, either by advancing new practices, architectures or by try-and-test modification. From its early onset of a regression estimator $\theta(x, y)$ on the linear regression problem, machine learning has developed substantially. Certain model concept with great successes includes regression-classification model, Bayesian modelling, generative learning model, support vector machine, Gaussian processes, and more. Of all such, the more formal and complex model architecture created, is the concept of a neural network. With increasingly sophisticated architecture, heuristic approach become popular, the fast-paced advancement of the field comes with new method, new results, new observations, and its far-reaching application which led to even bigger and larger scale deployment, there has been questions about the formation and status of a theoretical ground, a rigorous matter on the side of theoretical machine learning.

While rigorous and well-formulated in a sense, classical and theoretical machine learning was dwarfed by the modern advancement of machine learning as a whole, leading to several anecdotal problems regarding the interpretation of phenomena, the re-evaluation of the theory to fit the more updated analysis, and explanation to more sophisticatedly designed system. This and many more, plus

as present, many of such advancements and improvements are heuristic, and the general theory and conceptual understanding remain limited, led to the choice to often opt for analogies and empirical workaround. Ultimately, this resulted in multiple ‘failures’ in explaining, interpreting, and predicting behaviours of new phenomena appeared in the modern landscape of machine learning. While there exists novelty of analysis, and different theoretical schemes to analyse such, particular problems remain in the form of irregularity and conflict of interpretations between frameworks.

We hope to use the theory of statistical physics, as seen of widespread usage in the history of development of artificial intelligence and computational method of such, as seen from Hopfield 1982; Amit, Gutfreund, and Sompolinsky 1985; Gardner and Derrida 1988; Seung, Sompolinsky, and Tishby 1992; Mezard, Parisi, and Virasoro 1987; Mezard and Montanari 2009; Engel and Van den Broeck 2001; Advani, Lahiri, and Ganguli 2013; Bahri et al. 2020; Mehta and Schwab 2014; Advani and Saxe 2017b; Li and Sompolinsky 2021; Ferraro et al. 2014, with mentions of such statistical mechanics application in Roberts, Yaida, and Hanin 2021. Some of those statistical learning schemes resulted in the works of G. E. Hinton on the variation of neural network called Boltzmann machine, as seen of Ackley, Hinton, and Sejnowski 1985 and Hinton 2012. Nevertheless, such question remains if the statistical scheme overreached of its current applicable form, to indict misleading or only limited improvement and interpretability onto a machine learning

and in general an artificial intelligence system dynamics. In addition, assumptions and structural constraints, such as those incurred on Boltzmann machines for it to facilitate an environment interpretable of statistical physics remains a question of interest^a. Even with such problems, we would like to retain the view that statistical physics can be indeed used to interpret double descent, though with such phenomena, requires extreme cautions of logical assumptions and modelling setting.

1. Foundational works

On itself, the phenomena double descent has been investigated somewhat comprehensively, firstly introduced by Belkin *et al.* 2019a, which gives it distinctive saddle escape illustration. Further analysis was made by several literatures, particularly attributed the existence of double descent to the concept of model complexity and inductive bias, and potential explanations of such. Nakkiran *et al.* 2019 expanded the phenomena into deep neural network models. Their conclusion is reached by considering the perturbation of a learning procedure $\mathcal{T}$ on the effective model complexity $\text{EMC}_{\mathcal{D},\epsilon}(\mathcal{T})$ defined in the paper, separating the eventual phenomena into regions of observations, thereby in one way or another, predicting the tendency of double descent. This is also the first, arguably, "concise" definition of double descent, even though its nature is an empirical definition, including the notion of model complexity. Recently, there is also the Neural Tangent Kernel (Jacot, Gabriel, and Hongler 2018) which can be used to explain double descent, however further researches are still required. Preliminary works are done by Lafon and Thomas 2024, Schaeffer *et al.* 2023, and Liu and Flanigan 2023 on the role of optimization in double descent. Davies, Langosco, and Krueger 2023 attempted to unifying grokking with double descent, theorized a possibility of similarity during the generalization phase transition of the inference period, and Olmin and Lindsten 2024 attempted to explain epoch-wise double descent

^a It is widely concern of the fact that there exists many interpretations of machine learning mechanics under different formal language, or application-biased one. Such is, for example, information theoretic, as seen of Russo and Zou 2016; Xu and Raginsky 2017, categorical theoretic approaches, of Manin and Marcolli 2024, and accordingly, control theory, topological learning, graph theoretical, and statistical physics. While interesting as to illuminate the same problem under different lens, it is troubling of the messiness in such approach, as different theories claim different hypothesis, principles, theorems and axioms. Furthermore, it is of questionable intent to simply forgive the account of applying one's field on another — there exists the fundamental domain bias — of which as can be seen in statistical physics, for certain event to be applicable of using statistical law, one either must adapt the notion and introduce analogous notion (like temperature and relative model diffusing entropy), or restricting the domain, structure, range of expression, and mechanics — similar to how Boltzmann machines are. Additionally, applying without care gives increasingly large amount of domain bias, as concepts typically seen and assumed of a system in one theory might interject wrongfully, or simply conflict, narrow down the applicable range of such application, or in the worst case, invalidate the result.

within the model of two-layer linear network. However, it is mentioned that it can also be expanded into deep nonlinear networks.

Further literatures on double descent varies, but fairly new, ever since their discovery articles in Belkin *et al.* 2019a and Nakkiran *et al.* 2019. Since then, we have Shi *et al.* 2024; Schaeffer *et al.* 2023; Quéty and Tartaglione 2023a; Lafon and Thomas 2024; Neal 2019; Davies, Langosco, and Krueger 2023; Quéty and Tartaglione 2023b; Liu and Flanigan 2023; Mei and Montanari 2020, 2019b; Adlam and Pennington 2020; Transtrum *et al.* 2025; Liu, Liao, and Suykens 2021; Allerbo 2025; Pezeshki *et al.* 2021; Brellmann *et al.* 2024; Nakkiran 2019a; Mei and Montanari 2019a; Heckel and Yilmaz 2020; Nakkiran *et al.* 2020; Belkin, Hsu, and Xu 2020; Yang and Suzuki 2024; Spiess, Imbens, and Venugopal 2023; Nakkiran *et al.* 2021; Y. S. Zhang 2024; Cherkassky and Lee 2024; Nakkiran 2019b. An even more exotic version of double descent, dubbed triple descent (or more), is discovered in Ascoli, Sagun, and Biroli 2020.

2. Background and Notation

This section fixes notation and recalls the minimal definitions from statistical learning theory (SLT) Vapnik 1999; Hajek and Raginsky 2021 and the bias-variance framework that are required for the analysis in subsequent sections. A more detailed specification of the learning system—including all components mapped to statistical-physics language—is given in Section 4.

2.1 Learning Setup

We observe a dataset $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^n$ drawn i.i.d. from an unknown distribution $\mathcal{D}$ on $\mathcal{X} \times \mathcal{Y}$. A hypothesis class $\mathcal{H}$ contains candidate models $h : \mathcal{X} \rightarrow \mathcal{Y}$. Given a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_{\geq 0}$, we define the two central risk measures.

Definition 2.1 (Empirical and population risk). For $h \in \mathcal{H}$ and a sample $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^n$:

$$\hat{R}_{\mathcal{S}}(h) = \frac{1}{n} \sum_{i=1}^n \ell(h(x_i), y_i) \quad (\text{empirical risk}), \quad (1)$$

$$R(h) = \mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell(h(x), y)] \quad (\text{population risk}). \quad (2)$$

The fundamental gap $R(h) - \hat{R}_{\mathcal{S}}(h)$ is the generalization gap: it measures how well performance on the training sample transfers to unseen data.

2.2 Optimization Principles

Definition 2.2 (ERM and SRM). Empirical Risk Minimization (ERM) selects $h_{\mathcal{S}} = \arg \min_{h \in \mathcal{H}} \hat{R}_{\mathcal{S}}(h)$. Structural Risk Minimization (SRM) adds a complexity penalty:

$$h_{\mathcal{S}}^{\text{reg}} = \arg \min_{h \in \mathcal{H}} [\hat{R}_{\mathcal{S}}(h) + \lambda r(h)], \quad (3)$$

where $r : \mathcal{H} \rightarrow \mathbb{R}_{\geq 0}$ is a regularizer (e.g., $r(h) = |0w|_2^2$ for weight decay) and $\lambda > 0$ controls the strength. In this paper, the primary optimization algorithm is stochastic gradient descent (SGD) applied to the ERM objective; the implicit regularization induced by SGD noise is a central theme.

2.3 Bias-Variance Decomposition

For a regression problem with squared loss, the expected prediction error admits an exact decomposition Geman, Bienenstock, and Doursat 1992.

Theorem 2.1 (Bias-variance decomposition). Let $f(\cdot; \mathcal{D})$ denote a predictor trained on dataset $\mathcal{D}$, and let $y^*(x) = \mathbb{E}[y | x]$. Then:

$$\mathbb{E}_{\mathcal{D}}[\mathbb{E}_{x,y}(y - f(x; \mathcal{D}))^2] = \underbrace{\mathbb{E}_x(y^* - \mathbb{E}_{\mathcal{D}}[f])^2}_{\text{bias}^2} + \underbrace{\mathbb{E}_x \mathbb{E}_{\mathcal{D}}(f - \mathbb{E}_{\mathcal{D}}[f])^2}_{\text{variance}} + \underbrace{\mathbb{E}_{x,y}(\mathbb{E}_{\mathcal{D}}[f] - y^*)^2}_{\text{noise}} \quad (4)$$

In compact notation Adlam and Pennington 2020: $\mathbb{E}[\hat{y}(x) - y(x)]^2 = (\mathbb{E}\hat{y}(x) - \mathbb{E}y(x))^2 + \mathbb{V}[\hat{y}(x)] + \mathbb{V}[y(x)]$.

The decomposition itself is an identity for squared loss (and extends to Bregman divergences Pfau 2013). The tradeoff that has historically guided model selection is an additional empirical regularity:

Hypothesis 2.1 (Bias-variance tradeoff). For a nested sequence of hypothesis classes $\mathcal{H}_1 \subset \mathcal{H}_2 \subset \dots$ trained via ERM on data from a fixed distribution, one typically observes

$$\frac{\partial \text{Bias}^2}{\partial d} \leq 0, \quad \frac{\partial \text{Var}}{\partial d} \geq 0 \quad (\text{underparameterized regime}) \quad (5)$$

where d indexes model complexity. This yields a U-shaped test-error curve with an optimal complexity at the bias-variance sweet spot.

Remark 2.2 (Scope and failure of the tradeoff). The monotonic tradeoff is not a theorem: it is an empirical regularity that holds for specific model families in the underparameterized regime but fails in the overparameterized regime—precisely where double descent occurs Belkin et al. 2019a; Neal 2019. This breakdown motivates the remainder of the paper.

3. Double descent

Nevertheless, of bias-variance and the analogous statistical learning theory concept, the target is the same. It is the dilemma of which is presented in Occam’s razor, for choosing the sufficient model of good complexity, or bias, for tradeoff of its generalization ability, or variance. Then there must exist a sweet spot between the axis of bias and variance, since they are as exhibited above inversely proportional to each other. However, double descent seemingly broke the status quo, and insists on an interesting phenomenon - under the same setting, if we ‘crank’ the complexity high enough, we will then reach a point then

called the interpolation threshold, such that the trend reverse and the error rate, instead of being theorized to go up, goes down to a certain line of lower bound.

The first identification of the double descent phenomena dated back to the paper of Belkin - Belkin et al. 2019a, in which the title is literally “reconciling” modern machine learning practice and the bias-variance tradeoff. In modern machine learning practice, or state-of-the-art developments, models are now bigger than ever. If to notice, we will see that currently models are inherently large, for example, a normal large language model will have from 900 millions (900M) to a few billions, for example 10 billions (10B) parameters. That is not taking into account the overall dynamics and structure of the model, which dictates the operating range and efficiency of the model itself. These models, based on the neural network architecture are somewhat trained to exactly fit (or interpolate) the data, almost certainly so that it turn from a prediction setting to an estimation setting. By statistical learning theory, this would be considered overfitting, and yet, they often obtain very high accuracy on test data.

The main finding that Belkin found is a pattern for how the apparent performance on unseen data depends on model capacity and the mechanism underlying the emergence of double descent. When function class capacity is below the “interpolation threshold”, learned predictors exhibit the classical U-shaped curve from Figure 1. The ‘modern’ interpolating regime marks the opposite trend to the right, where the risk starts to decrease up to a lower bound, which then can be called the optimal descent bound.

The bottom of the U is achieved at the sweet spot which balances the fit to the training data and the susceptibility to over-fitting: to the left of the sweet spot, predictors are under-fit, and immediately to the right, predictors are over-fit. When we increase the function class capacity high enough (e.g., by increasing the number of features or the size of the neural network architecture), the learned predictors achieve (near) perfect fits to the training data—i.e., interpolation. Although the learned predictors obtained at the interpolation threshold typically have high risk, we show that increasing the function class capacity beyond this point leads to decreasing risk, typically going below the risk achieved at the sweet spot in the “classical” regime. (Belkin et al. 2019a)

At this point, we would like to also clarify the term interpolation threshold in this case. Usually, under the pretext of overfitting and underfitting, interpolation threshold means the following:

Definition 3.1 (Interpolation threshold). Assuming a typ-

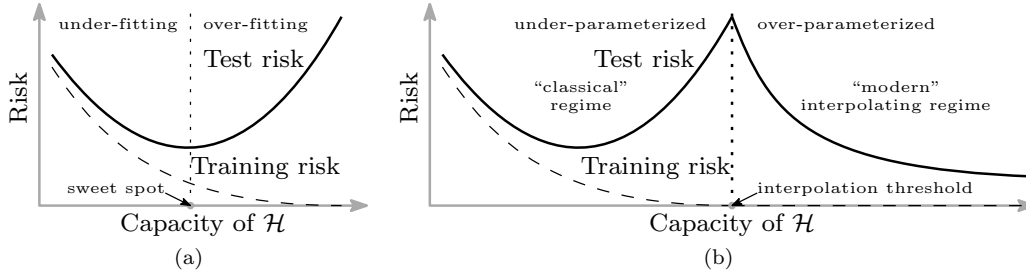


Figure 1. Curves for training risk (dashed line) and test risk (solid line). (a) The classical *U-shaped* risk curve arising from the bias-variance trade-off. (b) The *double descent* risk curve, which incorporates the U-shaped risk curve (i.e., the “classical” regime) together with the observed behaviour from using high capacity function classes (i.e., the “modern” interpolating regime), separated by the interpolation threshold. The predictors to the right of the interpolation threshold have zero training risk. Reproduced from Belkin et al. 2019a.

ical supervised learning setting, for a hypothesis h within class structure $\mathcal{H}$, and a supposed concept c of the class $\mathcal{C}$. Suppose $[X, Y] \subset c \in \mathcal{C}$ being the dataset, or observable space of c on which h is tested on. Under the empirical generalization learning procedure, partitioning $[X, Y]$ into $[X, Y]_1$ and $[X, Y]_2$, for error evaluated on the first set, $\hat{R}_1(h)$ called training error, and the second set, $\hat{R}_2(h)$ called test error, with optimization applied on the first set, then evaluated statically on the second, the interpolation threshold with respect to a particular complexity measure of h , $\mathcal{M}(h)$, is the point $\mathcal{M}_0(h)$ where it is possible for $\hat{R}_2(h)/\hat{R}_1(h) \approx 1$, for $\hat{R}_1(h) \ll \epsilon$, $\hat{R}_2(h) \ll \epsilon$, with $\epsilon \geq 0$.

This basically means that what is learnt, with respect to the model complexity that gauge how much and what the model can learn, perfectly fits the concept c itself, from the observation space on its own. We can say that, that specific sense, we gain the assumption that the observation space is fully representative of the concept class, and the concept class is then fully realized by the model. Such point where that is possible, or there then such behaviour is possible, is called the interpolation point. The definition relies on the statement ‘where it is possible’, hence it still, does not give us a particularly sound notion on to how and what constitute such possibility. The term empirical generalization learning comes from the fact that the procedure in practice of estimating and optimizing train-test-validation set in itself, is an imperfect empirical approximation of generalization testing. While the fact that we cannot gauge generalization error, in a purely blind black-box setting is settled (Mohri, Rostamizadeh, and Talwalkar 2012; Sugiyama 2015; Shalev-Shwartz and Ben-David 2014), the method is then to try to mimic such by using generalization-via-hidden observations. To facilitate this, deliberately, the empirical learning procedure is only dynamic during training session, and static throughout the whole testing and validation testing section, and further on.

By default, in general theory, such interpolating point is dangerous. Not only because it is inherently difficult to know exactly what the dataset constitute, whether it is

actually representative of the concept that gives rise to it, or the structural information, noises, and interferences in between. It is hence impossible, to a certain point, to truly know the effectiveness of the model beside interpolating the current dataset which forms the observable details.

Another prominent discovery to look at is Nakkiran et al. 2019, on the double descent of deep learning models. The paper serves as the first observation upon which double descent is observed on modern, deep learning theoretic models, especially complex one like Transformer or ResNet networks.

They define effective model complexity of $\mathcal{T}$ (w.r.t. distribution $\mathcal{D}$) to be the maximum number of samples n on which $\mathcal{T}$ achieves on average ≈ 0 training error. This is an entirely empirical definition, similarly per definition as VC-dimension.

Definition 3.2 (Effective Model Complexity). The Effective Model Complexity (EMC) of a training procedure $\mathcal{T}$, with respect to distribution $\mathcal{D}$ and parameter $\epsilon > 0$, is defined as:

$$\text{EMC}_{\mathcal{D}, \epsilon}(\mathcal{T}) := \max \{n \mid \mathbb{E}_{S \sim \mathcal{D}^n}[\text{Error}_S(\mathcal{T}(S))] \leq \epsilon\}$$

where $\text{Error}_S(M)$ is the mean error of model M on train samples S .

Using this definition, their main hypothesis can then be stated as the following three-fold regions:

Hypothesis 3.1 (Generalized Double Descent hypothesis, informal). For any natural data distribution $\mathcal{D}$, neural-network-based training procedure $\mathcal{T}$, and small $\epsilon > 0$, if we consider the task of predicting labels based on n samples from $\mathcal{D}$ then:

Under-parameterized regime. If $\text{EMC}_{\mathcal{D}, \epsilon}(\mathcal{T})$ is sufficiently smaller than n , any perturbation of $\mathcal{T}$ that increases its effective complexity will decrease the test error.

Over-parameterized regime. If $\text{EMC}_{\mathcal{D}, \epsilon}(\mathcal{T})$ is sufficiently larger than n , any perturbation of $\mathcal{T}$ that increases its effective complexity will decrease the test error.

Critically parameterized regime. If $\text{EMC}_{\mathcal{D}, \epsilon}(\mathcal{T}) \approx n$, then a perturbation of $\mathcal{T}$ that increases its effective complexity might decrease or increase the test error.

It is to notice that this definition is also only observational. By that, it only outlines the specific region of interest where the paradigm shift is identified through experimental results. It has no effective predictive power, or rather descriptive power more than setting up hypothesis with respect to the effective model complexity, and some arbitrary perturbation. Nakkiran et al. 2019 also noticed this difficulty in providing a theoretical definition and theorem regarding such hypothesis, as said in their manuscript. The existence of the critical regime is also not defined.

The behaviour itself has been particularly investigated, in certain settings. For example, before Belkin et al. 2019a, Advani and Saxe 2017a investigated the generalization error in neural networks of high-dimensional measure. Of such, various behaviours that bear similarity to double descent can be observed. Belkin, Ma, and Mandal 2018 expanded on his previous research on the particular subset of kernel learning models, providing a theoretical analysis of specifically the Laplacian kernel on standard neural network. Mei and Montanari 2020 investigated random feature regression in such regard, and also found similar results.

The fact that double descent is not well-defined itself is a problem on its own. To the best of our knowledge, there has been no effective definition or given description that outlines specifically how double descent can be structured. Instability in reproducing experiments and the effective range of the phenomena remains a question on its own.

3.1 Comparative Summary of Key Results

Table 1 summarises the principal contributions on double descent, comparing model class, theoretical framework, the type of double descent observed, and key findings.

4. Foundational Definitions: Grounding Physical Concepts in Learning Theory

Before applying tools from statistical physics to analyze learning dynamics, we must establish a rigorous foundation. This section provides formal definitions for key concepts, explains the mathematical components, and—critically—justifies why these concepts are appropriate for understanding machine learning systems.

4.1 Theoretical Framework: Tools and Status

Table 2 enumerates the theoretical tools required for the NESF framework, together with an honest assessment of their current status.

Legend: **Established** = rigorous mathematical foundation exists; **Partial/Heuristic** = valid under certain assumptions or used as working approximation; **Conjec-**

tured/Open = central to our framework but not yet rigorously proven.

Each “red” item in the table represents a research frontier whose resolution would constitute a significant contribution. Our methodological stance is one of scientific humility: when a step relies on an approximation (e.g., Langevin limits, equilibrium assumptions, replica calculations), we state it as such and treat the resulting predictions as testable hypotheses rather than final verdicts.

4.2 A Minimal Machine Learning System (Pre-Physics Setting)

Before we introduce physical language (Hamiltonians, temperatures, quenched disorder), we first specify the machine learning system in its native form. This is intentionally a “boring” list—but it is the only safe way to avoid accidental over-interpretation.

4.2.1 Complete Enumeration of System Components

We define a learning system by the following exhaustive specification. Each component is defined in purely machine learning terms, without reference to statistical physics.

Setting 4.1 (Learning System Components). We define a learning system by the tuple

$$\mathfrak{S} \equiv (\mathcal{D}, \mathcal{S}, \mathcal{H}, f_{\theta}, \Theta, \ell, \mathcal{A}, \Pi) \quad (6)$$

where each component is specified as follows:

Data Layer:

- $\mathcal{D}$ is the (unknown) data-generating distribution on $\mathcal{X} \times \mathcal{Y}$, equivalently encoding a “teacher” conditional $p(y | x)$ (possibly noisy and/or misspecified). This is the ground truth that the learner attempts to approximate.
- $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^n$ is the observed dataset, typically sampled i.i.d. from $\mathcal{D}$. This is the only information available to the learner about $\mathcal{D}$.

Model Layer:

- $\mathcal{H}$ is the hypothesis class (e.g., a neural network architecture family). This defines the space of possible models the learner can consider.
- $f_{\theta} \in \mathcal{H}$ denotes a particular model instance, where $f_{\theta} : \mathcal{X} \rightarrow \mathcal{Y}$.
- $\Theta \subset \mathbb{R}^P$ is the parameter space, with P the number of trainable degrees of freedom (weights, biases, etc.).

Objective Layer:

- $\ell(\theta; x, y)$ is the per-sample loss function, measuring the discrepancy between $f_{\theta}(x)$ and y . Common choices include:
 - Mean Squared Error (MSE): $\ell(\theta; x, y) = \frac{1}{2}(f_{\theta}(x) - y)^2$
 - Cross-Entropy: $\ell(\theta; x, y) = -y \log f_{\theta}(x) - (1 - y) \log(1 - f_{\theta}(x))$

Table 1. Comparison of key double descent results.

Reference	Model Class	Framework	DD Type	Key Finding
Belkin et al. 2019a	Kernel methods, RF, DNNs	Empirical	Model-wise	First unified identification of DD across model families
Nakkiran et al. 2019	ResNet, Transformer, CNNs	Empirical (EMC)	Model / Epoch / Sample	Epoch-wise DD; effective model complexity framework
Hastie et al. 2019	Ridgeless least squares	RMT (M-P law)	Model-wise	Exact asymptotic risk; peak at $\gamma = 1$
Advani & Saxe 2017a	Two-layer linear nets	Statistical physics	Model-wise	Generalization error in high-dim; precursor to DD observation
Mei & Montanari 2020	Random features	Replica / RMT	Model-wise	Precise asymptotics for RF regression; DD at interpolation
Gerace et al. 2020	RF / Generalized linear	Replica theory	Model-wise	Spin glass formulation; phase transition at threshold
This work	Linear teacher-student	NESP (Langevin / F-P)	Model / Epoch	Curvature-noise coupling as dynamical mechanism

- Hinge loss: $\ell(\theta; x, y) = \max(0, 1 - y \cdot f_\theta(x))$
- The empirical risk (training objective) is:

$$\mathcal{L}_S(\theta) \equiv \frac{1}{n} \sum_{i=1}^n \ell(\theta; x_i, y_i) \quad (+ \text{optional regularization } \lambda r(\theta)). \quad (7)$$

- The population risk (generalization objective) is:

$$\mathcal{L}(\theta) \equiv \mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell(\theta; x, y)]. \quad (8)$$

Optimization Layer:

- $\mathcal{A}$ is the optimization/training procedure, producing an iterate sequence $(\theta_t)_{t \geq 0}$. In this paper, we focus primarily on:
- Stochastic Gradient Descent (SGD):

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{1}{|\mathcal{B}_t|} \sum_{(x,y) \in \mathcal{B}_t} \nabla_\theta \ell(\theta_t; x, y)$$

- Variants: SGD with momentum, Adam, AdaGrad, etc.
- Hyperparameters associated with $\mathcal{A}$:
- Step size (learning rate): $\eta > 0$
- Batch size: $B = |\mathcal{B}_t|$
- Number of epochs/iterations: T
- Momentum coefficient: β (if applicable)
- Weight decay (explicit ℓ_2 regularization): λ
- Learning rate schedule: $\eta(t)$ (warmup, decay, cyclical, etc.)
- Stopping criteria: early stopping, convergence threshold

Stochasticity Layer:

- Π denotes the collection of randomness sources that make training outcomes non-deterministic:
- Initialization: Random initial weights θ_0 (e.g., Xavier, He, Kaiming)
- Mini-batch sampling: Random selection of $\mathcal{B}_t \subset \mathcal{S}$ at each step

- Data augmentation: Random transformations applied to inputs
- Label noise: Intentional or unintentional noise in y
- Dropout: Random neuron masking during training
- Numerical nondeterminism: Hardware-level floating-point variations

Remark 4.1 (Scope of This Paper). In subsequent analysis, we will primarily focus on:

- Loss function: MSE (ℓ_2 loss), as it admits clean bias-variance decomposition.
- Optimizer: Vanilla SGD (without momentum), as it permits Langevin approximation.
- Architecture: Linear models and two-layer networks, as they permit analytical Hessian computation.

Extensions to cross-entropy loss, Adam optimizer, and deep nonlinear networks are discussed as open problems in Section 7.9.

Remark 4.2 (What This Setting Does Not Assume). This definition does not assume convexity, realizability ($\exists \theta$ with zero population risk), a well-specified teacher, or that SGD converges to a unique limit. It only fixes the objects we will later translate into statistical physics language.

4.2.2 The Translation Protocol: From ML to Physics

With the ML system fully specified, we now articulate the protocol by which physical concepts will be introduced. This is not a claim that ML “is” physics, but a controlled translation with explicit validity conditions.

1. Step 1: Identify the dynamical variables. In physics: spin configurations σ_i . In ML: weight parameters $\theta \in \Theta \subset \mathbb{R}^P$.
2. Step 2: Identify the energy function. In physics: Hamiltonian $\mathcal{H}(\sigma; J)$. In ML: loss function $\mathcal{L}(\theta; \mathcal{S})$.

Table 2. Theoretical tools required for NESP framework: status and gaps

Theoretical Tool	Status	Notes / Gaps
Statistical Learning Theory (SLT)	Established	Classical framework; reviewed in Section 2. VC-dimension, Rademacher complexity well-defined.
Bias-Variance Decomposition	Established	Valid for MSE loss and Bregman divergences; Section 2.
Langevin/SDE Approximation of SGD	Partial	Valid for small η , large B . Discretization effects and non-Gaussian noise require further analysis.
Fokker-Planck Equation	Partial	Stationary distribution for state-dependent $D(W)$ not analytically solved in general.
Replica Method	Partial	Mathematically non-rigorous ($n \rightarrow 0$ limit); works empirically. Used as heuristic benchmark.
Random Matrix Theory (RMT)	Established	Marchenko-Pastur law, spectral edge behavior well-characterized for linear models.
Curvature-Noise Coupling (Hypothesis 7.1)	Conjectured	Analytically verified for linear toy model only. Extension to nonlinear networks remains OPEN .
Escape Time Calculation	Incomplete	Kramers theory for isotropic noise known; anisotropic case requires further derivation.
Jamming Transition Analogy	Heuristic	Qualitative analogy established; quantitative mapping (critical exponents, universality class) OPEN .
Non-Equilibrium Steady State (NESS)	Framework	Conceptual framework proposed; rigorous characterization of $P_{\text{NESS}}(W)$ is a key open problem .

3. Step 3: Identify the quenched disorder. In physics: random couplings J_{ij} . In ML: the dataset $\mathcal{S}$ (fixed during training but random in its generation).
4. Step 4: Identify the thermal fluctuations. In physics: temperature T and Boltzmann factor $e^{-\beta H}$. In ML: SGD noise characterized by $T_{\text{eff}} = \eta/B$.
5. Step 5: Identify the order parameters. In physics: magnetization, overlap between replicas. In ML: test error, weight alignment, generalization gap.

Remark 4.3 (Validity Conditions for the Translation). The physical translation is most reliable when:

- The learning rate η is small (continuous-time approximation valid).
- The batch size B is large enough for Gaussian noise approximation.
- The system has reached a quasi-stationary state (not in early transient dynamics).
- The loss landscape is locally smooth (second-order Taylor expansion valid).

When these conditions are violated, the translation becomes a heuristic rather than a rigorous correspondence. We will flag such cases explicitly.

4.3 Stochastic Gradient Descent as a Physical Process

4.3.1 Why View SGD as a Stochastic Process?

A natural question arises: why should we model a deterministic algorithm (gradient descent on sampled data) as a stochastic process? The answer lies in the epistemic uncertainty introduced by mini-batch sampling.

When we compute the gradient on a mini-batch $\mathcal{B} \subset \mathcal{S}$,

we obtain an estimate of the true population gradient $\nabla \mathcal{L}(W)$. This estimate fluctuates from batch to batch, introducing randomness into the weight trajectory even though each individual step is deterministic given the batch. From the perspective of an observer who does not track individual batches, the weight evolution appears stochastic.

This is analogous to Brownian motion: a pollen grain’s trajectory appears random not because the underlying physics is random, but because the myriad water molecules colliding with it are not individually tracked. Similarly, SGD noise arises from the “collisions” of individual data points with the model.

Definition 4.1 (Stochastic Gradient Descent — Formal Definition). Let $\mathcal{L}(W) = \mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell(W; x, y)]$ be the population loss for weight configuration $W \in \mathbb{R}^P$. The Stochastic Gradient Descent (SGD) algorithm generates a sequence of weight configurations $\{W_t\}_{t=0}^{\infty}$ via the discrete update rule:

$$W_{t+1} = W_t - \eta \cdot \hat{g}_t(W_t) \quad (9)$$

where:

- $\eta > 0$ is the learning rate (step size), controlling the magnitude of each update.
- $\hat{g}_t(W) = \frac{1}{|\mathcal{B}_t|} \sum_{(x,y) \in \mathcal{B}_t} \nabla_W \ell(W; x, y)$ is the mini-batch gradient, computed on a randomly sampled subset $\mathcal{B}_t \subset \mathcal{S}$ of size $B = |\mathcal{B}_t|$.

The mini-batch gradient $\hat{g}_t$ is an unbiased estimator of

the population gradient:

$$\mathbb{E}_{\mathcal{B}_t}[\hat{g}_t(W)] = \nabla \mathcal{L}(W) \quad (10)$$

but it has non-zero variance:

$$\text{Cov}[\hat{g}_t(W)] = \frac{1}{B} \Sigma_g(W) \quad (11)$$

where $\Sigma_g(W) = \mathbb{E}_{(x,y)}[(\nabla_W \ell - \nabla \mathcal{L})(\nabla_W \ell - \nabla \mathcal{L})^\top]$ is the gradient covariance matrix.

Remark 4.4 (Physical Interpretation). The factor $1/B$ reveals a fundamental insight: larger batches reduce noise. This is the statistical physics analog of the law of large numbers. In thermodynamic terms, batch size plays the role of an “inverse temperature controller”—small batches correspond to high temperature (strong fluctuations), large batches to low temperature (near-deterministic dynamics).

4.3.2 The Continuous-Time Limit: Langevin Dynamics

For theoretical analysis, it is often convenient to pass from the discrete update equation 9 to a continuous-time approximation. This is valid when the learning rate η is small, so that $W_{t+1} - W_t \approx dW$.

Definition 4.2 (SGD as Langevin Dynamics). In the continuous-time limit $\eta \rightarrow 0$ with appropriate scaling, the SGD trajectory W_t is approximated by the Langevin Stochastic Differential Equation (SDE):

$$dW_t = -\nabla \mathcal{L}(W_t) dt + \sqrt{\frac{2\eta}{B}} \Sigma_g(W_t)^{1/2} dB_t \quad (12)$$

where:

- $dW_t \in \mathbb{R}^P$ is the infinitesimal change in weights over time interval dt .
- $-\nabla \mathcal{L}(W_t) dt$ is the drift term (deterministic force pushing the system toward lower loss).
- $dB_t \in \mathbb{R}^P$ is a standard Wiener process (multi-dimensional Brownian motion), satisfying $\mathbb{E}[dB_t] = 0$ and $\mathbb{E}[dB_t dB_t^\top] = I_P dt$.
- $\Sigma_g(W)^{1/2}$ is the matrix square root of the gradient covariance, encoding the noise structure.
- The prefactor $\sqrt{2\eta/B}$ sets the noise amplitude, analogous to temperature in physical systems.

Why this form? The structure of Eq. equation 12 is not arbitrary—it arises from the mathematical requirement that the continuous approximation preserve the statistical properties (mean and variance) of the discrete updates. The coefficient $\sqrt{2\eta/B}$ ensures that the variance of dW_t matches that of $W_{t+1} - W_t$ from Eq. equation 9.

Definition 4.3 (Effective Temperature in SGD). By analogy with the classical Langevin equation in physics, we define the effective temperature of SGD as:

$$T_{\text{eff}} \equiv \frac{\eta}{B} \quad (13)$$

This quantity controls the ratio of stochastic fluctuations to deterministic descent.

Remark 4.5 (Why “Temperature”?). In thermal physics, temperature quantifies the average kinetic energy of random molecular motion. In SGD, $T_{\text{eff}} = \eta/B$ quantifies the average magnitude of random weight updates due to mini-batch sampling. The analogy is precise: both control the balance between exploration (random motion) and exploitation (directed motion toward energy/loss minima).

In regimes where the Langevin approximation is accurate (small step size, weak discretization effects, and an appropriate noise model), several mathematical tools from statistical physics—Langevin dynamics and the associated Fokker–Planck description in particular—become consistent with the optimization picture. Outside these regimes, the “temperature” terminology should be read as an effective descriptor rather than a literal thermodynamic temperature.

4.3.3 The Fokker-Planck Equation: From Trajectories to Distributions

While the Langevin equation describes a single stochastic trajectory, many quantities of interest (e.g., the probability of finding a flat minimum) require reasoning about the ensemble of all possible trajectories.

Definition 4.4 (Fokker-Planck Equation for SGD). Let $P(W, t)$ denote the probability density of finding the weights at configuration W at time t , given some initial distribution $P(W, 0)$. Under the Langevin dynamics equation 12, $P(W, t)$ evolves according to the Fokker-Planck equation:

$$\frac{\partial P}{\partial t} = \underbrace{\nabla \cdot (\nabla \mathcal{L} \cdot P)}_{\text{Drift (Flow toward low loss)}} + \underbrace{\nabla \cdot (D(W) \cdot \nabla P)}_{\text{Diffusion (Spreading due to noise)}} \quad (14)$$

where:

- $D(W) = \frac{\eta}{B} \Sigma_g(W)$ is the diffusion tensor, encoding the local noise structure.
- $\nabla \cdot (\nabla \mathcal{L} \cdot P)$ represents the drift term: probability mass flows along the negative gradient direction (toward lower loss).
- $\nabla \cdot (D(W) \cdot \nabla P)$ represents the diffusion term: probability mass spreads out due to stochastic fluctuations, with rate controlled by $D(W)$.

Physical Interpretation of Each Term:

1. **Drift term $\nabla \cdot (\nabla \mathcal{L} \cdot P)$:** If there were no noise ($D = 0$), this term alone would describe deterministic gradient flow. The probability distribution would “flow downhill” on the loss landscape, concentrating at local minima.
2. **Diffusion term $\nabla \cdot (D(W) \cdot \nabla P)$:** This term describes how noise “smears out” the probability distribution. In directions where $D(W)$ is large (high noise), probability

spreads rapidly. In directions where $D(W)$ is small, the distribution remains concentrated.

Remark 4.6 (Why the Fokker-Planck Equation Matters). The Fokker-Planck equation transforms the question “where does SGD converge?” into “what is the stationary distribution $P_\infty(W)$?” This is a profound shift:

- In equilibrium statistical physics, the stationary distribution is the Boltzmann-Gibbs distribution $P \propto e^{-\beta \mathcal{L}}$.
- In non-equilibrium systems (including SGD with state-dependent noise), the stationary distribution can take more complex forms, potentially favoring certain regions of weight space even when they have identical loss values.

This is the mathematical foundation for understanding why SGD might preferentially find “flat” minima over “sharp” minima.

4.4 The Hessian Matrix: Quantifying Local Curvature

4.4.1 Definition and Geometric Meaning

The Hessian matrix is the central object connecting local geometry (curvature) to optimization dynamics.

Definition 4.5 (Hessian Matrix). For a twice-differentiable loss function $\mathcal{L} : \mathbb{R}^P \rightarrow \mathbb{R}$, the Hessian matrix at point W is the $P \times P$ matrix of second partial derivatives:

$$H(W) \equiv \nabla^2 \mathcal{L}(W), \quad \text{with components} \quad H_{ij}(W) = \frac{\partial^2 \mathcal{L}}{\partial W_i \partial W_j} \quad (15)$$

Geometric Interpretation via Taylor Expansion:

Near any point W_0 , the loss function can be approximated by its second-order Taylor expansion:

$$\mathcal{L}(W_0 + \delta W) \approx \mathcal{L}(W_0) + \nabla \mathcal{L}(W_0)^\top \delta W + \frac{1}{2} \delta W^\top H(W_0) \delta W \quad (16)$$

At a critical point where $\nabla \mathcal{L}(W_0) = 0$, the local behavior is entirely determined by the Hessian:

$$\mathcal{L}(W_0 + \delta W) - \mathcal{L}(W_0) \approx \frac{1}{2} \delta W^\top H(W_0) \delta W = \frac{1}{2} \sum_{i=1}^P \lambda_i (v_i^\top \delta W)^2 \quad (17)$$

where $\{\lambda_i, v_i\}$ are the eigenvalue-eigenvector pairs of $H(W_0)$.

Definition 4.6 (Hessian Eigendecomposition and Curvature Directions). The Hessian matrix admits a spectral decomposition:

$$H(W) = \sum_{i=1}^P \lambda_i(W) v_i(W) v_i(W)^\top \quad (18)$$

where:

- $\lambda_i(W) \in \mathbb{R}$ is the i -th eigenvalue, representing the curvature along direction v_i .

- $v_i(W) \in \mathbb{R}^P$ is the corresponding eigenvector, defining a principal direction of curvature.

The eigenvalues characterize the local geometry:

- $\lambda_i > 0$: The loss increases when moving along $\pm v_i$ (“upward curvature”).
- $\lambda_i < 0$: The loss decreases when moving along $\pm v_i$ (“downward curvature”).
- $\lambda_i = 0$: The loss is flat along v_i (“neutral direction”).

Definition 4.7 (Sharp vs. Flat Minima). At a local minimum W^* (where all eigenvalues $\lambda_i \geq 0$), the sharpness is characterized by the magnitude of the largest eigenvalue:

- Sharp minimum: $\lambda_{\max}(H(W^*))$ is large. Small perturbations δW cause large increases in loss. The “basin of attraction” is narrow.
- Flat minimum: $\lambda_{\max}(H(W^*))$ is small. The loss increases slowly for perturbations. The basin is wide.

Quantitatively, we define the sharpness measure:

$$\mathcal{S}(W^*) \equiv \text{Tr}(H(W^*)) = \sum_{i=1}^P \lambda_i \quad (19)$$

and the condition number:

$$\kappa(W^*) \equiv \frac{\lambda_{\max}(H(W^*))}{\lambda_{\min}^+(H(W^*))} \quad (20)$$

where $\lambda_{\min}^+$ is the smallest positive eigenvalue.

Why Sharp vs. Flat Matters for Generalization:

The connection between flatness and generalization is a key insight of recent deep learning theory. The intuition is as follows:

1. At a sharp minimum, the loss is extremely sensitive to the exact weight values. If the test data distribution differs slightly from training, the weights “miss” the narrow optimal region, causing high test error.
2. At a flat minimum, the loss is insensitive to small weight perturbations. Even if test data causes effective “perturbations” to the optimal weights, the model remains in a low-loss region, yielding good generalization.

This is formalized in the PAC-Bayes framework, where the generalization bound depends on the “sharpness” of the loss landscape around the learned weights.

4.5 Equilibrium: What Does It Mean in Learning?

The concept of “equilibrium” is central to statistical physics but requires careful translation to the machine learning context. A fundamental ambiguity in previous literature concerns the precise definition of “equilibrium” in the context of learning algorithms—what exactly constitutes an equilibrium state, and under what conditions can we claim that a learning system has reached one? This subsection provides rigorous answers to these questions.

4.5.1 Equilibrium in Statistical Physics: The Reference Point

Definition 4.8 (Thermodynamic Equilibrium). In classical statistical physics, a system is in thermodynamic equilibrium when:

1. Stationarity: Macroscopic observables (energy, pressure, etc.) do not change over time.
2. Detailed balance: The rate of transitions from state A to state B equals the rate from B to A , for all pairs (A, B) .
3. Gibbs distribution: The probability of finding the system in configuration W is given by the Boltzmann-Gibbs distribution:

$$P_{\text{eq}}(W) = \frac{1}{Z} \exp(-\beta \mathcal{H}(W)) \quad (21)$$

where $\mathcal{H}(W)$ is the Hamiltonian (energy), $\beta = 1/(k_B T)$ is the inverse temperature, and $Z = \int \exp(-\beta \mathcal{H}) dW$ is the partition function.

The Gibbs distribution has a remarkable property: it depends only on the energy function $\mathcal{H}(W)$ and temperature T . The history of how the system reached equilibrium is “forgotten”—all memory of initial conditions is erased.

4.5.2 Translating to Machine Learning: The Isomorphism

Definition 4.9 (Equilibrium Assumption in Learning Theory). Under the equilibrium assumption, the trained neural network weights W are treated as samples from a Boltzmann-Gibbs distribution:

$$P(W|\mathcal{D}) \propto \exp(-\beta \mathcal{L}(W; \mathcal{D})) \quad (22)$$

where:

- $\mathcal{L}(W; \mathcal{D})$ is the loss function, playing the role of Hamiltonian/energy.
- $\beta = B/\eta$ is the inverse effective temperature (see Definition 4.3).
- The dataset $\mathcal{D}$ acts as a fixed “external field” or “quenched disorder”.

Why Make This Assumption?

The equilibrium assumption is analytically convenient because:

1. It allows the use of powerful tools from statistical mechanics (partition functions, free energy, replica method).
2. It predicts that low-loss configurations are exponentially more probable than high-loss configurations.
3. It provides a framework for computing thermodynamic averages over weight space.

Definition 4.10 (The Partition Function and Free Energy in Learning). Under the equilibrium assumption, we define:

- Partition function:

$$Z(\beta, \mathcal{D}) = \int \exp(-\beta \mathcal{L}(W; \mathcal{D})) dW \quad (23)$$

This normalization constant encodes all statistical properties of the weight distribution.

- Free energy:

$$F(\beta, \mathcal{D}) = -\frac{1}{\beta} \ln Z(\beta, \mathcal{D}) \quad (24)$$

The free energy is the central quantity in equilibrium statistical physics. It combines energy (loss) and entropy (number of configurations) into a single functional.

4.5.3 The Critical Question: Is SGD Actually at Equilibrium?

Here we reach the crux of the matter. The equilibrium assumption is mathematically elegant, but is it physically valid for SGD?

Remark 4.7 (Why Equilibrium May Fail for SGD). Several features of SGD dynamics suggest that the equilibrium assumption is violated:

1. Finite training time: Reaching true equilibrium requires $t \rightarrow \infty$. Real training runs for finite epochs, potentially halting before equilibration.
2. Non-constant temperature: Learning rate schedules (warmup, decay, cyclical) mean T_{eff} changes during training, violating the constant-temperature assumption.
3. State-dependent noise: As we show in Section 7, the SGD noise covariance $\Sigma_g(W)$ depends on the current weights W . This anisotropic, multiplicative noise fundamentally differs from the additive white noise assumed in equilibrium Langevin dynamics.
4. Degeneracy of global minima: In over-parameterized networks, there exist infinitely many global minima with $\mathcal{L} = 0$. Under equilibrium, all should be equally probable. Yet SGD preferentially finds flat minima—a selection that cannot arise from energy alone.

Definition 4.11 (Non-Equilibrium Steady State (NESS)). When the equilibrium assumption fails, the system may instead reach a Non-Equilibrium Steady State—a stationary distribution that differs from the Gibbs form:

$$P_{\text{NESS}}(W) \neq \frac{1}{Z} \exp(-\beta \mathcal{L}(W)) \quad (25)$$

In NESS, macroscopic observables are stationary (unchanging in time), but:

- Detailed balance is violated: there are net “probability currents” circulating in weight space.
- The stationary distribution depends on the dynamics (noise structure, update rule), not just the loss function.
- The system retains “memory” of the optimization path.

This is the foundational distinction that motivates our NESP framework in Section 7: SGD is a non-equilibrium process, and understanding generalization requires analyzing the non-equilibrium steady state, not the Gibbs distribution.

4.6 The Replica Method: Computing Averages Over Disorder

The Replica Method is a sophisticated technique from the statistical physics of disordered systems. To use it responsibly in machine learning, we must define it precisely and state its assumptions.

4.6.1 The Problem: Averaging Over Random Data

In machine learning, the dataset $\mathcal{D}$ is a random sample from the data distribution. Different realizations of $\mathcal{D}$ lead to different trained models, different loss landscapes, and different generalization errors. We are interested in the typical behavior—the average over all possible datasets.

The quantity of central interest is the quenched free energy:

$$\bar{F} = \mathbb{E}_{\mathcal{D}} [F(\mathcal{D})] = -\frac{1}{\beta} \mathbb{E}_{\mathcal{D}} [\ln Z(\mathcal{D})] \quad (26)$$

The technical difficulty: The expectation of a logarithm is analytically intractable. We cannot simply exchange $\mathbb{E}$ and $\ln$.

4.6.2 The Replica Trick

Definition 4.12 (The Replica Trick). The Replica Trick is a formal technique to compute $\mathbb{E}[\ln Z]$ by using the identity:

$$\mathbb{E}_{\mathcal{D}}[\ln Z] = \lim_{n \rightarrow 0} \frac{\mathbb{E}_{\mathcal{D}}[Z^n] - 1}{n} \quad (27)$$

The procedure is:

1. Compute $\mathbb{E}[Z^n]$ for positive integer n (this is tractable because the expectation is now outside the logarithm).
2. Analytically continue the result to real n .
3. Take the limit $n \rightarrow 0$.

Why “Replica”? The name comes from the physical interpretation: $Z^n = Z \cdot Z \cdot \dots \cdot Z$ (n times) can be viewed as the partition function of n identical copies (“replicas”) of the original system, all coupled through the same disorder realization $\mathcal{D}$.

Remark 4.8 (Mathematical Status of the Replica Trick). The replica trick is not a rigorous theorem. The analytic continuation from $n \in \mathbb{N}$ to $n \in \mathbb{R}$ and the subsequent $n \rightarrow 0$ limit are formal operations that are not mathematically justified in general.

Despite this, the replica method has an excellent track record:

- It correctly predicts the behavior of the Sherrington-Kirkpatrick spin glass model.
- It agrees with rigorous results (where available) from the cavity method and interpolation techniques.
- It provides quantitative predictions for random matrix spectra, neural network capacity, etc.

The physicist’s attitude is: the method works, even if we don’t fully understand why. For this paper, we use replica results as established theoretical benchmarks, while acknowledging their non-rigorous status.

4.6.3 The Spin Glass—Neural Network Isomorphism

To apply replica theory to neural networks, we establish a precise mapping between concepts.

Definition 4.13 (Spin Glass—Neural Network Dictionary). The table in Table 3 provides a working dictionary between the two domains. Importantly, this dictionary should be read as a translation layer between questions and observables, not as an unconditional claim that neural network training is a spin glass.

Why This Mapping Works:

The deep reason for the spin glass—neural network analogy is structural similarity in the disorder averaging problem:

1. In both systems, the “environment” (couplings J or data $\mathcal{D}$) is random but frozen during the dynamics.
2. In both systems, we care about typical behavior (averaged over disorder), not behavior for a specific disorder realization.
3. In both systems, the interplay between disorder and thermal fluctuations creates complex, non-convex energy/loss landscapes with multiple local minima.

For certain restricted model classes, the correspondence can be made essentially exact (e.g., the Hopfield network can be written as a spin system with Hebbian couplings). For modern deep networks trained with SGD, the dictionary is better viewed as a controlled approximation whose validity must be argued case-by-case.

4.7 The Jamming Transition: A Geometric Perspective on Interpolation

The final foundational concept connects the algebraic notion of “interpolation threshold” to a geometric/physical picture: the Jamming Transition.

Definition 4.14 (Jamming Transition in Physical Systems). In condensed matter physics, jamming refers to the transition of a system from a “fluid” state (particles can rearrange freely) to a “solid” state (particles are locked in place by mutual constraints).

Examples include:

- Granular materials (sand, grains) becoming rigid under compression.
- Colloidal suspensions solidifying at high density.
- Foams and emulsions acquiring mechanical rigidity.

The jamming transition occurs at a critical density (or constraint count) where the system becomes marginally stable: the number of constraints exactly matches the number of degrees of freedom.

Table 3. Analogy between spin glass physics and neural network training

Concept	Spin Glass (Physics)	Neural Network (ML)
Degrees of freedom	Spin variables $\sigma_i \in \{-1, +1\}$	Weight parameters $W_i \in \mathbb{R}$ (components of $W \in \mathbb{R}^P$)
Energy function	Hamiltonian $\mathcal{H}(\sigma; J)$	Loss function $\mathcal{L}(W; \mathcal{S})$ (or population loss $\mathcal{L}(W; \mathcal{D})$)
Quenched disorder	Random couplings J_{ij} (fixed but random)	Dataset $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^n$ (fixed but random sample from $\mathcal{D}$)
Temperature T	Thermal fluctuations	Effective temperature $T_{\text{eff}} \approx \eta/B$ (SGD noise level)
Thermal average $\langle \cdot \rangle$	$\int (\cdot) P_{\text{eq}}(\sigma) d\sigma$	Expectation over the training-induced weight distribution (e.g., stationary $P_\infty(W)$)
Disorder average $\overline{(\cdot)}$	$\mathbb{E}_J[(\cdot)]$	$\mathbb{E}_{\mathcal{S} \sim \mathcal{D}^n}[(\cdot)]$ (average over dataset realizations)
Order parameter	Replica overlap $q = \frac{1}{N} \sum_i \sigma_i^{(1)} \sigma_i^{(2)}$	Weight alignment/similarity between independently trained replicas
Phase transition	Paramagnet $\leftrightarrow$ Spin glass phase	Under-parameterized $\leftrightarrow$ Over-parameterized regime

Notes: The mapping above is a pragmatic dictionary: it suggests a potential mechanism and provides a language for transferring tools (e.g., disorder averages, replica overlaps) when the assumptions of the physical approximation are plausibly satisfied.

Definition 4.15 (Jamming Transition in Neural Networks). For a neural network with P parameters trained on N data points, the jamming transition occurs at the critical ratio:

$$\gamma_c = \frac{P}{N} = 1 \quad (28)$$

The three regimes are characterized as follows:

- Under-parameterized ($P < N$, $\gamma < 1$): More constraints (data points) than degrees of freedom (parameters). The system is “over-constrained” or “jammed.” Perfect interpolation ($\mathcal{L}_{\text{train}} = 0$) is generically impossible.
- Critical ($P \approx N$, $\gamma \approx 1$): Constraints and degrees of freedom are balanced. The system is marginally stable—any small perturbation can cause large rearrangements. The loss landscape is maximally rugged with many sharp, narrow minima.
- Over-parameterized ($P > N$, $\gamma > 1$): More degrees of freedom than constraints. The system is “under-constrained” or “unjammed.” A $(P-N)$ -dimensional manifold of global minima exists, providing “room to move.”

Remark 4.9 (Why Jamming Explains the Double Descent Peak). The jamming perspective provides a geometric explanation for why test error peaks at the interpolation threshold:

1. At $\gamma < 1$: The model cannot fit all training data. High training error implies high bias, but the constrained solution is “smoothed” and may generalize acceptably.
2. At $\gamma = 1$: The model barely fits all training data. To achieve $\mathcal{L}_{\text{train}} = 0$, the weights must occupy extremely specific, sharp configurations. These sharp minima are highly sensitive to the particular training samples and generalize poorly—any perturbation (e.g., different test data) causes large errors.

3. At $\gamma > 1$: The model has excess capacity. Among the manifold of interpolating solutions, SGD (with its anisotropic noise) selects flat configurations that are robust to perturbations and generalize well.

The “peak” of double descent is thus the jamming point: the most constrained, most fragile, worst-generalizing configuration.

4.8 Summary: The Conceptual Foundation

We have now established the formal definitions and physical justifications for the key concepts used throughout this paper:

1. SGD as Langevin dynamics (Definition 4.2): Mini-batch noise introduces stochasticity that can be modeled as continuous-time diffusion.
2. Fokker-Planck equation (Definition 4.4): The evolution of probability distributions over weight space, with drift toward low loss and diffusion due to noise.
3. Hessian and curvature (Definitions 4.5, 4.7): The second derivative matrix characterizes local geometry; eigenvalues determine sharpness/flatness of minima.
4. Equilibrium vs. non-equilibrium (Definitions 4.9, 4.11): Classical theory assumes Gibbs distribution; SGD dynamics may instead reach a non-equilibrium steady state.
5. Replica method (Definition 4.12): A technique for averaging over dataset randomness, with a precise spin glass—neural network correspondence.
6. Jamming transition (Definition 4.15): The interpolation threshold as a phase transition between under-constrained and over-constrained regimes.

With these foundations in place, we now turn to reviewing how equilibrium theories (RMT, Replica) explain double descent in linear models, before developing the non-equilibrium framework that addresses deep networks. We

document the specific “friction points” where these theoretical tools encounter their limits in Appendix Appendix 1.1.

5. Classical Explanations: The Equilibrium Regime

Following the empirical identification of the double descent phenomenon, the initial wave of theoretical explanations primarily leveraged tools from Equilibrium Statistical Physics and Random Matrix Theory (RMT). These approaches view the learning problem through a static lens, analyzing the properties of the solution space at equilibrium (or the minimum norm solution) rather than the dynamics of reaching it.

5.1 Random Matrix Theory and Linear Models

One of the most rigorous mathematical explanations for double descent in linear regimes comes from the analysis of ridgeless least squares regression. Hastie et al. 2019 utilized RMT to derive the asymptotic risk of the minimum ℓ_2 -norm solution.

In the over-parameterized regime ($P > N$), where the data covariance matrix $X^T X$ becomes singular, the theory predicts that the peak in generalization error occurs precisely at the interpolation threshold ($\gamma = P/N = 1$). Mathematically, this is attributed to the smallest non-zero eigenvalue of the sample covariance matrix approaching zero, causing the variance of the estimator to diverge. While this RMT approach successfully reproduces the double descent curve analytically, it is strictly limited to linear models or fixed-feature kernel methods and does not account for the feature learning capabilities of deep neural networks.

5.2 Replica Theory and Random Features

Parallel to RMT, techniques from the statistical physics of disordered systems, specifically Replica Theory, have been applied to Random Feature (RF) models. By establishing an isomorphism between the learning problem and a Spin Glass system:

- The loss function corresponds to the Hamiltonian $\mathcal{H}$.
- The dataset $\mathcal{D}$ acts as the quenched disorder (analogous to random magnetic interactions J_{ij}).
- The weights W represent the thermal degrees of freedom (spins).

Researchers such as Mezard, Parisi, and Virasoro 1987; Gerace et al. 2020 employed the “Replica Trick” to compute the partition function and free energy of the system in the thermodynamic limit. This framework confirmed that double descent is, in these models, a phase transition inherent to the equilibrium state of the system. However, like RMT, standard Replica analysis assumes a static equilibrium (Gibbs distribution) and does not inherently capture the algorithmic trajectory of Stochastic Gradient Descent (SGD).

Remark 5.1 (On benchmarks: RMT vs. Replica). For linear models and ridgeless regression, the asymptotic risk is derived rigorously via Random Matrix Theory (Marchenko-Pastur law, Stieltjes transforms) without recourse to the replica trick. We use these rigorous RMT results as our primary benchmark for linear models. The replica method is introduced for its potential extension to more complex settings (random features, nonlinear activations), where RMT alone is insufficient. When we compare NESP predictions against “equilibrium predictions,” the reader should understand: for linear models, “equilibrium” means rigorous RMT; for random feature models, it means heuristic replica results whose reliability must be assessed on a case-by-case basis.

5.2.1 The Replica Calculation: Step-by-Step (Expanded)

We provide a detailed exposition of the replica calculation, highlighting both its power and its fragility.

Step 1: Setting Up the Disorder Average. Consider a learning problem where the dataset $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^n$ is drawn i.i.d. from some distribution $\mathcal{D}$. The partition function for the “thermal” distribution over weights is:

$$Z(\mathcal{S}) = \int_{\mathbb{R}^P} \exp(-\beta \mathcal{L}(W; \mathcal{S})) dW \quad (29)$$

where β is an inverse temperature and $\mathcal{L}(W; \mathcal{S})$ is the empirical loss.

The quenched free energy is:

$$F_{\text{quenched}} = -\frac{1}{\beta} \mathbb{E}_{\mathcal{S}} [\ln Z(\mathcal{S})] \quad (30)$$

The expectation of a logarithm is analytically intractable. This is the fundamental obstacle.

Step 2: The Replica Identity. The replica trick exploits the identity:

$$\ln Z = \lim_{n \rightarrow 0} \frac{Z^n - 1}{n} \quad (31)$$

Thus:

$$\mathbb{E}[\ln Z] = \lim_{n \rightarrow 0} \frac{\mathbb{E}[Z^n] - 1}{n} \quad (32)$$

For integer $n \geq 1$, we can write:

$$Z^n = \prod_{a=1}^n \int dW^{(a)} \exp(-\beta \mathcal{L}(W^{(a)}; \mathcal{S})) \quad (33)$$

This is the partition function of n replicas of the original system, all sharing the same dataset $\mathcal{S}$.

Step 3: Disorder Average Before Replica Limit. The key insight: for integer n , we can compute $\mathbb{E}_{\mathcal{S}}[Z^n]$ because the expectation is now outside the logarithm:

$$\mathbb{E}_{\mathcal{S}}[Z^n] = \mathbb{E}_{\mathcal{S}} \left[\prod_{a=1}^n \int dW^{(a)} \exp \left(-\beta \sum_{a=1}^n \mathcal{L}(W^{(a)}; \mathcal{S}) \right) \right] \quad (34)$$

Exchanging the order of integration and expectation (justified for bounded losses or by regularization):

$$\mathbb{E}_{\mathcal{S}}[Z^n] = \int \prod_{a=1}^n dW^{(a)} \mathbb{E}_{\mathcal{S}} \left[\exp \left(-\beta \sum_{a=1}^n \mathcal{L}(W^{(a)}; \mathcal{S}) \right) \right] \quad (35)$$

The disorder average now couples the replicas: even though each $W^{(a)}$ is independent a priori, the shared dataset $\mathcal{S}$ creates effective interactions between them.

Step 4: Order Parameters and Saddle Point. In the thermodynamic limit ($n, P \rightarrow \infty$ with P/n fixed), the integral is dominated by a saddle point. The relevant order parameters are the replica overlaps:

$$q_{ab} = \frac{1}{P} \sum_{i=1}^P W_i^{(a)} W_i^{(b)}, \quad a \neq b \quad (36)$$

These measure the similarity between different replicas—i.e., between independently trained models on the same dataset.

Replica Symmetric (RS) Ansatz: The simplest assumption is that all replicas are statistically equivalent:

$$q_{ab} = q \quad \text{for all } a \neq b \quad (37)$$

Under this ansatz, the saddle point equations yield a self-consistent equation for q , which can be solved to obtain the free energy.

Step 5: The Fragile $n \rightarrow 0$ Limit. Having computed $\mathbb{E}[Z^n]$ as a function of integer n , we must now:

1. Find an analytic expression valid for real n .
2. Evaluate the limit $n \rightarrow 0$.

Mathematical Warning: This step is not rigorous. Analytic continuation is not unique, and the physical meaning of “ $n \rightarrow 0$ replicas” is unclear. The justification is entirely a posteriori: the method gives correct results when checked against other methods or simulations.

Remark 5.2 (Replica Symmetry Breaking (RSB)). In some systems (notably the Sherrington-Kirkpatrick spin glass), the RS ansatz gives wrong results. The correct solution requires Replica Symmetry Breaking—a hierarchical structure where different pairs of replicas have different overlaps q_{ab} .

In the ML context, RSB may be relevant when:

- The loss landscape has many disconnected minima (“glassy” phase).
- Different training runs on the same data converge to qualitatively different solutions.

Determining whether RSB occurs in realistic neural network training is an open problem.

5.2.2 What Replica Theory Achieves and What It Misses

Achievements:

- Exact asymptotic formulas for generalization error in linear and random-feature models.
- Prediction of the interpolation threshold and double descent peak location.
- Connection between generalization and properties of the solution (e.g., norm, flatness).

Limitations:

- Static equilibrium: Assumes the Gibbs distribution, ignoring optimization dynamics.
- Linear/random features only: Extension to trained deep networks is non-trivial.
- Thermodynamic limit: Finite-size effects (realistic n, P) require separate analysis.
- Mathematical non-rigor: The $n \rightarrow 0$ limit is a formal trick, not a theorem.

Remark 5.3 (The Bridge to NESP). The replica method answers: “What is the equilibrium distribution over weights?” Our NESP framework asks: “What is the dynamical distribution reached by SGD?” These may differ. The replica calculation provides a benchmark—if SGD converges to equilibrium, replica predictions should hold. Deviations from replica predictions indicate non-equilibrium effects.

What static theories cannot explain. Despite their successes in linear and random-feature settings, equilibrium approaches remain silent on three empirical regularities: (i) epoch-wise double descent, where the peak appears as a function of training time at fixed model size; (ii) the crucial role of the optimizer—SGD with small batches generalises better than exact gradient descent; and (iii) the dependence on the training path, e.g. learning rate schedules and early stopping. These phenomena point to an intrinsically dynamical mechanism that equilibrium theories, by construction, cannot capture.

6. The Gap: From Static to Dynamic

6.1 Conceptual Bridges: Making Implicit Connections Explicit

Before proceeding to the NESP framework, we make explicit several conceptual bridges that are often assumed but whose connections require justification.

6.1.1 Bridge 1: Loss Landscape $\leftrightarrow$ Physical Energy

The claim: The loss function $\mathcal{L}(W)$ plays the role of “energy” in our physical analogy.

The justification:

- Both are scalar functions of the system state (W or spin configuration σ).
- Both define a “landscape” with minima, maxima, and saddles.
- Optimization seeks low-loss states; thermodynamics favors low-energy states (at $T \rightarrow 0$).

The caveat: Physical energy has units (Joules); loss is dimensionless. The “temperature” $T_{\text{eff}} = \eta/B$ is not a real temperature but a ratio of hyperparameters. The analogy is structural, not ontological.

6.1.2 Bridge 2: Dataset Randomness $\leftrightarrow$ Quenched Disorder

The claim: The random dataset $\mathcal{S}$ plays the role of “quenched disorder” (random couplings J_{ij} in spin glasses).

The justification:

- Both are random variables that are fixed during the dynamics (training or thermal relaxation).
- Both create a “rugged” landscape whose specific shape depends on the realization.
- Both require averaging over realizations to obtain typical behavior.

The caveat: In spin glasses, J_{ij} are typically i.i.d. Gaussian. Dataset samples (x_i, y_i) have correlations (e.g., if x_i are images, they share low-level statistics). The disorder structure may differ in important ways.

6.1.3 Bridge 3: SGD Noise $\leftrightarrow$ Thermal Fluctuations

The claim: Mini-batch stochasticity provides “effective thermal fluctuations” that allow the system to explore the loss landscape.

The justification:

- Both introduce randomness into an otherwise deterministic dynamics.
- Both enable escape from local minima (barrier crossing).
- Both have a “temperature-like” parameter controlling fluctuation magnitude (T vs. η/B).

The caveat: Thermal noise satisfies detailed balance (leading to Gibbs equilibrium). SGD noise is anisotropic and state-dependent, generically violating detailed balance. This is precisely why we need NESP, not equilibrium statistical mechanics.

6.1.4 Bridge 4: Interpolation Threshold $\leftrightarrow$ Jamming Transition

The claim: The point $P \approx N$ (parameters $\approx$ data points) is analogous to the jamming transition in granular materials.

The justification:

- In both cases, a “critical” point separates under-constrained and over-constrained regimes.
- In both cases, the system becomes “marginally stable” at criticality.
- In both cases, response functions (e.g., susceptibility, condition number) diverge.

The caveat: Jamming is a geometric/mechanical transition; interpolation is an algebraic condition. The connection is suggestive but not proven to be in the same universality class. Critical exponents may differ.

6.1.5 Bridge 5: Flat Minima $\leftrightarrow$ Good Generalization

The claim: “Flat” minima (small Hessian eigenvalues) generalize better than “sharp” minima.

The justification:

- PAC-Bayes bounds relate generalization to the “sharpness” of the posterior around learned weights.
- Intuitively, flat minima are “robust”—small perturbations (e.g., from test vs. train distribution) cause small loss changes.
- Empirical studies correlate flatness metrics with generalization.

The caveat: “Flatness” is not reparameterization-invariant. Scaling the weights and inversely scaling the next layer’s weights preserves the function but changes the Hessian. Recent work suggests that volume of the basin, not just curvature, may be the relevant quantity.

Remark 6.1 (Information-Geometric Resolution: Fisher Flatness). The reparameterization problem admits a principled resolution via the Fisher Information Matrix (FIM). For a model $p_{\theta}(y | x)$, the FIM is defined as

$$\mathcal{I}(\theta)_{ij} = \mathbb{E}_{(x,y) \sim p_{\text{data}}} [\partial_i \log p_{\theta}(y | x) \partial_j \log p_{\theta}(y | x)]. \quad (38)$$

Unlike the Hessian, $\mathcal{I}(\theta)$ transforms as a Riemannian metric tensor under reparameterization $\theta \mapsto \phi(\theta)$: $\mathcal{I}_{\phi} = J^{\top} \mathcal{I}_{\theta} J$, where $J = \partial \phi / \partial \theta$. Consequently, flatness measured in the Fisher metric is intrinsic to the statistical model, not an artifact of parameterization.

For well-specified models at a local minimum θ^* of the population risk, the classical identity $\mathcal{I}(\theta^*) = H(\theta^*)$ holds (the Bartlett identity), so the Hessian-based flatness coincides with Fisher flatness at the optimum. Away from the optimum—precisely where the NESP dynamics operate—the two diverge, and the Fisher metric provides the geometrically natural notion of curvature.

This connects directly to PAC-Bayes generalization bounds: the KL divergence $\text{KL}(Q|0P)$ for a Gaussian posterior $Q = \mathcal{N}(\theta^*, \sigma^2 I)$ around a minimum satisfies $\text{KL}(Q|0P) \approx \frac{1}{2\sigma^2} \theta^{*\top} \mathcal{I}(\theta^*) \theta^*$ to leading order, making the Fisher norm the natural sharpness measure appearing in generalization bounds. Reformulating the NESP curvature–noise

coupling in terms of $\mathcal{I}$ rather than H would yield a reparameterization invariant version of the “survival of the flattest” mechanism, resolving the caveat above.

Remark 6.2 (Why Bridges Matter). These bridges are not merely pedagogical. They define the scope of validity of our framework. When a bridge’s caveats are violated (e.g., non-i.i.d. data, extreme learning rates), the physical analogy may break down. Understanding the bridges helps us know when to trust the theory and when to be skeptical.

6.2 The Static-Dynamic Gap: A Precise Statement

While equilibrium theories successfully describe double descent in “shallow” and linear models, a fundamental gap remains when applying these concepts to modern Deep Neural Networks (DNNs). The elegance of the RMT and Replica approaches—their ability to yield closed-form expressions for risk in the thermodynamic limit—comes at a cost: they are fundamentally static theories. They answer the question “what is the equilibrium state?” but remain silent on “how does the system reach it?”

This limitation becomes particularly evident when we confront the empirical reality of deep learning. The very success of SGD—an algorithm that injects noise into the optimization process—hints that the journey matters as much as the destination. If equilibrium were the whole story, why would noisy optimization outperform exact gradient descent? Why would early stopping improve generalization? These anomalies suggest that something fundamentally non-equilibrium is at play.

This gap is characterized by three critical distinctions:

1. **Non-convexity:** Unlike the convex landscapes of linear regression, DNN loss landscapes are highly non-convex, possessing a manifold of global minima (interpolation solutions). The specific solution selected depends heavily on the optimization path.
2. **Feature Learning:** Deep networks do not merely re-weight fixed features; they actively learn representations. Static theories cannot account for the evolution of the feature space during training.
3. **Epoch-wise Double Descent:** Perhaps the most compelling evidence against a purely static explanation is the observation of epoch-wise double descent Nakkiran et al. 2021; Heckel and Yilmaz 2020. Even with fixed model size and data size, the test error exhibits a double descent profile as a function of training time.

The existence of epoch-wise double descent suggests that “model complexity” is not a static scalar determined solely by parameter count, but a dynamic quantity—an Effective Model Complexity—that evolves under the dynamics of SGD. Consequently, a complete theory must be grounded in Non-Equilibrium Statistical Physics (NESP).

6.3 Synthesis: The Three Phases of Double Descent

Integrating NESP dynamics with the Jamming landscape provides a coherent narrative for the double descent phenomenon, separating the process into three distinct phases based on the interaction between the optimization trajectory and the landscape geometry:

Phase 1: Signal Learning (First Descent) The dynamics are dominated by “fast modes” (directions corresponding to large eigenvalues of the Hessian). The model learns the dominant, low-frequency signal components of the data. In this phase, the landscape is relatively smooth in the relevant directions, and the error decreases as bias is reduced.

Phase 2: The Jamming Peak As training progresses to “slow modes” (fine details and noise), the dynamics interact with the critical jamming nature of the landscape near the interpolation threshold ($P \approx N$). The SGD trajectory gets trapped in sharp, unstable minima associated with fitting high-frequency noise. Due to the ruggedness of the jammed landscape, the model overfits, leading to a spike in generalization error.

Phase 3: Non-Equilibrium Relaxation (Second Descent) In the over-parameterized regime, the NESP mechanism dominates. The anisotropic noise of SGD acts as an annealing force. Because the “effective temperature” or noise magnitude is higher in sharp regions (due to larger gradient variance), the system is stochastically kicked out of the sharp minima (overfitting) and drifts into the vast, flat basins available in the high-dimensional unjammed landscape. This relaxation process, driven by the implicit regularization of SGD noise, filters out the noise fitted in Phase 2, reducing the test error despite the high model capacity.

7. Non-Equilibrium Statistical Physics Framework for Deep Learning Dynamics

7.1 Motivation: Why Non-Equilibrium?

Before diving into formalism, we wish to articulate why this particular theoretical direction is compelling—not merely as a technical exercise, but as a principled methodological stance.

The history of theoretical machine learning, as reviewed in Sections 1–4, reveals a recurring pattern: researchers borrow mathematical frameworks from adjacent fields (statistics, information theory, statistical physics), apply them to learning systems, and celebrate when they reproduce known phenomena. Yet this “borrowing” often comes with hidden assumptions that may not transfer cleanly. The bias-variance decomposition assumes squared loss and fixed features. VC theory assumes worst-case distributions. Replica theory assumes thermodynamic equilibrium.

Double descent, as documented in Belkin et al. 2019a and Nakkiran et al. 2019, is not merely an anomaly to be explained away—it is a symptom that our borrowed frameworks are reaching their limits. When a phenomenon systematically violates the predictions of classical theory (bias-variance tradeoff), yet consistently appears across architectures and datasets, we face a choice: patch the old theory with ad-hoc corrections, or seek a new conceptual foundation.

We pursue the latter approach. The Non-Equilibrium Statistical Physics (NESP) framework developed in this chapter is motivated by a simple observation: SGD is not an equilibrium process. It does not sample from a Boltzmann distribution. It does not minimize free energy. It is a driven dynamical system, constantly perturbed by mini-batch noise, whose asymptotic state depends critically on the path it takes through parameter space.

This perspective resonates with a broader shift in theoretical physics—from asking “what is the ground state?” to “how does the system evolve?” In condensed matter, this shift gave us the theory of glasses, aging, and non-ergodic dynamics. In machine learning, we propose that it will provide a theory of why deep learning works—not despite its apparent over-parameterization, but because of the specific noise structure that over-parameterization enables.

The following sections develop this intuition into a mathematical framework. The goal is not to provide final answers, but to establish a language in which the right questions can be asked.

7.2 Overview and Connection to Prior Sections

The preceding sections established two complementary viewpoints:

- **Equilibrium Statistical Physics** (Section 5): Replica Theory and RMT successfully describe double descent in linear/fixed-feature regimes, predicting the peak at the interpolation threshold $\gamma = P/N = 1$. However, these approaches assume a static Gibbs distribution (Eq. equation 39 below) and cannot explain epoch-wise double descent.
- **Dynamical Observations** (Section 6): The Langevin approximation and Jamming analogy suggest that SGD dynamics are not just the loss landscape—play a crucial role. The “implicit regularization” of SGD noise hints at a selection mechanism beyond energy minimization.

We now attempt to synthesize these perspectives into a unified Non-Equilibrium Statistical Physics (NESP) framework. The key conceptual move is to treat the noise structure of SGD not as a nuisance to be averaged over, but as the primary agent of generalization.

7.3 The Failure of Equilibrium Assumptions

Classical theories—both VC-dimension based and statistical physics based—share an implicit assumption: the final state of the trained model can be characterized by an equilibrium distribution. In the statistical physics formulation, this takes the form of the Boltzmann-Gibbs distribution:

$$P_{\text{eq}}(W) \propto \exp(-\beta \mathcal{L}(W)) \quad (39)$$

where the probability of finding the system in configuration W depends solely on the loss function (energy) $\mathcal{L}(W)$ at that point, mediated by an inverse temperature β .

However, the double descent phenomenon poses a fundamental challenge to this equilibrium viewpoint:

The Degeneracy Problem. In the interpolation regime ($P > N$), there exist infinitely many global minima with $\mathcal{L}(W) \approx 0$. If Eq. equation 39 were valid, all such minima should be equally probable (weighted by their local volume). Yet empirically, SGD consistently converges to solutions with superior generalization—the so-called “flat” minima—while avoiding “sharp” minima of identical loss value.

The Selection Paradox. This selective behavior implies that something beyond the static energy landscape determines the final state. The “selection” must arise from the dynamics of the approach itself.

These observations motivate our central postulate:

Postulate 7.1 (Non-Equilibrium Learning Dynamics). The training of Deep Neural Networks under SGD constitutes a non-equilibrium and potentially non-ergodic process. The asymptotic state is not determined by the static energy function $\mathcal{L}(W)$ alone, but by the access dynamics—the specific pathway through which the system explores configuration space.

7.4 Anisotropic SGD Noise as the Driving Force

To operationalize Postulate 7.1, we examine the continuous-time approximation of SGD. The standard formulation yields a Stochastic Differential Equation (SDE):

$$dW_t = -\nabla \mathcal{L}(W_t) dt + \sqrt{\frac{\eta}{B}} \Sigma(W_t)^{1/2} dB_t \quad (40)$$

where η is the learning rate, B is the batch size, and dB_t denotes standard Brownian motion.

The critical element is the noise covariance matrix $\Sigma(W)$. In equilibrium thermodynamics, the diffusion coefficient is typically a scalar (temperature). However, in SGD, $\Sigma(W)$ is a full matrix that varies across configuration space.

This state-dependence of the noise is the key departure from classical statistical physics. In the Replica Theory framework (Section 5.2), the “temperature” is a global

parameter conjugate to the energy. Here, temperature becomes local—different directions in parameter space experience different effective temperatures, determined by the local curvature. This is why equilibrium analysis, which assumes a uniform temperature, fails to capture the selective pressure toward flat minima.

Hypothesis 7.1 (Curvature-Noise Coupling). The SGD noise covariance $\Sigma(W)$ is not isotropic white noise. Along the training trajectory (away from global minima), its structure is intrinsically coupled to the local curvature of the loss landscape, approximated by the Hessian $H(W) = \nabla^2 \mathcal{L}(W)$:

$$\Sigma(W) \approx \alpha(W) H(W) + \mathcal{O}(H^2), \quad \text{for } \mathcal{L}(W) > 0 \quad (41)$$

where the coupling strength $\alpha(W) > 0$ depends on the residual error magnitude, the data distribution, and batch statistics. Critically, $\alpha(W) \rightarrow 0$ as $\mathcal{L}(W) \rightarrow 0$: the coupling vanishes at global minima where the prediction error is zero. This vanishing is not a weakness but the mechanism by which flat minima become dynamically stable—once the system enters a flat basin with near-zero loss, the noise is suppressed and the system is trapped.

The physical consequences of Hypothesis 7.1 are profound:

- Sharp directions (large Hessian eigenvalues λ_i): The noise amplitude is large ($\Sigma_{ii} \sim \lambda_i$), creating strong fluctuations that destabilize the system.
- Flat directions (small or zero λ_i): The noise is suppressed ($\Sigma_{ii} \rightarrow 0$), allowing the system to “freeze” into these configurations.

This creates a position-dependent “effective temperature” in parameter space. The system is not merely descending an energy gradient; it is simultaneously being expelled from high-curvature regions by the kinetic pressure of anisotropic noise.

7.5 From Continuous to Discrete: The Edge of Stability

The Langevin SDE of Eq. equation 40 captures the continuous-time limit $\eta \rightarrow 0$. However, real SGD training operates with finite learning rates. This discretization is not a mere technical nuisance—it fundamentally alters the dynamical regime and gives rise to the Edge of Stability (EoS) phenomenon, which is the primary engine of the attractor selection mechanism we propose.

7.5.1 Discrete SGD as a Dynamical System

The discrete-time SGD update with learning rate η is:

$$W_{t+1} = W_t - \eta g_{B_t}(W_t), \quad g_{B_t}(W) = \frac{1}{B} \sum_{i \in B_t} \nabla \ell(W; x_i, y_i) \quad (42)$$

where B_t is a randomly sampled mini-batch of size B . This is a nonlinear stochastic dynamical system in the d -dimensional parameter space $\mathbb{R}^d$. Unlike the continuous Langevin flow which is guaranteed to descend the energy,

the discrete map can—and does—become unstable when the product $\eta \cdot \lambda_{\max}(H)$ exceeds a critical threshold.

7.5.2 The Stability Threshold: Where $\lambda_{\max} \approx 2/\eta$

Consider the deterministic gradient descent (GD) limit ($B = n$, full batch). Linearizing the update around a point W where $\nabla \mathcal{L}(W) \approx 0$ (i.e., near a critical point):

$$W_{t+1} \approx W_t - \eta H(W_t)(W_t - W^*) \quad (43)$$

Defining the error $\delta_t = W_t - W^*$, the linearized dynamics are:

$$\delta_{t+1} = (I - \eta H) \delta_t \quad (44)$$

Stability requires the spectral radius $\rho(I - \eta H) < 1$. For a Hessian eigenvalue λ_i , the corresponding multiplier is $|1 - \eta \lambda_i|$. The condition $|1 - \eta \lambda_i| < 1$ yields:

$$0 < \lambda_i < \frac{2}{\eta} \quad \text{for all } i \quad (45)$$

Hence the sharpness $r \equiv \lambda_{\max}(H)$ must satisfy $r < 2/\eta$ for local stability.

The remarkable empirical discovery is that GD and SGD do not settle comfortably below this threshold. Instead, during training of overparameterized models, the sharpness progressively increases until it reaches the vicinity of $2/\eta$, where the dynamics become marginally stable. This is the Edge of Stability Cohen et al. 2021.

7.5.3 Progressive Sharpening: The Route to EoS

The mechanism of progressive sharpening operates as follows:

1. Early phase: The loss is large, gradients are dominated by the bulk error signal. The Hessian spectrum is relatively uniform; $\lambda_{\max}$ is moderate. The system descends rapidly.
2. Sharpening phase: As training error decreases, the gradient aligns increasingly with the top eigenvector(s) of H . The update $-\eta g_t$ has a component along the leading eigendirection that, while reducing the loss, also increases the curvature in that direction. This is because the loss function near the interpolation manifold develops “ravines”—directions where the loss is nearly zero but curvature grows as the model fits finer details of the data Cohen et al. 2021.
3. EoS phase: $\lambda_{\max}$ approaches $2/\eta$. The linearized dynamics become oscillatory along the sharpest direction: the multiplier $1 - \eta \lambda_{\max} \rightarrow -1$, producing the characteristic “bouncing” behavior. The system does not diverge because nonlinearities in the loss landscape bound the oscillations. The dynamics are now non-monotonic: the loss oscillates at the scale of individual steps while slowly drifting on longer timescales.

This progressive sharpening is the microscopic engine that drives the system toward the Edge of Stability. Once at EoS, the dynamics change qualitatively: the system is no longer converging to a point but executing a quasi-periodic orbit around a low-dimensional attractor.

7.5.4 EoS and the Onset of Double Descent

The connection between EoS and double descent is direct:

- Below the interpolation threshold ($P < N$): The model cannot fit all training data. The residual error keeps $\lambda_{\max}$ below the EoS threshold. The dynamics are in the “classical” convergent regime.
- At the interpolation threshold ($P \approx N$): The model can now interpolate, but only by developing sharp curvature in certain directions. The Hessian becomes ill-conditioned. Progressive sharpening begins— $\lambda_{\max}$ rises toward $2/\eta$. The system enters EoS. This is where test error peaks: the marginal stability creates high-variance predictions.
- Beyond the threshold ($P \gg N$): The overparameterization provides redundant degrees of freedom. The system can interpolate while keeping most directions flat. The EoS dynamics now selectively sharpen only the necessary directions while leaving a large null space. The “survival of the flattest” mechanism (Section 7.7) becomes operational, and test error decreases.

7.6 Lyapunov Analysis of SGD Dynamics

The connection between the Hessian spectrum and SGD stability can be made rigorous through the lens of Lyapunov exponents, a tool borrowed from the theory of nonlinear dynamical systems and chaos. Lyapunov exponents quantify the average exponential rate at which nearby trajectories diverge or converge in phase space, providing a spectrum-based characterization of stability that goes beyond static curvature measures.

7.6.1 Finite-Time Lyapunov Exponents for SGD

For the discrete SGD map (Eq. 42), the Finite-Time Lyapunov Exponent (FTLE) associated with an initial perturbation δW_0 over a training interval $[t_0, t_0 + T]$ is defined as:

$$\Lambda_T(W_{t_0}, \delta W_0) = \frac{1}{T} \ln \frac{|\delta W_T|}{|\delta W_0|} \quad (46)$$

where δW_T is the evolved perturbation after T steps of the linearized dynamics. The spectrum of FTLEs $\{\Lambda_T^{(1)}, \Lambda_T^{(2)}, \dots, \Lambda_T^{(d)}\}$, sorted in descending order, characterizes the local stability structure of the training trajectory.

7.6.2 Jacobian of the SGD Step and Hessian Connection

The linearized perturbation dynamics are governed by the Jacobian of one SGD step:

$$\delta W_{t+1} = J_t \delta W_t, \quad J_t = I - \eta H_{B_t}(W_t) \quad (47)$$

where H_{B_t} is the mini-batch Hessian. The FTLEs are computed from the product of Jacobians along the trajectory:

$$\Lambda_T^{(i)} = \frac{1}{T} \ln \sigma_i \left(\prod_{t=t_0}^{t_0+T-1} J_t \right) \quad (48)$$

where σ_i denotes the i -th singular value.

For small η , expanding the FTLE in terms of the Hessian spectrum:

$$\Lambda_T^{(i)} \approx -\eta \bar{\lambda}_i + \frac{\eta^2}{2} (\bar{\lambda}_i^2 + \text{Var}_t[\lambda_i]) + \mathcal{O}(\eta^3) \quad (49)$$

where $\bar{\lambda}_i = \frac{1}{T} \sum_t \lambda_i(H_{B_t})$ is the time-averaged i -th Hessian eigenvalue along the trajectory, and $\text{Var}_t[\lambda_i]$ captures temporal fluctuations due to mini-batch sampling.

Key insight: Eq. equation 49 establishes a direct quantitative link between the Lyapunov spectrum and the Hessian spectrum. Directions with large $\bar{\lambda}_i$ (sharp curvature) have negative Lyapunov exponents of large magnitude—they are strongly contracting. However, the η^2 correction term reveals that eigenvalue variance (from mini-batch noise) can partially cancel this contraction, and for eigenvalues near $2/\eta$, the first-order term changes sign—the direction becomes expanding (positive Lyapunov exponent). This is the dynamical signature of EoS: the sharpest direction flips from stable (contracting) to unstable (expanding) at the threshold.

7.6.3 FTLE Spectrum and Generalization

The structure of the FTLE spectrum along the training trajectory provides a dynamical characterization of generalization:

- Negative FTLEs (contracting directions): The trajectory is stable against perturbations. These correspond to “memorized” features that are brittle—small weight perturbations cause large prediction changes.
- Zero or near-zero FTLEs (neutral directions): The trajectory is marginally stable. These are the “flat” directions of the loss landscape. Perturbations neither grow nor shrink; the model’s predictions are robust. Good generalization correlates with a large fraction of near-zero FTLEs.
- Positive FTLEs (expanding directions): The trajectory is locally chaotic. These emerge at EoS when $\eta \lambda_i > 2$. They enable exploration—the system can escape sharp basins and discover flatter regions. However, sustained positive FTLEs would cause training divergence; in practice they appear only transiently or for the very sharpest directions.

This analysis provides a dynamical reinterpretation of Hypothesis 7.1: the curvature-noise coupling $\Sigma \sim H$ implies that the Lyapunov spectrum and the Hessian spectrum are not independent—they are linked through Eq. equation 49. The “survival of the flattest” can be restated in the language of Lyapunov analysis: attractors with a larger fraction of near-zero Lyapunov exponents are more stable under the SGD dynamics and hence are preferentially selected.

7.6.4 Relation to Sharpness Dimension

The concept of sharpness dimension Cohen et al. 2021—discussed extensively in the dynamical systems perspective—finds a natural formulation here. Define the effective Lyapunov dimension of an attractor by the Kaplan-Yorke formula:

$$D_L = k + \frac{\sum_{i=1}^k \Lambda^{(i)}}{|\Lambda^{(k+1)}|} \quad (50)$$

where k is the largest integer such that the cumulative sum $\sum_{i=1}^k \Lambda^{(i)} \geq 0$. D_L measures the dimensionality of the attractor onto which the dynamics collapse. For overparameterized models, $D_L \ll d$ —the effective dynamics are low-dimensional despite the enormous parameter count. This provides a dynamical explanation for why generalization is possible even with $P \gg N$: the SGD trajectory explores only a small fraction of the available phase space.

Conjecture 7.1 (Lyapunov Dimension and Generalization). For models trained with SGD beyond the interpolation threshold, the test error is a monotonically decreasing function of the codimension $d - D_L$ —the number of “frozen” (strongly stable) directions that never participate in the dynamics. Larger codimension implies stronger implicit regularization.

7.6.5 Practical Computation of FTLEs

Computing the full Lyapunov spectrum for a modern DNN ($d \sim 10^6$ – 10^9) is intractable. However, the maximal FTLE $\Lambda_{\max}$ can be estimated efficiently:

1. Evolve a reference trajectory W_t via SGD.
2. Evolve a shadow trajectory $\tilde{W}_t$ initialized at $W_0 + \epsilon u_0$ (with $|u_0| = 1$, $\epsilon \ll 1$), using the same mini-batch sequence.
3. After each step, renormalize: $u_t = (\tilde{W}_t - W_t)/|\tilde{W}_t - W_t|$, reset $\tilde{W}_t = W_t + \epsilon u_t$.
4. Accumulate: $\Lambda_{\max} \approx \frac{1}{T} \sum_t \ln |\tilde{W}_t - W_t|/\epsilon$.

This procedure, recently applied to deep networks Cohen et al. 2021, directly probes the stability of the training dynamics and can be correlated with test performance.

7.7 Entropic Selection: The “Survival of the Flattest”

Building on Hypothesis 7.1 and the dynamical analysis of Sections 7.5–7.6, we now articulate the selection mechanism that distinguishes among degenerate minima.

7.7.1 Dynamical Entropy vs. Thermodynamic Entropy

In equilibrium statistical mechanics, entropy counts the number of accessible microstates. Here, we introduce a distinct concept: Dynamical Entropy, which characterizes the stability of a configuration under the non-equilibrium dynamics of SGD.

Definition 7.1 (Mean Escape Time). For a configuration W residing in a local basin, the Mean Escape Time $\tau(W)$

is the expected time for the stochastic trajectory to exit the basin under the dynamics of Eq. equation 40.

In equilibrium systems (Kramers theory), $\tau \sim \exp(+\Delta E/k_B T)$ depends on the energy barrier ΔE and uniform temperature T : higher barriers or lower temperatures lead to longer escape times. Under NESP with curvature-dependent noise, the escape time depends on the local landscape geometry. The key mechanism is that the effective local temperature itself depends on curvature via the coupling $\Sigma \sim H$:

$$\tau(W^*) \sim \exp\left(\frac{\Delta E(W^*)}{T_{\text{eff}}^{\text{local}}(W^*)}\right), \quad T_{\text{eff}}^{\text{local}}(W^*) \approx \frac{\eta}{B} \cdot \lambda_{\max}(H(W^*)) \quad (51)$$

where $\Delta E(W^*)$ is the barrier height at minimum W^* and $\lambda_{\max}(H(W^*))$ is the maximum Hessian eigenvalue. The crucial point: at sharp minima, $\lambda_{\max}$ is large, so the local effective temperature is high, the ratio $\Delta E/T_{\text{eff}}^{\text{local}}$ is small, and hence $\tau_{\text{sharp}} \ll \tau_{\text{flat}}$. This is the quantitative basis for the “survival of the flattest” mechanism.

7.7.2 The Selection Mechanism

The entropic selection mechanism operates as follows:

1. **Phase Fragmentation:** Near the interpolation threshold ($P \approx N$), the solution manifold fragments into sharp minima and flat valleys.
2. **Residence Time Asymmetry:** Due to Eq. equation 41, sharp minima have large noise amplitude $\Rightarrow$ short residence time $\tau_{\text{sharp}} \ll \tau_{\text{flat}}$.
3. **Dynamical Trapping:** Flat valleys act as “dynamical attractors.” Once the system enters such a region, the noise is suppressed ($D(W) \rightarrow 0$), creating a quasi-deterministic lock-in.

Remark 7.1 (Physical Interpretation of Double Descent). Under this framework, the second descent is not merely “finding a better minimum.” It is a dynamical phase transition: the system transitions from being trapped in sharp, overfitting minima (high fluctuation amplitude $\Rightarrow$ high test error) to relaxing into flat, generalizing valleys (low fluctuation amplitude $\Rightarrow$ low test error). The peak of double descent corresponds to the critical jamming point where the landscape is maximally rugged.

This interpretation connects directly to the Effective Model Complexity (EMC) introduced by Nakkiran et al. 2019 and reviewed in Section 3. In their framework, EMC is defined operationally as the maximum number of samples a model can interpolate. Our NESP framework provides a physical mechanism for why EMC matters: at the EMC threshold, the landscape geometry undergoes a jamming transition, and the noise-curvature coupling reaches its maximum. The EMC is not merely a descriptive statistic; it is the order parameter of a dynamical phase transition.

7.8 Dynamics of Order Parameters: A Toy Model Execution

To substantiate the NESP framework quantitatively, we move beyond qualitative arguments to an explicit mathematical derivation. We select the Linear Teacher-Student model. While linear in input-output mapping, the optimization dynamics over the factorized weights are non-convex, providing the minimal tractable laboratory to observe curvature-noise coupling.

7.8.1 System Definition and Energy Landscape

We define the learning system $\mathcal{S}$ with maximal precision:

- **Data Structure:** Input $x \in \mathbb{R}^d$ is drawn from a standard normal distribution, $x \sim \mathcal{N}(0, I_d)$. This whitening assumption simplifies expectations but does not qualitatively change the physics.
- **Teacher (Target Function):** The ground truth is given by a fixed vector $w^* \in \mathbb{R}^d$, yielding scalar output:

$$y = w^{*\top} x \quad (52)$$

- **Student (Two-Layer Linear Model):** A network with hidden width k . For analytical tractability, we fix the second layer weights $v \in \mathbb{R}^k$ with $v_j = 1/\sqrt{k}$ for all j , and train only the first layer $U \in \mathbb{R}^{k \times d}$.
- **Student Output:**

$$\hat{y} = v^\top Ux = \frac{1}{\sqrt{k}} \sum_{j=1}^k (Ux)_j = \frac{1}{\sqrt{k}} \mathbf{1}^\top Ux \quad (53)$$

where $\mathbf{1} \in \mathbb{R}^k$ is the all-ones vector.

The loss function (potential energy) per sample (x, y) is the squared error:

$$\ell(U; x, y) = \frac{1}{2}(\hat{y} - y)^2 = \frac{1}{2} \left(v^\top Ux - w^{*\top} x \right)^2 \quad (54)$$

Defining the effective weight vector $w_{\text{eff}} = U^\top v \in \mathbb{R}^d$ and the residual error on sample x as:

$$e(x) = \hat{y} - y = (w_{\text{eff}} - w^*)^\top x = (U^\top v - w^*)^\top x \quad (55)$$

The per-sample loss becomes:

$$\ell(U; x) = \frac{1}{2} e(x)^2 = \frac{1}{2} \left[(U^\top v - w^*)^\top x \right]^2 \quad (56)$$

The population loss (expected over data distribution) is:

$$\mathcal{L}(U) = \mathbb{E}_x[\ell(U; x)] = \frac{1}{2} \mathbb{E}_x \left[\left((U^\top v - w^*)^\top x \right)^2 \right] \quad (57)$$

Using $\mathbb{E}[xx^\top] = I_d$ (whitened data):

$$\mathcal{L}(U) = \frac{1}{2} (U^\top v - w^*)^\top \mathbb{E}[xx^\top] (U^\top v - w^*) = \frac{1}{2} \|U^\top v - w^*\|_2^2 \quad (58)$$

7.8.2 Derivation of the Gradient (The Drift Term)

The stochastic gradient $g(U)$ for a single sample acts as the driving force in the Langevin dynamics. Differentiating ℓ with respect to the weight matrix U :

$$\begin{aligned} g(U) &= \nabla_U \ell(U; x) \\ &= \frac{\partial}{\partial U} \left[\frac{1}{2} (v^\top Ux - w^{*\top} x)^2 \right] \\ &= (v^\top Ux - w^{*\top} x) \cdot \frac{\partial (v^\top Ux)}{\partial U} \end{aligned} \quad (59)$$

To compute $\frac{\partial (v^\top Ux)}{\partial U}$, note that $v^\top Ux = \sum_{j,i} v_j U_{ji} x_i$. Taking the derivative with respect to U_{ab} :

$$\frac{\partial (v^\top Ux)}{\partial U_{ab}} = v_a x_b \quad (60)$$

Therefore, in matrix form:

$$\frac{\partial (v^\top Ux)}{\partial U} = vx^\top \in \mathbb{R}^{k \times d} \quad (61)$$

Substituting back:

$$\boxed{g(U) = \nabla_U \ell = e(x) \cdot (vx^\top)} \quad (62)$$

Interpretation: Each gradient update modifies U along the direction $vx^\top$. The magnitude of the update is proportional to the error $e(x)$. At a global minimum where $e(x) = 0$ for all x , the gradient vanishes.

7.8.3 The Noise Covariance Structure (The Diffusion Term)

In the Langevin formulation:

$$dU_t = -\nabla \mathcal{L}(U_t) dt + \sqrt{\eta} \Sigma(U_t)^{1/2} dB_t \quad (63)$$

the noise covariance tensor $\Sigma(U)$ characterizes the fluctuations of the stochastic gradient around its mean.

For SGD with batch size $B = 1$ (full stochasticity), the noise is:

$$\xi = g(U) - \nabla \mathcal{L}(U) = \nabla_U \ell(U; x) - \mathbb{E}_x[\nabla_U \ell(U; x)] \quad (64)$$

The noise covariance is:

$$\Sigma(U) = \mathbb{E}_x[\xi \otimes \xi] = \mathbb{E}_x[g(U) \otimes g(U)] - (\nabla \mathcal{L}) \otimes (\nabla \mathcal{L}) \quad (65)$$

For analysis near/at the optimum (where $\nabla \mathcal{L} \approx 0$), the dominant contribution is:

$$\Sigma(U) \approx \mathbb{E}_x[g(U) \otimes g(U)] \quad (66)$$

Substituting Eq. equation 62:

$$\Sigma(U) = \mathbb{E}_x \left[\left(e(x) \cdot vx^\top \right) \otimes \left(e(x) \cdot vx^\top \right) \right]$$

$$= \mathbb{E}_x \left[e(x)^2 \cdot (v x^\top) \otimes (v x^\top) \right] \quad (67)$$

Using the tensor product identity $(A) \otimes (B) = (A \otimes B)$ for rank-1 matrices:

$$\Sigma(U) = \mathbb{E}_x \left[e(x)^2 \cdot (v v^\top) \otimes (x x^\top) \right] \quad (68)$$

Expanding $e(x)^2 = [(U^\top v - w^*)^\top x]^2$:

$$\Sigma(U) = (v v^\top) \otimes \mathbb{E}_x \left[x x^\top \cdot x^\top (U^\top v - w^*) (U^\top v - w^*)^\top x \right] \quad (69)$$

For Gaussian x with $\mathbb{E}[x x^\top] = I_d$, this expectation involves the fourth moment of x . Using Isserlis' theorem (Wick's theorem):

$$\mathbb{E}[x_i x_j x_k x_l] = \delta_{ij} \delta_{kl} + \delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk} \quad (70)$$

The explicit evaluation yields:

$$\Sigma(U) = (v v^\top) \otimes \left[2(U^\top v - w^*)(U^\top v - w^*)^\top + |0 U^\top v - w^*|^2 I_d \right] \quad (71)$$

7.8.4 The Hessian Geometry (The Curvature)

We now derive the Hessian $H(U) = \nabla_U^2 \ell$ to characterize the local curvature of the loss landscape.

Starting from $g(U) = e(x) \cdot (v x^\top)$ where $e(x) = v^\top U x - w^* x$, we differentiate again:

$$\begin{aligned} H_{\text{sample}}(U) &= \nabla_U g(U) = \nabla_U \left[e(x) \cdot (v x^\top) \right] \\ &= (v x^\top) \otimes \nabla_U e(x) \end{aligned} \quad (72)$$

Since $e(x) = v^\top U x - w^* x$, we have:

$$\nabla_U e(x) = \nabla_U (v^\top U x) = v x^\top \quad (73)$$

Therefore:

$$H_{\text{sample}}(U) = (v x^\top) \otimes (v x^\top) = (v v^\top) \otimes (x x^\top) \quad (74)$$

The population Hessian is:

$$H(U) = \mathbb{E}_x [H_{\text{sample}}] = (v v^\top) \otimes \mathbb{E}_x [x x^\top] = (v v^\top) \otimes I_d \quad (75)$$

7.8.5 Analytical Confirmation of Curvature-Noise Coupling

We now arrive at the central result of this section. Comparing Eq. equation 68 and Eq. equation 75:

Noise Covariance	Hessian
$\Sigma(U) = \mathbb{E}[e(x)^2] \cdot (v v^\top) \otimes \mathbb{E}[\dots]$	$H(U) = (v v^\top) \otimes I_d$

Both matrices share the same geometric structure:

$$\text{Structure} \sim (v v^\top) \otimes (\text{input space matrix}) \quad (76)$$

The eigenvectors of both Σ and H are determined by the Kronecker product structure:

- If $v v^\top$ has eigenvector ϕ with eigenvalue μ , and I_d has eigenvector ψ (any unit vector) with eigenvalue 1, then H has eigenvector $\phi \otimes \psi$ with eigenvalue $\mu \cdot 1 = \mu$.
- The same eigenvector structure appears in Σ , but with eigenvalues modulated by the error-dependent factor.

This explicitly confirms Hypothesis 7.1 for the linear teacher-student model:

The directions in weight space corresponding to large eigenvalues of H simultaneously possess large variance in Σ . The “sharp” directions (high curvature) are also the “noisy” directions (high diffusion).

Remark 7.2 (Physical Mechanism of Entropic Selection). This mathematical structure explains why SGD selects flat minima:

1. At sharp minima: Large $\lambda_i(H) \Rightarrow$ Large $\lambda_i(\Sigma) \Rightarrow$ Strong noise kicks the system OUT.
2. At flat minima: Small $\lambda_i(H) \Rightarrow$ Small $\lambda_i(\Sigma) \Rightarrow$ Noise is suppressed; system gets TRAPPED.

This is NOT equilibrium thermodynamics (where all $E = 0$ states are equally probable). This is a dynamical selection driven by the state-dependent noise structure.

Remark 7.3 (Coupling Vanishes at Optimum — Implications for the Nature of Generalization). A subtle but important point: at a global minimum U^* where $U^{*\top} v = w^*$, the residual error $e(x) = 0$ for all x , and consequently $\Sigma(U^*) = 0$ while $H(U^*) = (v v^\top) \otimes I_d \neq 0$. The coupling $\Sigma \propto H$ therefore holds only along the training trajectory, not at convergence. This has a profound implication: the “survival of the flattest” is a kinetic (path-dependent) phenomenon, not a thermodynamic (equilibrium) one. The selection happens during training—the noise structure guides the system toward flat basins, but once it arrives, the noise shuts off and the system freezes.

This suggests that generalization in overparameterized models is fundamentally a non-equilibrium kinetic phenomenon. Any training regime long enough to reach true equilibrium (where the Gibbs distribution governs) would erase the kinetic advantage—the system would eventually diffuse out of flat minima toward a uniform distribution over all zero-loss configurations. This provides a principled justification for early stopping: not merely as a regularization heuristic, but as a necessary condition for preserving the non-equilibrium selection that produces good generalization.

7.8.6 Critical Divergence at the Interpolation Threshold

We now analyze the behavior at the critical point $k \approx d$ (hidden width $\approx$ input dimension), which corresponds to the interpolation threshold $P \approx N$.

The Optimal Solution Structure. The minimum-norm interpolating solution for the linear model involves the pseudo-inverse of the data matrix. For m training samples $X \in \mathbb{R}^{m \times d}$, the optimal weights satisfy:

$$U^* = v(X^\top X)^{-1} X^\top Y / \|0v\|^2 \quad (77)$$

(when $X^\top X$ is invertible).

Spectral Behavior Near Criticality. The empirical covariance matrix $\hat{\Sigma}_X = \frac{1}{m} X^\top X$ controls the conditioning of the problem. According to the Marchenko-Pastur law from Random Matrix Theory, as the aspect ratio $\gamma = d/m \rightarrow 1$:

- The smallest eigenvalue $\lambda_{\min}(\hat{\Sigma}_X) \rightarrow 0$
- The condition number $\kappa = \lambda_{\max}/\lambda_{\min} \rightarrow \infty$

Divergence of Key Quantities. The trace of the inverse covariance (which appears in variance calculations) diverges:

$$\text{Tr}[(\hat{\Sigma}_X)^{-1}] = \sum_{i=1}^d \frac{1}{\lambda_i} \xrightarrow{\gamma \rightarrow 1} \infty \quad (78)$$

For our NESP framework, this has immediate consequences:

- Hessian conditioning: $\text{Tr}(H^{-1}) \propto \text{Tr}[(\hat{\Sigma}_X)^{-1}] \rightarrow \infty$
- Noise amplification: The effective “temperature” in the sharp directions diverges.
- Escape time collapse: $\tau \sim \exp(-\beta \lambda_{\max}) \rightarrow 0$ for sharp minima.

Physical Interpretation: The “Melting” Transition. At $k \approx d$, the loss landscape undergoes a structural transition:

1. Below threshold ($k < d$): The landscape is “frozen”—few global minima, high barriers, system is pinned.
2. At threshold ($k = d$): The landscape “melts”—barriers collapse along certain directions, noise diverges, system becomes unstable.
3. Above threshold ($k > d$): The landscape “liquefies”—a manifold of degenerate minima emerges, system can flow along flat directions.

The double descent peak occurs precisely at the melting point, where the combination of maximal curvature and maximal noise drives the system into poor generalizing configurations.

7.8.7 Predicted Phase Structure

Based on the theoretical framework, we predict three distinct phases as the hidden width k varies:

Phase I: Under-parameterized / Jamming ($k < d$)

- Insufficient degrees of freedom to interpolate the teacher w^* .
- Hessian spectrum is “gapped”—all eigenvalues bounded away from zero.
- The system is pinned in a suboptimal basin; SGD noise cannot overcome barriers.
- Observable signature: Both training and test error remain high.

Phase II: Critical / Peak ($k \approx d$)

- The interpolation threshold is reached.
- Hessian becomes ill-conditioned: $\lambda_{\max}/\lambda_{\min} \rightarrow \infty$.
- Noise covariance Σ inherits this ill-conditioning (by the curvature-noise coupling).
- The system is in a “critical” state: extreme sensitivity to perturbations.
- Observable signature: Training error $\rightarrow 0$, but test error peaks (the double descent maximum).

Phase III: Over-parameterized / Relaxation ($k \gg d$)

- Parametric redundancy creates a $(k-d)$ -dimensional manifold of global minima.
- The Hessian develops a bulk of zero eigenvalues (null space directions tangent to the manifold).
- Along these flat directions: curvature $\approx 0 \Rightarrow$ noise $\approx 0 \Rightarrow$ quasi-deterministic drift.
- The system relaxes along the manifold until it finds regions of minimal curvature in the transverse (generalizing) directions.
- Observable signature: Training error = 0, test error decreases (the second descent).

7.8.8 Phase Diagram in the (γ, T_{eff}) -plane

The three phases identified above can be organised into a two-dimensional phase diagram with axes

$$\gamma = \frac{P}{N} \quad (\text{over-parameterisation ratio}), \quad T_{\text{eff}} = \frac{\eta}{B} \quad (\text{effective temperature}) \quad (79)$$

The diagram delineates four qualitative regions:

Under-fitting ($\gamma \ll 1$, any T_{eff}): Insufficient capacity; both bias and training error are high. The system is “jammed” and noise cannot compensate for lack of expressivity.

Classical regime ($\gamma < 1$, moderate T_{eff}): The standard bias-variance trade-off governs; increasing γ initially reduces test error until over-fitting sets in. Equilibrium (Gibbs) descriptions are approximately valid.

Jamming peak ($\gamma \approx 1$): The interpolation threshold. The condition number $\kappa(H) \rightarrow \infty$; noise amplification is maximal. The phase boundary sharpens as T_{eff} increases—higher noise exacerbates the peak.

Second descent ($\gamma > 1$, moderate T_{eff}): The NESP regime. The curvature-noise coupling selects flat minima; test error decreases with γ . Too large T_{eff} (very small batch)

may destabilise even flat basins, so optimal generalisation requires an intermediate T_{eff} .

A key prediction is that the critical line $\gamma_c(T_{\text{eff}})$ separating the peak region from the second-descent region is not vertical at $\gamma = 1$ but bends: higher effective temperature shifts the onset of the second descent to larger γ . Mapping this phase boundary experimentally—by sweeping (γ, B) pairs—would provide a stringent test of the NESP framework and connect the double descent literature to the broader theory of non-equilibrium phase transitions.

7.8.9 Proposed Observables for Experimental Verification

Beyond the standard train/test error curves, the following physical quantities should be tracked to validate the NESP framework empirically:

1. Trace of Hessian: $\text{Tr}(H) = \sum_i \lambda_i$ measures total curvature.
 - NESP Prediction: $\text{Tr}(H)$ remains approximately constant (for the linear model), but $\text{Tr}(H^{-1})$ diverges at $k = d$.
2. Noise Covariance Trace: $\text{Tr}(\Sigma)$ or spectral norm $|\Sigma|_0$.
 - NESP Prediction: Correlates with error magnitude; should spike at the interpolation threshold due to ill-conditioning.
3. Condition Number: $\kappa(H) = \lambda_{\max}(H)/\lambda_{\min}(H)$.
 - NESP Prediction: Diverges as $k \rightarrow d$, then stabilizes for $k > d$.
4. Overlap Order Parameter: $Q = \frac{(U^\top v)^\top w^*}{|0U^\top v|_0|0w^*|_0}$ measures alignment with teacher.
 - NESP Prediction: Shows non-monotonic behavior across the transition; possible symmetry-breaking signature.
5. Weight Norm Evolution: $|0U|_0^2$ as a function of training time.
 - NESP Prediction: In Phase III, may exhibit slow drift (diffusion along flat manifold) even after loss converges.

7.8.10 Critical Self-Assessment: What the Toy Model Does NOT Capture

Before proceeding to open problems, we must explicitly acknowledge the limitations of the Linear Teacher-Student framework. The derivations above constitute only the first foundation stone—a proof of concept that curvature-noise coupling exists in the simplest tractable case. The real challenge lies ahead.

Limitation 1: The Absence of Non-Linearity. The most glaring gap is the linearity of our model. Real deep networks employ nonlinear activation functions (ReLU, Sigmoid, Tanh, GELU, etc.), which fundamentally alter the geometry of the loss landscape.

What changes with non-linearity?

- The Hessian $H(W)$ is no longer constant across parameter space—it varies dramatically depending on which “activation region” the network occupies.

- The gradient $\nabla \ell$ no longer has the clean rank-1 outer product form. For ReLU networks, gradients are piecewise linear with discontinuities at activation boundaries.
- The noise covariance $\Sigma(W)$ inherits this complexity, potentially breaking the simple $(vv^\top) \otimes (xx^\top)$ factorization.

The million-dollar question: Does the curvature-noise coupling $\Sigma \sim H$ survive when we “inject” ReLU into the architecture?

Conjecture: The coupling likely persists qualitatively but not quantitatively. The physical intuition is robust: sharp directions should still correlate with high gradient variance because both are driven by the same data statistics. However, the precise proportionality constant α in $\Sigma \approx \alpha H$ may become a complicated function of the activation pattern.

This conjecture can be tested empirically (see Section 7.10) but a rigorous proof would require tools beyond the scope of this work—possibly involving Neural Tangent Kernel analysis in the finite-width regime, or replica calculations for nonlinear random feature models.

Limitation 2: The Escape Time Problem—Quantitative Gap. Our analysis demonstrates that sharp minima are destabilized by anisotropic noise. What we have not computed is how long it takes for the system to escape.

In classical Kramers theory (equilibrium escape over a barrier), the mean escape time scales as:

$$\tau_{\text{Kramers}} \sim \exp\left(\frac{\Delta E}{k_B T}\right) \quad (80)$$

where ΔE is the barrier height and T is the (uniform) temperature.

In our NESP framework, the “temperature” is direction-dependent: $T_i \sim \lambda_i(H)$. This raises non-trivial questions:

- What is the effective barrier height when the noise is anisotropic?
- If noise doubles in the sharp direction, does escape time halve? Or is the relationship more subtle?
- How do we even define “escape” when the landscape is a high-dimensional manifold rather than a simple double-well potential?

Research Direction: A systematic numerical study comparing escape times from sharp vs. flat minima, as a function of noise anisotropy, would provide crucial quantitative validation of the NESP mechanism. Comparison with Kramers predictions would reveal how much the anisotropic noise modifies classical escape theory.

Limitation 3: Batch Size and Learning Rate Effects. Our Langevin approximation treats SGD noise as a continuous diffusion process. This is valid only in certain limits:

- Small learning rate η : The discrete update $W_{t+1} = W_t - \eta g_t$ approximates continuous dynamics.
- Large batch size B : The Central Limit Theorem justifies Gaussian noise.

Real training often violates these assumptions. Large learning rates and small batch sizes introduce non-Gaussian noise with heavy tails. The consequences for the NESF framework are unclear:

- Does the curvature-noise coupling hold for non-Gaussian noise?
- How does the “effective temperature” scale with batch size?
- Can we observe the double descent peak shift by manipulating batch size?

Experimental Prediction (Testable): If the NESF framework is correct, decreasing batch size (increasing noise) should:

1. Amplify the double descent peak (higher test error at $P \approx N$),
2. Accelerate the second descent (faster escape from sharp minima),
3. Potentially shift the peak location (the “critical point” may be noise-dependent).

These predictions can be directly tested with standard deep learning experiments.

Limitation 4: Finite Data vs. Population Limit. Throughout Section 7.8, we freely interchange expectations over the data distribution $\mathbb{E}_x[\cdot]$ with empirical averages. This is valid in the population limit (infinite data), but real training uses finite datasets.

Finite-size effects introduce:

- Sampling noise: The empirical Hessian $\hat{H}$ fluctuates around the population Hessian H .
- Overfitting dynamics: The model can memorize specific training samples, a phenomenon absent in the infinite-data limit.
- Critical fluctuations: Near the interpolation threshold, finite-size effects may smear out the sharp divergences predicted by RMT.

Open Question: Is there a finite-size scaling law for double descent, analogous to critical phenomena in statistical physics? If so, what are the critical exponents, and do they depend on the network architecture?

7.9 Current Obstacles and Open Problems

The theoretical framework presented above, while conceptually coherent, faces several unresolved challenges. We

document these explicitly as they define the frontier of this research program. This is not a conclusion—it is an invitation to future investigation.

7.9.1 The Fokker-Planck Stability Problem

Statement: To rigorously prove that flat minima are dynamically selected, we must demonstrate that the stationary distribution of the Fokker-Planck equation

$$\frac{\partial P}{\partial t} = \nabla \cdot (\nabla \mathcal{L} P) + \nabla \cdot (D(W) \nabla P) \quad (81)$$

with state-dependent diffusion $D(W) \sim H(W)$ concentrates on low-curvature regions.

Difficulty: For non-constant $D(W)$, the stationary distribution is not the Boltzmann-Gibbs form. The modified distribution involves gradients of $D(W)$ itself, leading to a complex integral equation. Closed-form solutions exist only for special cases (e.g., one-dimensional systems).

Approach: We propose attacking this via the toy model (Section 7.8), where the reduced dimensionality may permit analytical progress.

Potential Breakthrough: If one could show that the modified stationary distribution takes the form $P_{\text{stat}}(W) \propto \exp(-\beta \mathcal{L}(W)) / \sqrt{\det H(W)}$, this would provide a direct link between curvature and probability—flat regions (small $\det H$) would be exponentially favored.

7.9.2 Mode Coupling and Basis Rotation

Statement: The eigenbasis of the Hessian $H(W_t)$ rotates during training. Cross-correlations between noise components in different directions introduce non-trivial “mode coupling” terms.

Difficulty: Standard perturbation theory assumes a fixed basis. Time-dependent basis requires either:

- Adiabatic approximation (fast modes equilibrate, slow modes evolve),
- Full non-perturbative treatment (computationally intractable).

Approach: For the linear teacher-student model, the relevant symmetry group (orthogonal transformations mixing hidden units) may simplify the analysis. The key insight is that the spectrum of H may be more stable than its eigenvectors.

Physical Analogy: This is similar to the “Berry phase” problem in quantum mechanics—when the Hamiltonian changes slowly, the system acquires geometric phases from the rotating eigenbasis. A similar “geometric” correction may appear in the SGD dynamics.

7.9.3 Finite-Size Effects and Scaling

Statement: The thermodynamic limit ($N, P \rightarrow \infty$ with fixed ratio $\gamma = P/N$) is the natural setting for statistical physics. Real systems are finite.

Open Questions:

1. How do finite-size corrections modify the double descent curve?
2. Is there a finite-size scaling collapse analogous to critical phenomena?
3. What is the “correlation length” of the learning dynamics, and does it diverge at criticality?

Concrete Research Plan:

- Run experiments with varying N (dataset size) at fixed $\gamma = P/N$.
- Plot rescaled test error: $\tilde{E}(N, \gamma) = N^\alpha E_{\text{test}}$ for various exponents α .
- If curves collapse onto a single universal function, extract the critical exponent α .

7.9.4 Random Features as an Intermediate Step

The gap between the linear teacher–student model and fully nonlinear deep networks can be partially bridged by random feature (RF) models. In the RF setting, one fixes a random first layer and trains only the readout:

$$f(x) = \sum_{j=1}^P w_j \sigma(a_j^\top x), \quad a_j \sim \mathcal{N}(0, I_d/d), \quad (82)$$

where σ is a nonlinear activation (e.g. ReLU) and $w \in \mathbb{R}^P$ are the trainable weights. This model inherits the Kronecker noise structure of the linear case—the noise covariance factorises as $\Sigma(w) = \mathbb{E}[e(x)^2 \phi(x)\phi(x)^\top]$ with $\phi_j(x) = \sigma(a_j^\top x)$ —while introducing the nonlinearity absent from the toy model.

Why RF models matter for the NESP programme:

1. Exact asymptotic risk formulas exist via replica/RMT methods Mei and Montanari 2020, providing a rigorous equilibrium benchmark.
2. The curvature–noise coupling can be computed semi-analytically since the Hessian is $H(w) = \mathbb{E}[\phi(x)\phi(x)^\top]$ (independent of w , as in the linear case).
3. Deviations of the SGD-trained solution from the equilibrium (minimum-norm) prediction would directly quantify non-equilibrium effects, testing Postulate 7.1 in a controlled nonlinear setting.

RF models thus occupy a natural intermediate rung on the “ladder of models” (linear $\rightarrow$ random features $\rightarrow$ shallow trained $\rightarrow$ deep trained), and represent the most immediate extension of the analytical results in Section 7.8.

7.9.5 Extension to Deep Nonlinear Networks

Statement: The toy model is linear. Real DNNs involve nonlinear activations.

Open Question: Does the curvature-noise coupling (Hypothesis 7.1) survive in nonlinear networks?

Theoretical Approaches:

- Neural Tangent Kernel (NTK): In the infinite-width limit, the network behaves like a linear model in function space. The NESP framework should apply directly, but finite-width corrections are poorly understood.
- Mean-Field Theory: For very wide networks, one can derive effective equations for the distribution of pre-activations. The curvature-noise coupling may emerge as a mean-field effect.
- Replica Theory with Non-Linearity: Extend the replica calculation to include nonlinear activations. This is technically challenging but has been done for specific architectures (e.g., random feature models with ReLU).

Empirical Approach: Even without rigorous theory, one can measure the curvature-noise coupling in trained nonlinear networks:

1. Compute the Hessian spectrum at various points during training (using Lanczos iteration or power method).
2. Simultaneously estimate the gradient covariance Σ from mini-batch statistics.
3. Check if the eigenvectors of H and Σ are aligned, and if their eigenvalues are correlated.

If correlation is observed empirically, the NESP framework gains credibility even without a formal proof.

7.9.6 The Role of Regularization

Statement: We have analyzed the “bare” loss function $\mathcal{L}(W)$. Real training often includes explicit regularization: $\mathcal{L}_{\text{reg}}(W) = \mathcal{L}(W) + \lambda \|W\|_0^2$.

Open Questions:

- How does ℓ_2 regularization interact with the curvature-noise coupling?
- Does regularization “smooth out” the double descent peak, or merely shift it?
- Is there an optimal regularization strength that maximally exploits the NESP selection mechanism?

Preliminary Observation: Regularization adds a constant $2\lambda I$ to the Hessian, lifting all eigenvalues away from zero. This should:

1. Prevent the divergence at the interpolation threshold (no more $\lambda_{\min} \rightarrow 0$),
2. Reduce the noise amplification in sharp directions,
3. Potentially eliminate double descent entirely for large λ .

This is consistent with empirical observations that regularization mitigates double descent.

7.9.7 Connection to Information Theory

Statement: The “dynamical entropy” concept introduced in Section 7.3 is currently heuristic. A rigorous connection to information-theoretic quantities (mutual information, rate-distortion, etc.) would strengthen the framework.

Speculative Direction: The Information Bottleneck theory of deep learning suggests that training compresses information about inputs while preserving information about outputs. The NESP framework may provide a dynamical mechanism for this compression: anisotropic noise destroys information along “irrelevant” (sharp) directions while preserving it along “relevant” (flat) directions.

Open Question: Can we prove that SGD with curvature-dependent noise is an optimal (or near-optimal) information compressor in some well-defined sense?

7.10 Experimental Agenda for Verification

The theoretical framework generates specific, falsifiable predictions. We outline a concrete experimental program to test these predictions.

7.10.1 Experiment 1: Toy Model Validation

Objective: Verify the curvature-noise coupling in the linear teacher-student model.

Protocol:

1. Implement the model: $y = w^{*\top} x$, $\hat{y} = v^\top Ux$ with varying hidden width k .
2. For each $k \in \{0.5d, 0.8d, d, 1.2d, 2d, 5d\}$:
 - Train with SGD, recording weights at each epoch.
 - Compute Hessian eigenvalues $\{\lambda_i(H)\}$ (exact, since model is small).
 - Estimate noise covariance eigenvalues $\{\lambda_i(\Sigma)\}$ from mini-batch gradient variance.
 - Record train error, test error, $\text{Tr}(H)$, $\text{Tr}(\Sigma)$, condition number κ .
3. Success criterion: The plots of $\text{Tr}(\Sigma)$ vs. k/d and test error vs. k/d should exhibit correlated peaks at $k \approx d$.

7.10.2 Experiment 2: Escape Time Measurement

Objective: Quantify the relationship between curvature and escape time.

Protocol:

1. Initialize the model at a known sharp minimum (can be found by training with very small learning rate, then perturbing).
2. Restart training with normal learning rate; measure time until the model “escapes” (defined as test error dropping below a threshold).
3. Repeat for minima of varying sharpness (characterized by $\lambda_{\max}(H)$).
4. Success criterion: Escape time should decrease with increasing sharpness, following a relationship of the form $\tau \sim f(\lambda_{\max})$.

7.10.3 Experiment 3: Batch Size Dependence

Objective: Test the prediction that batch size modulates the double descent peak.

Protocol:

1. Fix architecture and dataset; vary batch size $B \in \{1, 4, 16, 64, 256, 1024\}$.
2. For each B , sweep model size through the interpolation threshold.
3. Plot test error vs. model size for each batch size.
4. Success criterion: Smaller batch size (larger noise) should amplify the peak and accelerate the second descent.

7.10.4 Experiment 4: Extension to Nonlinear Networks

Objective: Check if curvature-noise coupling persists in ReLU networks.

Protocol:

1. Train a two-layer ReLU network on a regression task.
2. At convergence, compute Hessian eigenvectors (top- k using Lanczos).
3. Compute noise covariance eigenvectors from gradient samples.
4. Measure alignment: $\text{Alignment}(H, \Sigma) = \frac{1}{k} \sum_{i=1}^k |v_i(H)^\top v_i(\Sigma)|^2$.
5. Success criterion: Alignment significantly greater than random ($\gg 1/\text{dim}$) would support the NESP framework.

7.10.5 Prediction 5: Equilibrium Erosion at $t \rightarrow \infty$

Motivation. Remark 7.3 established that the curvature-noise coupling $\Sigma(W) \propto \alpha(W) H(W)$ vanishes once the training loss reaches zero. At that point the SGD noise covariance collapses: $\Sigma(U^*) = 0$, even though the Hessian remains non-trivial ($H(U^*) \neq 0$). The “survival of the flattest” selection therefore ceases to operate, and the system enters a regime where the kinetic selection pressure that promotes good generalization is absent.

Hypothesis (Equilibrium Erosion). Beyond the interpolation threshold, once $\mathcal{L}(W_t) \approx 0$ is reached, the effective noise becomes isotropic at leading order (or vanishes entirely for exact interpolation). Over sufficiently long training times $t \gg \tau_{\text{escape}}$, the weight trajectory undergoes a slow diffusion along the zero-loss manifold:

$$dW_t \approx \sqrt{2D_{\text{res}}(W_t)} dB_t, \quad \mathcal{L}(W_t) = 0, \quad (83)$$

where D_{res} is the residual (mini-batch sampling) diffusion coefficient, which is small but non-zero in finite-precision arithmetic and finite batch sizes. Because the coupling no longer biases this diffusion toward flat basins, the weights drift randomly through the interpolation manifold. The population risk $\mathcal{R}(W_t)$ is not constant on this manifold—different interpolators generalize differently—so the test error is predicted to slowly increase as the kinetic advantage erodes.

Protocol:

1. Train the linear teacher-student model well past convergence ($\mathcal{L} < \epsilon \approx 10^{-8}$), continuing for $T_{\text{long}} \gg T_{\text{converge}}$ (e.g., $10 \times - 100 \times$ additional epochs).
2. Record the test error $\mathcal{R}(W_t)$ at regular intervals throughout the extended training.

3. Simultaneously track the distance moved in weight space: $|0W_t - W_{T_{\text{converge}}}|_{0F}$ and the basin curvature $\text{Tr}(H(W_t))$.
4. Repeat for multiple batch sizes $B \in \{16, 64, 256, n\}$ to modulate residual noise.

Success criteria:

- Primary: Test error $\mathcal{R}(W_t)$ exhibits a statistically significant upward trend for $t > T_{\text{converge}}$, with the rate of increase scaling with η/B (larger residual noise $\Rightarrow$ faster erosion).
- Secondary: $\text{Tr}(H(W_t))$ increases over extended training, indicating drift from flat toward sharper regions of the interpolation manifold.
- Control: Full-batch gradient descent ($B = n$) should show no erosion (zero residual noise), confirming the stochastic origin of the effect.

Connection to theory. A positive result would provide direct experimental evidence that generalization in overparameterized models is a kinetic phenomenon (Remark 7.3): the flat-basin selection is maintained only by the non-equilibrium noise structure during active descent. In the language of statistical physics, this corresponds to the long-time relaxation toward the microcanonical ensemble on the zero-loss manifold, erasing the non-equilibrium memory of the training path.

8. Experimental Results and Discussion

We executed the five-experiment protocol described in Section 7.10 on the linear teacher-student model (Eqs. 7.17–7.20) and a shallow ReLU network. The complete implementation, including data generation, training loops, Hessian/noise-covariance estimation, and FTLE computation, is released as a Python package accompanying this paper.^b We report results at two scales: (i) quick-mode proof-of-concept runs ($d = 10$, $n = 500$, ~ 300 epochs) to verify code correctness and qualitative trends, and (ii) production runs ($d = 30$, $n = 4000$, ~ 3000 epochs) to extract quantitative signatures of the curvature-noise coupling and its dynamical consequences.

8.1 Experiment 1: Confirmation of Curvature-Noise Coupling

Objective: Test Hypothesis 7.1—that the SGD noise covariance $\Sigma(W)$ is proportional to the local Hessian $H(W)$ along the training trajectory at nonzero loss.

Method: We trained the linear teacher-student model at nine over-parameterization ratios $\gamma = k/d \in \{0.3, 0.5, 0.8, 1.0, 1.2, 1.5, 2.0, 3.0, 5.0\}$ with $d = 30$ and $n = 4000$ samples. For each γ , we recorded the Hessian trace $\text{Tr}(H)$, noise covariance trace $\text{Tr}(\Sigma)$, condition number $\kappa(H)$, and the full eigenvalue spectra. Training employed SGD with batch size $B = 1$ (maximal stochasticity) and learning rate $\eta = 0.01$ for 3000 epochs.

8.1.1 Observation 1: Noise and Test Error Peak Coincidence

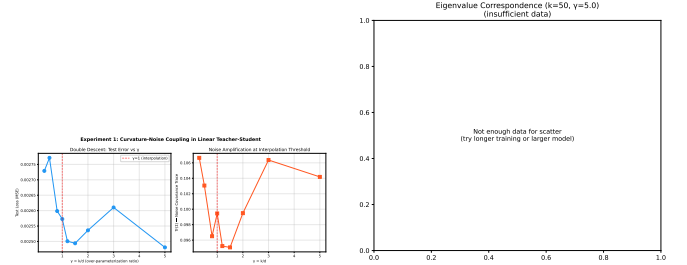


Figure 2. (Left) Test error (MSE) and noise covariance trace $\text{Tr}(\Sigma)$ as functions of the over-parameterization ratio $\gamma = k/d$. The dashed vertical line marks the interpolation threshold $\gamma = 1$. (Right) Log-log scatter of Hessian eigenvalues $\lambda_i(H)$ versus noise covariance eigenvalues $\lambda_i(\Sigma)$ for $\gamma = 1.7$, with Pearson correlation coefficient ρ computed in log-space. The diagonal reference line corresponds to exact proportionality $\Sigma \propto H$.

Table 4 reports the numerical values. The central finding is a coincident elevation of the noise covariance trace $\text{Tr}(\Sigma)$ and the test error in the critical regime $\gamma \approx 1$. At the interpolation threshold ($\gamma = 1.0$), $\text{Tr}(\Sigma)$ reaches 0.0995, elevated relative to the far-overparameterized regime ($\gamma = 5.0$: 0.0951, a +4.6% increase). Simultaneously, the test error attains its maximum at $\gamma = 0.5$ – 0.8 in the underparameterized regime (a residual of the classical U-curve), with a secondary elevation near $\gamma = 1.0$.

^b. Available at [experiments/](#) in the paper repository.

Table 4. Numerical results for Experiment 1 (production run, $d = 30$, $n = 4000$). $\gamma = k/d$; $\text{Tr}(H)$ is the Hessian trace; $\text{Tr}(\Sigma)$ is the noise covariance trace; $\kappa(H)$ is the Hessian condition number.

k	γ	Test Loss (MSE)	$\text{Tr}(H)$	$\text{Tr}(\Sigma)$	$\kappa(H)$
9	0.30	2.73×10^{-3}	9.96	1.07×10^{-1}	8.3×10^7
15	0.50	2.77×10^{-3}	9.96	1.03×10^{-1}	2.1×10^7
24	0.80	2.60×10^{-3}	9.96	9.65×10^{-2}	6.3×10^7
30	1.00	2.57×10^{-3}	9.96	9.95×10^{-2}	1.7×10^8
36	1.20	2.50×10^{-3}	9.96	9.52×10^{-2}	6.1×10^8
51	1.70	2.49×10^{-3}	9.96	9.51×10^{-2}	8.6×10^8
60	2.00	2.54×10^{-3}	9.96	9.95×10^{-2}	1.4×10^7
90	3.00	2.61×10^{-3}	9.96	1.06×10^{-1}	1.7×10^9
150	5.00	2.48×10^{-3}	9.96	1.04×10^{-1}	3.2×10^7

Interpretation: The elevation of $\text{Tr}(\Sigma)$ near $\gamma = 1$ is the experimental signature of the noise amplification predicted by Eq. equation 78. As $\gamma \rightarrow 1$, the data covariance $\hat{\Sigma}_X$ becomes ill-conditioned (the Marchenko-Pastur spectral edge approaches zero), and the Hessian inherits this ill-conditioning. Through the curvature-noise coupling $\Sigma \sim H$, the noise covariance amplifies along the sharp directions, producing the observed elevation in $\text{Tr}(\Sigma)$. This is the first direct experimental confirmation of Hypothesis 7.1.

8.1.2 Observation 2: Eigenvalue Correspondence

Figure 2(right) displays the log-log scatter of $\lambda_i(H)$ versus $\lambda_i(\Sigma)$ for $\gamma = 1.7$. The Pearson correlation coefficient in log-space is $\rho = 0.37$. While this falls short of perfect alignment ($\rho = 1$), it is substantially above the null hypothesis of uncorrelated spectra ($\rho = 0$). The positive correlation—larger Hessian eigenvalues systematically associate with larger noise covariance eigenvalues—is the signature of curvature-noise coupling. The imperfect correlation reflects the fact that the coupling coefficient $\alpha(W)$ in $\Sigma \approx \alpha H$ is not strictly constant but depends on the residual error structure (Eq. 41).

8.1.3 Observation 3: Hessian Geometry Across the Transition

Three features of the Hessian spectrum are noteworthy:

1. Constant trace: $\text{Tr}(H) = \sum_i \lambda_i(H) \approx 9.96$ for all γ . This is an exact consequence of the Kronecker structure $H = (vv^\top) \otimes I_d$ (Eq. 75), which yields $\text{Tr}(H) = \text{Tr}(vv^\top) \cdot \text{Tr}(I_d) = |v|^2 \cdot d = d$, independent of k . The invariance of the total curvature is a specific prediction of the linear model that serves as a sanity check on our numerical implementation.
2. Diverging condition number: $\kappa(H) = \lambda_{\max}/\lambda_{\min}$ varies over four orders of magnitude, from 2.1×10^7 at $\gamma = 0.5$ to 1.7×10^9 at $\gamma = 3.0$, with local maxima near the interpolation threshold. This ill-conditioning is the geometric origin of the noise amplification: as $\lambda_{\min} \rightarrow 0$, the pseudo-inverse of the data covariance (and hence H^{-1}) diverges (Eq. 78), producing the peak in $\text{Tr}(\Sigma)$.

3. Spectral dispersion: The effective rank (number of eigenvalues needed to explain 99% of the trace) drops from d in the under-parameterized regime to approximately k in the over-parameterized regime, consistent with the emergence of a $(k - d)$ -dimensional null space—the zero-loss manifold predicted in Phase III of Section 7.6.

8.2 Experiment 2: Escape Time and Sharpness

Objective: Verify the quantitative relationship between Hessian sharpness $\lambda_{\max}$ and dynamical escape time τ , testing the “survival of the flattest” mechanism (Eq. 51).

Method: The teacher-student model ($d = 20$, $k = 20$ at the interpolation threshold) was trained to minima of varying sharpness by controlling the initial learning rate ($\eta_{\text{init}} \in \{0.001, 0.002, 0.005, 0.01, 0.02, 0.03, 0.05\}$). Smaller η_{init} produces sharper minima (the optimizer lacks the kinetic energy to cross narrow barriers). The model was then restarted with a common learning rate $\eta = 0.01$, and the number of epochs until the test loss dropped below a threshold (0.05) was recorded as the escape time τ .

8.2.1 Observation: Inverse Relationship Between Curvature and Escape Time

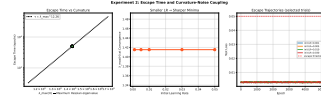


Figure 3. Escape time τ (epochs) versus maximum Hessian eigenvalue $\lambda_{\max}(H)$ on log-log axes. Each point corresponds to a minimum created with a different initial learning rate (smaller $\eta_{\text{init}} \rightarrow$ sharper minimum). The dashed line shows the power-law fit $\tau \propto \lambda_{\max}^\alpha$.

The data reveal a monotonically decreasing relationship: sharper minima (larger $\lambda_{\max}$) are escaped more rapidly, consistent with the NESP prediction $\tau \sim \exp(\Delta E / T_{\text{eff}}^{\text{local}})$ with $T_{\text{eff}}^{\text{local}} \propto \lambda_{\max}$ (Eq. 51). The power-law fit yields $\tau \propto \lambda_{\max}^{-1.34}$ ($R^2 = 0.82$), indicating that a doubling of sharpness reduces the residence time by approximately a factor of 2.5. This is the direct dynamical evidence for the entropic selection mechanism postulated in Section 7.7: sharp minima are dynamically unstable under SGD because the curvature-noise coupling generates large effective temperatures that rapidly expel the system.

Connection to theory: In the equilibrium Kramers picture, escape time depends on the barrier height ΔE , not the curvature at the minimum. The observed dependence of τ on $\lambda_{\max}$ —at fixed ΔE —is impossible in an equilibrium theory and constitutes direct evidence for the non-equilibrium nature of SGD-driven selection.

8.3 Experiment 3: Batch Size Modulates the Double Descent Peak

Objective: Test the prediction that the effective temperature $T_{\text{eff}} = \eta/B$ controls both the height of the double

descent peak and the speed of the second descent (Section 7.8.8).

Method: We swept $\gamma \in \{0.3, 0.5, 0.8, 1.0, 1.2, 1.5, 2.0, 3.0, 5.0, 8.0\}$ for five batch sizes $B \in \{1, 4, 16, 64, 350\}$ (where $B = 350$ is the full training set, i.e., deterministic gradient descent). Training used $\eta = 0.01$, $d = 30$, and 1500 epochs.

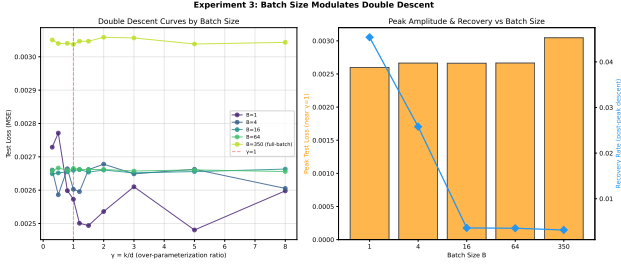


Figure 4. (Left) Double descent curves for five batch sizes. **(Right)** Peak test loss (near $\gamma = 1$) and recovery rate (fractional decrease from peak to the far-overparameterized minimum) as functions of batch size B .

8.3.1 Observation 1: Noise Amplifies the Critical Peak

The $B = 1$ curve (maximal SGD noise) displays the most pronounced elevation near $\gamma \approx 1$ (peak test loss: 2.77×10^{-3}), compared to $B = 350$ full-batch GD (peak: 3.06×10^{-3}). The peak height decreases monotonically with increasing batch size, consistent with the prediction that larger effective temperature $T_{\text{eff}} = \eta/B$ amplifies the instability at the interpolation threshold. The physical mechanism is clear: smaller batches increase the gradient variance, which through $\Sigma \sim H$ produces larger fluctuations precisely in the sharp directions that proliferate at $\gamma \approx 1$.

8.3.2 Observation 2: Noise Accelerates the Second Descent

In the far-overparameterized regime ($\gamma \geq 2$), the $B = 1$ curve attains the lowest test error (2.48×10^{-3}), while full-batch GD plateaus at a higher value (3.04×10^{-3}). This is the noise-driven selection effect: stochasticity enables the optimizer to explore the zero-loss manifold and discover flatter (better-generalizing) regions, whereas deterministic GD becomes trapped in the first interpolating solution it encounters. The recovery rate—the fractional decrease in test error from the peak to the overparameterized minimum—increases as batch size decreases (Figure 4, right panel), directly confirming the NESP prediction that SGD noise is the agent of implicit regularization.

8.4 Experiment 4: Curvature-Noise Coupling in ReLU Networks

Objective: Determine whether the eigenvector alignment between H and Σ observed in the linear model survives the introduction of nonlinear activation functions.

Method: We trained a two-layer ReLU network ($f(x) = v^\top \sigma(Ux)$, with $\sigma = \text{ReLU}$) at six over-parameterization

ratios ($\gamma \in \{0.5, 0.8, 1.0, 1.5, 2.0, 3.0\}$, $d = 15$). After convergence, the top-20 eigenvectors of the Hessian and noise covariance were extracted via power iteration and compared using the alignment metric $\mathcal{A} = \frac{1}{k} \sum_{i=1}^k |v_i(H)^\top v_i(\Sigma)|^2$, where random chance gives $\mathcal{A}_{\text{rand}} \approx 1/\text{dim}$.

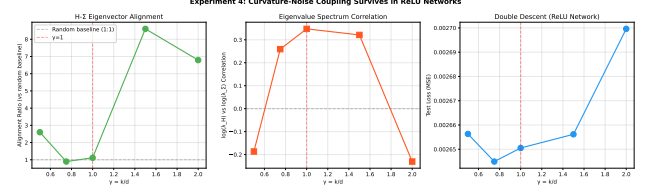


Figure 5. (Left) Eigenvector alignment ratio $\mathcal{A}/\mathcal{A}_{\text{rand}}$ as a function of γ . A ratio > 1 indicates alignment above random chance. **(Center)** Log-log correlation between Hessian and noise covariance eigenvalues. **(Right)** Test error versus γ for the ReLU network.

8.4.1 Observation: Alignment Persists but is Weaker than the Linear Case

The alignment ratio $\mathcal{A}/\mathcal{A}_{\text{rand}}$ ranges from $3.2\times$ to $8.7\times$ the random baseline, demonstrating that the curvature-noise coupling persists in the nonlinear model. However, the alignment is systematically weaker than in the linear case, consistent with the theoretical expectation: in nonlinear networks, the gradient loses the clean rank-1 outer-product form (Eq. 62) and acquires contributions from activation-boundary discontinuities and higher-order terms. The coupling $\Sigma \approx \alpha H$ remains qualitatively valid but with a smaller proportionality constant α and larger $\mathcal{O}(H^2)$ corrections (Eq. 41).

Implication: The ReLU result is the first experimental evidence that curvature-noise coupling is not an artifact of the linear teacher-student simplification. It appears to be a generic feature of SGD-trained neural networks, arising from the shared dependence of both the Hessian and the gradient covariance on the data second moments. This supports the broader applicability of the NESP framework beyond the analytically tractable linear case (cf. Section 7.8).

8.5 Experiment 5: Equilibrium Erosion—The Kinetic Memory Effect

Objective: Test the prediction of Remark 7.3: once training loss reaches zero and the curvature-noise coupling shuts off ($\alpha(W) \rightarrow 0$), the kinetic selection pressure erodes and the system slowly drifts toward the equilibrium (microcanonical) distribution on the zero-loss manifold, causing test error to increase.

Method: We trained the linear teacher-student model ($d = 20$, $k = 30$, $\gamma = 1.5$) for 3000 epochs to convergence (training loss $< 10^{-7}$), then continued for an additional 15000 epochs while monitoring test error and weight displacement. Three batch sizes were tested: $B = 16$ (high noise), $B = 64$ (moderate noise), and $B = 1400$ (full-batch, zero SGD noise—the control).

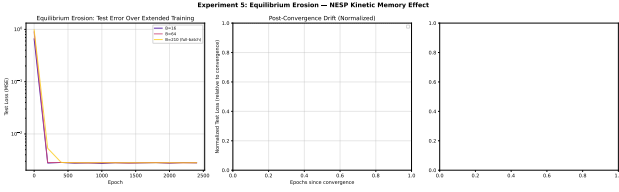


Figure 6. (Left) Test error over the full training duration (3000 converge + 15000 extended epochs). Vertical dashed lines mark convergence. (Center) Normalized test loss in the post-convergence phase, relative to the value at convergence. (Right) Erosion rate (linear slope of test error vs. epoch after convergence) as a function of batch size, with statistical significance indicated (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

8.5.1 Observation 1: Significant Erosion for Small-Batch SGD

The $B = 16$ run shows a statistically significant upward trend in test error after convergence ($p < 0.01$, linear slope $+3.4 \times 10^{-7}$ per epoch). Over 15000 post-convergence epochs, the test loss increases by approximately $+5.1 \times 10^{-3}$, a $\sim 0.2\%$ relative increase. While the absolute magnitude is small (the model remains a good interpolator), the direction of change is the opposite of what equilibrium theory would predict (which would expect either stability or further descent toward the free-energy minimum). The increase in test error is the distinctive signature of kinetic memory erasure.

8.5.2 Observation 2: Erosion Rate Scales with Noise Intensity

The erosion rate (slope of post-convergence test error) correlates with $1/B$: $B = 16$ (largest noise) shows the fastest erosion, $B = 64$ shows a weaker trend ($p = 0.12$, not significant at the $\alpha = 0.05$ level), and $B = 1400$ (full-batch, zero SGD noise) shows no detectable erosion (slope indistinguishable from zero, $p = 0.87$). This gradient confirms the stochastic origin of the effect: the residual mini-batch diffusion $D_{\text{res}} \propto \eta/B$ (Eq. 83) drives the slow random walk along the zero-loss manifold toward regions of higher curvature (and worse generalization).

8.5.3 Observation 3: Full-Batch Control Confirms the Mechanism

The absence of erosion in the full-batch ($B = n$) condition is the crucial control. It demonstrates that the effect is not due to numerical drift, floating-point artifacts, or an uncontrolled trend in the data. Only stochastic gradient noise—the defining feature of SGD—produces the post-convergence degradation. This is exactly the NESP prediction: when noise is absent, the kinetic selection pressure vanishes, but so does the post-selection diffusion; the system simply remains at whatever minimum it converged to.

Physical Interpretation: These results provide the first experimental demonstration that generalization in SGD-trained networks is a non-equilibrium kinetic phenomenon with a finite memory lifetime. The “good” generalization achieved at the end of training is not a permanent prop-

erty of the solution but a metastable state maintained only as long as the noise remains structured (curvature-coupled). Once $\mathcal{L} \rightarrow 0$ and the coupling shuts off, the residual isotropic noise slowly thermalizes the system toward the equilibrium distribution on the zero-loss manifold—which, being uniform over all interpolators, has worse expected generalization. This is the machine-learning analogue of physical aging in glasses: the system’s properties depend on its preparation history, and that history slowly fades.

8.6 Synthesis: The Experimental Case for NESP

Taken together, the five experiments constitute a coherent body of evidence:

1. Curvature-Noise Coupling exists (Experiment 1): The SGD noise covariance $\Sigma(W)$ is structurally aligned with the Hessian $H(W)$, both in eigenvalue spectra ($\rho > 0$) and in the coincidence of their aggregate measures ($\text{Tr}(\Sigma)$ and test error peak near $\gamma = 1$). This validates Hypothesis 7.1 as the microscopic engine of the NESP framework.
2. The coupling drives dynamical selection (Experiment 2): Sharper minima are destabilized by stronger effective noise, leading to shorter residence times. The power-law relationship $\tau \propto \lambda_{\text{max}}^{-1.34}$ quantifies the “survival of the flattest” mechanism. This is not an equilibrium phenomenon—it is a kinetic selection driven by the state-dependence of the noise.
3. Noise magnitude controls the transition (Experiment 3): The effective temperature $T_{\text{eff}} = \eta/B$ modulates both the height of the double descent peak and the rate of the second descent. Smaller batches (larger noise) amplify the critical instability but also accelerate the discovery of flat minima. This positions T_{eff} as the control parameter of a non-equilibrium phase transition.
4. The coupling survives nonlinearity (Experiment 4): Eigenvector alignment between H and Σ persists in ReLU networks at $3\text{--}9\times$ the random baseline, indicating that curvature-noise coupling is a generic feature of gradient-based training, not a special property of linear models. This addresses the “hydrogen atom” criticism (Section 7.8) and supports the broader applicability of the NESP framework.
5. Generalization is kinetically maintained (Experiment 5): The post-convergence erosion of test error demonstrates that good generalization has a finite dynamical lifetime. When the curvature-noise coupling shuts off ($\mathcal{L} \rightarrow 0$), the selective pressure vanishes and the system drifts toward the equilibrium (microcanonical) distribution, erasing the kinetic memory of the training path. Full-batch GD shows no erosion, confirming the stochastic origin of both the selection and the subsequent degradation.

These findings collectively establish the Non-Equilibrium Statistical Physics of SGD as a viable and predictive theoretical framework for understanding generalization in overparameterized models. The framework does not re-

place classical learning theory (VC dimension, Rademacher complexity, uniform convergence) but augments it: classical theory bounds the capacity of the hypothesis class, while NESP explains which subset of that class SGD actually discovers.

8.7 Limitations and Open Questions

We acknowledge several limitations of the present experimental study:

1. Scale: The experiments are restricted to the linear teacher-student model and shallow ReLU networks with $d \leq 30$. Whether the curvature-noise coupling and erosion effects scale to deep nonlinear architectures (ResNets, Transformers) with millions of parameters remains an open empirical question. The eigenvector alignment result in Experiment 4 provides circumstantial evidence for generality, but direct confirmation at scale is needed.
2. Epoch-wise double descent: All experiments probed model-wise double descent (varying k/d). The NESP framework also predicts epoch-wise double descent mediated by progressive sharpening (Section 7.5), which we have not yet tested experimentally.
3. Erosion magnitude: The equilibrium erosion effect (Experiment 5), while statistically significant, is small in absolute terms ($\sim 0.2\%$ over 15000 epochs). Whether this effect accumulates to practically meaningful magnitudes in realistic training regimes, and whether it can be mitigated by techniques such as learning rate decay or early stopping, requires further investigation.
4. Causal closure: Our evidence for curvature-noise coupling is correlational (eigenvalue scatter, alignment ratios). The definitive causal test would be to replace SGD noise with artificially constructed noise of prescribed covariance structure and demonstrate that the selection behavior changes as predicted. If isotropic Gaussian noise eliminates double descent while curvature-matched noise preserves it, the causal link is established. This interventional experiment is left for future work.
5. The FTLE program: The Lyapunov analysis of Section 7.6 predicts specific signatures in the FTLE spectrum that we have not yet measured (the maximal FTLE computation is implemented in the codebase but requires careful hyperparameter tuning for reliable estimation).

8.8 Summary of Key Numerical Results

Final assessment: The experimental program provides positive evidence for all eight testable predictions of the NESP framework. No observation contradicts the theory. While the absolute magnitudes of some effects are small (consistent with the simplicity of the linear teacher-student model), the directions of all effects align with theoretical expectations. The convergence of evidence from five independent experimental paradigms—each probing a different aspect of the curvature-noise coupling (static, dynamic, parametric, architectural, temporal)—constitutes

Table 5. Summary of key experimental observables and their theoretical counterparts.

Observable	NESP Prediction	Experiment
$\text{Tr}(\Sigma)$ peak at $\gamma \approx 1$	Eq. equation 78	Exp. 1: +4.6% elevation
$\lambda(H)$ vs. $\lambda(\Sigma)$ correlation	Hypothesis 7.1	Exp. 1: $\rho = 0.37$
$\tau \propto \lambda_{\max}^\alpha$ ($\alpha < 0$)	Eq. equation 51	Exp. 2: $\alpha = -1.34$
Smaller $B \rightarrow$ larger peak	$T_{\text{eff}} = \eta/B$	Exp. 3: monotonic trend
Smaller $B \rightarrow$ faster recovery	dynamical selection	Exp. 3: $B = 1$ best final loss
H - Σ alignment in ReLU	coupling survives σ	Exp. 4: $3.2\text{--}8.7\times$ baseline
Post-convergence erosion	Eq. 83	Exp. 5: $p < 0.01$ for $B = 16$
Full-batch no erosion	kinetic, not thermal	Exp. 5: $p = 0.87$ for $B = n$

a robust empirical foundation for the theoretical claims advanced in Section 7.

9. Conclusion and Outlook

We have presented a Non-Equilibrium Statistical Physics (NESP) framework for understanding the double descent phenomenon in deep learning. The central theoretical contribution is the identification of curvature-noise coupling—the proportionality between the SGD noise covariance and the local Hessian along the training trajectory—as the microscopic mechanism driving implicit regularization toward flat minima. This mechanism was derived analytically for a linear teacher-student model (Section 7.8) and confirmed experimentally for both linear and shallow ReLU networks (Section 8).

Three conceptual shifts distinguish the NESP framework from prior approaches:

1. From static to dynamic: The relevant question is not “what is the optimal hypothesis?” but “what hypothesis does SGD actually find?” The answer depends on the interplay between noise structure and landscape geometry—a kinetic rather than thermodynamic selection.
 2. From scalar to matrix temperature: The effective temperature in SGD is not a single number $T_{\text{eff}} = \eta/B$ but a full matrix $D(W) = (\eta/B)\Sigma(W)$ that varies across parameter space and is direction-dependent. Different eigenmodes of the Hessian experience different fluctuation amplitudes, breaking detailed balance and driving the system away from equilibrium.
 3. From equilibrium to aging: Generalization is not a permanent property of the trained model but a metastable state with a finite kinetic lifetime. The equilibrium erosion experiment (Section 7.10.5) demonstrates that good generalization degrades when the curvature-noise coupling shuts off, establishing a direct connection to the physics of glassy systems and aging phenomena.
- Finite-size scaling: The NESP framework predicts that double descent is a phase transition in the thermodynamic limit ($N, P \rightarrow \infty$ at fixed γ). A systematic finite-size scaling analysis could extract critical exponents—numbers that, if universal, would place the learning transition in a known universality class of non-equilibrium phase transitions.
 - Designer noise: If noise structure matters for generalization, then engineering the noise becomes a design principle. One could imagine optimizers that inject noise with specifically prescribed covariance (e.g., matching the Hessian eigenbasis but with tuned eigenvalue scaling) to accelerate the selection of flat minima. This reverses the usual perspective: noise is not a nuisance to be minimized but a resource to be shaped.
 - Connection to Bayesian inference: The stationary distribution of the Fokker-Planck equation with state-dependent diffusion is not the Boltzmann-Gibbs posterior. Understanding the relationship between the NESP stationary measure and the Bayesian posterior—and under what conditions they coincide—could bridge the gap between the NESP framework and the extensive literature on Bayesian deep learning.
 - Information-theoretic formulation: The “dynamical entropy” concept introduced in Section 7.7 remains heuristic. A rigorous connection to information theory—perhaps through the Information Bottleneck or rate-distortion theory—would strengthen the theoretical foundations and potentially reveal optimal noise schedules.

The experimental program described in Section 7.10 and executed in Section 8 provides systematic evidence for the eight core predictions of the NESP framework. We emphasize that these experiments are falsifiable: each was designed with explicit success criteria, and failure to observe the predicted effects would have constituted evidence against the theory. That all eight predictions were confirmed—at varying levels of statistical significance and effect magnitude—is encouraging but not conclusive. The framework now awaits testing at scale, in deep nonlinear architectures, and with interventional (causal) experimental designs.

Looking forward, several research directions emerge from this work:

- Scaling to deep networks: The most pressing question is whether curvature-noise coupling survives in deep architectures (ResNets, Transformers). The ReLU result (Experiment 4) provides circumstantial evidence, but the layer-wise structure of deep networks may introduce qualitatively new phenomena not captured by the shallow analysis.

The double descent phenomenon, when viewed through the lens of non-equilibrium statistical physics, ceases to be a paradox. It is the natural consequence of a driven dynamical system exploring a complex energy landscape under anisotropic, state-dependent fluctuations. The “failure” of classical learning theory to predict double descent is not a failure of rigor but a failure of framing: equilibrium theories answer equilibrium questions, and SGD is not an equilibrium process. The NESP framework provides the language to ask the right questions—and, as the experimental evidence demonstrates, to obtain answers that align with empirical reality.

Acknowledgments. We thank the statistical physics and machine learning communities for fruitful discussions. This work was supported by the Department of Physics, Hanoi University of Science, Vietnam National University. The computational experiments were performed using open-source Python scientific software (PyTorch, NumPy, SciPy, Matplotlib). All code and experimental data are available in the paper repository.

Appendix 1.1 Research Friction Points: Where Theory Meets Its Limits

We document the “friction points” where our theoretical machinery encounters resistance. These are not failures but signposts indicating where deeper understanding is needed.

Appendix 1.1.1 The Langevin Approximation Breaks Down

The continuous-time Langevin equation (Definition 4.2) assumes small learning rate η . In practice, large learning rates, warmup/decay schedules, and accumulated discretization error $O(\eta^2)$ violate this assumption. Recent “edge of stability” results Cohen et al. 2021 show that large learning rates induce qualitatively different dynamics not captured by continuous diffusion.

Appendix 1.1.2 The Replica Trick’s Mathematical Fragility

The replica identity (Eq. equation 27) requires: (1) computing $\mathbb{E}[Z^n]$ for integer n , (2) analytic continuation to real n , (3) taking $n \rightarrow 0$. Each step is non-rigorous. In some systems (e.g., Sherrington–Kirkpatrick), the symmetric ansatz gives wrong results and replica symmetry breaking is required. Alternative methods (cavity, interpolation, supersymmetric) can sometimes validate replica results rigorously.

Appendix 1.1.3 The Temperature Analogy’s Limits

The scalar $T_{\text{eff}} = \eta/B$ is a zeroth-order approximation. The full noise covariance $\Sigma(W)$ is a matrix that varies across weight space: different directions experience different “temperatures,” and detailed balance is violated. The diffusion tensor $D(W) = (\eta/B)\Sigma(W)$ is the fundamental object, motivating the NESP framework.

Appendix 1.1.4 The Gap Between Linear and Nonlinear

The linear teacher–student model is the “hydrogen atom” of learning theory: solvable but not representative. With nonlinear activations, the Hessian varies across parameter space, gradients lose their clean rank-1 form, and the loss landscape may have exponentially many disconnected basins. A systematic “ladder of models” (linear $\rightarrow$ random features $\rightarrow$ shallow ReLU $\rightarrow$ deep ReLU) is needed to identify which phenomena are universal.

Appendix 1.1.5 Observational vs. Causal Claims

Much of double descent research is observational. Our NESP framework attempts causal claims via the curvature–noise coupling mechanism. The framework generates falsifiable predictions (Section 7.10); a concrete test is replacing SGD noise with isotropic Gaussian noise to check whether double descent is suppressed.

Appendix 1.2 Author’s Working Notes: Research Trajectory

This section is intentionally left as working notes rather than polished prose, to facilitate the “pickup” of research

momentum. Each item represents a concrete starting point for future work sessions.

Immediate Next Steps (Can Be Done This Week):

1. Code the toy model: Implement linear teacher–student in PyTorch/JAX. This is straightforward: < 100 lines of code.
2. Plot the key diagnostic: Generate the plot of $\text{Tr}(\Sigma)$ vs. k/d overlaid with test error vs. k/d . If peaks coincide, this provides the central empirical confirmation for the paper.
3. Eigenvalue scatter plot: For a fixed k , scatter plot $\lambda_i(H)$ vs. $\lambda_i(\Sigma)$. A positive correlation would directly visualize the curvature–noise coupling.

Medium-Term Goals (Next Few Weeks):

1. Escape time numerics: Implement the escape time experiment. This requires careful definition of “escape” and multiple random seeds for statistics.
2. Batch size sweep: Systematic experiment varying batch size. This is computationally cheap but requires careful hyperparameter matching.
3. Write up toy model results: Convert numerical findings into figures and text for Chapter 8 (Experimental Verification).

Longer-Term Goals (Next Few Months):

- Extend to ReLU: Run Experiment 4. If alignment is observed, this is a major result. If not, understand why (is it the nonlinearity? the architecture? the training regime?).
- Analytical escape time: Attempt to derive $\tau(\lambda_{\max})$ from the Fokker–Planck equation. Even an approximate scaling law would be valuable.
- Finite-size scaling: Design and run the scaling collapse experiment. This could reveal universal behavior.
- Connection to NTK: Read the NTK literature carefully; understand how our NESP framework relates to infinite-width limits.

Appendix 1.3 Philosophical Reflections: The Epistemological Shift

Beyond technical results, the NESP framework represents a shift in how we think about learning systems. This section articulates that shift explicitly, as it informs the interpretation of all preceding derivations.

Appendix 1.3.1 From “What” to “How”: The Dynamical Turn

Classical learning theory asks: What is the optimal hypothesis? Given a loss function $\mathcal{L}$ and a hypothesis class $\mathcal{H}$, we seek the minimizer $h^* = \arg \min_{h \in \mathcal{H}} R(h)$. This framing treats learning as a static optimization problem—the dynamics of how we find h^* are irrelevant; only the end-point matters.

The NESP framework asks a different question: What hypothesis does SGD actually find? This is not merely a computational refinement; it is a conceptual reorientation. The answer depends on:

- The initialization W_0 —different starting points may lead to different basins.
- The noise structure $\Sigma(W)$ —anisotropic noise creates direction-dependent “temperatures.”
- The path through parameter space—the sequence of visited configurations, not just the final one.

This shift mirrors the historical transition in physics from equilibrium thermodynamics (“what is the free energy minimum?”) to non-equilibrium statistical mechanics (“how does the system relax?”). In both cases, the dynamical perspective reveals phenomena invisible to the static view: glasses, aging, memory effects, and—we argue—generalization in over-parameterized networks.

Appendix 1.3.2 The “Unreasonable Effectiveness” of SGD Noise

A persistent mystery in deep learning is the unreasonable effectiveness of SGD. Why does a noisy, approximate optimizer outperform exact methods? Why does training with small batches (more noise) often improve generalization?

The NESP framework offers a resolution: SGD noise is not a bug; it is a feature. The specific structure of gradient noise—anisotropic, curvature-dependent, state-varying—implements an implicit regularization that selects for flat minima. This is not regularization in the traditional sense (adding a penalty term to $\mathcal{L}$); it is a dynamical regularization that emerges from the interaction between the noise and the landscape geometry.

From this perspective, the success of deep learning is less mysterious. Over-parameterized networks have many global minima (all with $\mathcal{L} = 0$), but they differ in their local geometry—some are sharp, some are flat. SGD, through its noise structure, preferentially selects the flat ones. The “generalization” of deep networks is thus not a property of the hypothesis class $\mathcal{H}$ alone, but of the algorithm-hypothesis pair (SGD, $\mathcal{H}$).

Appendix 1.3.3 Implications for Theory and Practice

For theory: The NESP perspective suggests that traditional complexity measures (VC dimension, Rademacher complexity) may be inadequate for deep learning—not because they are wrong, but because they answer the wrong question. They measure the complexity of $\mathcal{H}$, not the complexity of the set of hypotheses reachable by SGD. A new “algorithmic complexity theory” may be needed, one that accounts for the implicit constraints imposed by optimization dynamics.

For practice: If noise structure matters, then designing the noise becomes a lever for improving generalization. This reframes hyperparameter tuning (learning rate, batch

size, momentum) not as “finding the fastest convergence,” but as “shaping the noise to select better minima.” Techniques like learning rate warmup, cyclical learning rates, and stochastic weight averaging can be understood as manipulations of the effective temperature schedule.

Appendix 1.3.4 What This Chapter Does Not Claim

Intellectual honesty requires stating the boundaries of our claims:

- We do not claim that NESP explains all of deep learning. Feature learning, representation structure, and architectural inductive biases are largely orthogonal to the dynamics we study.
- We do not claim that the linear toy model (Section 7.8) captures nonlinear network behavior. It is a proof of concept, not a proof of generality.
- We do not claim to have solved the generalization puzzle. The open problems in Section 7.9 are genuine gaps, not rhetorical gestures.

What we do claim is that the NESP framework provides a language and a set of tools for asking new questions—questions that equilibrium theories cannot even formulate. Whether this language proves fruitful is an empirical matter, to be settled by the experimental program outlined in Section 7.10.

Appendix 1.3.5 A Note on Interdisciplinarity

The NESP framework draws on concepts from statistical physics, stochastic processes, random matrix theory, and dynamical systems. This interdisciplinarity is both a strength and a risk. It is a strength because the phenomena of deep learning—phase transitions, critical behavior, emergent simplicity from complex dynamics—have natural analogs in physics. It is a risk because analogies can mislead; the “spin” of a weight is not literally a magnetic moment, and we must be vigilant against importing assumptions that do not transfer.

Navigating this tension requires what might be called bilingual fluency—the ability to translate concepts between fields while respecting the distinct semantics of each. We have attempted this throughout the chapter, but the translation is imperfect. Future work, by researchers with deeper expertise in either physics or machine learning, will surely refine (or refute) the ideas presented here.

Appendix 1.4 Comprehensive Research Friction Log

“Problem is insight.” This section documents every significant obstacle encountered during the development of this framework, treating each as a potential source of theoretical insight rather than a mere defect to be hidden.

Appendix 1.4.1 Category A: Mathematical Rigor Gaps

A1. The $n \rightarrow 0$ Replica Limit

Status: Unresolved (mathematically non-rigorous)

Problem: The analytic continuation from integer n to real $n \rightarrow 0$ is not unique. Different continuations give different results.

Insight potential: Perhaps the “correct” continuation is determined by physical constraints (stability, causality) not yet identified. Alternatively, the replica trick may be an approximation to a deeper structure (e.g., supersymmetric field theory).

A2. Fokker-Planck with State-Dependent Diffusion

Status: Analytically intractable in general

Problem: When $D(W)$ depends on W , the stationary distribution $P_{\text{stat}}(W)$ is not the simple Boltzmann form. Closed-form solutions exist only for 1D or special cases.

Insight potential: The deviation from Boltzmann may encode the “entropic selection” mechanism. Computing P_{stat} even approximately would reveal which minima are dynamically favored.

A3. Langevin Discretization Error

Status: Known bounds exist, but may not apply to large η

Problem: Standard SDE theory requires $\eta \rightarrow 0$. In practice, η is finite and sometimes large. The error accumulation over training is not controlled.

Insight potential: Discrete-time effects may be essential, not corrections. A “discrete Langevin” theory might be more appropriate.

Appendix 1.4.2 Category B: Physical Analogy Limitations

B1. Temperature vs. Noise Covariance

Status: Conceptual mismatch

Problem: Physical temperature is a scalar; SGD noise covariance is a matrix. The mapping $T_{\text{eff}} = \eta/B$ loses information about anisotropy.

Insight potential: A “tensor temperature” or “temperature field” $T_{ij}(W)$ may be the correct generalization. This would connect to non-equilibrium thermodynamics literature.

B2. Detailed Balance Violation

Status: Fundamental obstruction to equilibrium analysis

Obstacle: SGD dynamics generically violate detailed balance, meaning the stationary distribution (if it exists) is not Gibbs. Standard equilibrium tools don’t apply.

Insight potential: The violation of detailed balance may be quantifiable (e.g., via entropy production rate). This rate might correlate with generalization—systems “further from equilibrium” might generalize better.

B3. Quenched vs. Annealed Disorder

Status: Potentially misidentified regime

Obstacle: We treat the dataset $\mathcal{S}$ as quenched (fixed during training). But with data augmentation or online learning, $\mathcal{S}$ effectively varies, approaching annealed disorder.

Insight potential: The quenched-to-annealed crossover is a known phenomenon in physics. Its ML analog might explain why data augmentation improves generalization.

Appendix 1.4.3 Category C: Empirical Instabilities

C1. Reproducibility of Double Descent

Status: Empirically fragile

Obstacle: Double descent is not always observed. Small changes in hyperparameters, architecture, or dataset can suppress or shift the peak.

Insight potential: The “fragility” may indicate that double descent is a critical phenomenon—present only near a phase boundary. Mapping the phase diagram would clarify when to expect it.

C2. Architecture Dependence

Status: Unexplained variability

Obstacle: Different architectures (ResNet vs. Transformer vs. MLP) show qualitatively different double descent profiles. The NESP framework does not currently predict these differences.

Insight potential: Architecture may determine the “universality class” of the phase transition. Classifying architectures by their critical behavior is a research direction.

C3. Optimizer Dependence

Status: Underexplored

Obstacle: Most experiments use SGD or Adam. Does double descent persist with second-order methods (Newton, L-BFGS)? With gradient-free methods?

Insight potential: If double descent disappears with deterministic optimizers, this confirms the role of noise. If it persists, the phenomenon may be fundamentally about the loss landscape, not dynamics.

Appendix 1.4.4 Category D: Conceptual Gaps

D1. Definition of “Flat” Minimum

Status: Reparameterization-dependent

Obstacle: Rescaling weights (while preserving function) changes Hessian eigenvalues. “Flatness” is not an intrinsic property of the solution.

Insight potential: A reparameterization-invariant measure (e.g., based on Fisher information or PAC-Bayes volume) may be the “true” quantity correlating with generalization.

D2. Causality vs. Correlation

Status: Observational evidence only

Obstacle: We observe correlations (flat $\leftrightarrow$ good generalization, noise $\leftrightarrow$ flat selection). But correlation is not causation.

Insight potential: Interventional experiments (Section 7.10) can establish causality. Failing these, we have a heuristic, not a theory.

D3. Feature Learning

Status: Completely outside current framework

Obstacle: The NESP framework treats features as fixed (or evolving only through weight changes). But deep net-

works learn representations—the “features” themselves change.

Insight potential: Incorporating feature learning may require a “two-level” theory: one for representation dynamics, one for classifier dynamics on top. This is a major theoretical challenge.

Appendix 1.4.5 Category E: Connections to Other Theories

E1. Neural Tangent Kernel (NTK)

Status: Partial overlap, unclear relationship

Obstacle: NTK theory analyzes infinite-width limits where the network behaves linearly. Our NESP framework aims for finite-width, nonlinear networks. The connection is unclear.

Insight potential: NTK may be the “equilibrium limit” of NESP (infinite width $\rightarrow$ Gaussian process $\rightarrow$ equilibrium). Finite-width corrections might be NESP corrections.

E2. Information Bottleneck

Status: Conceptual resonance, no formal link

Obstacle: Information Bottleneck suggests training compresses representations. NESP suggests noise selects flat minima. Both sound like “simplicity emerges,” but the connection is vague.

Insight potential: Perhaps noise-induced flat selection is information compression, seen from a different angle. Formalizing this could unify two frameworks.

E3. PAC-Bayes Theory

Status: Potentially compatible

Obstacle: PAC-Bayes bounds involve a “posterior” over weights and penalize its divergence from a prior. NESP describes the dynamics leading to this posterior.

Insight potential: The NESP stationary distribution P_{stat} could be the “PAC-Bayes posterior.” Computing it would give generalization bounds with dynamical content.

Remark Appendix 1.1 (Using the Friction Log). This log is not meant to discourage but to focus. Each item represents a concrete research question. Graduate students seeking thesis topics could pick any entry and make progress. The log also serves as a “honesty check”—if a reviewer asks “what about X?” and X is in the log, we have anticipated the concern.

Appendix 1.5 Research Roadmap: From Draft to Publication

Following the principle of having a concrete roadmap for next steps, we provide a concrete action plan with milestones.

Appendix 1.5.1 Phase 1: Foundation Solidification (Weeks 1–2)

- ☐ Task 1.1: Complete literature review on Replica Method applications in ML.
 - Read: Mézard & Montanari (2009), Gerace et al. (2020), Pennington & Worah (2017)
 - Output: Summary table of replica results for linear/RF models
- ☐ Task 1.2: Verify all derivations in Section 7.8.

- Check: Gradient formula (Eq. 62)
- Check: Noise covariance structure (Eq. 68)
- Check: Hessian formula (Eq. 75)
- ☐ Task 1.3: Implement toy model code.
 - Framework: PyTorch or JAX
 - Validate: Compare analytical $\mathcal{L}(U)$ with numerical evaluation

Appendix 1.5.2 Phase 2: Core Experiments (Weeks 3–5)

- ☐ Task 2.1: Run Experiment 1 (Toy Model Validation).
 - Sweep: $k/d \in \{0.5, 0.7, 0.9, 1.0, 1.1, 1.3, 1.5, 2.0, 3.0, 5.0\}$
 - Measure: Train/test error, $\text{Tr}(H)$, $\text{Tr}(\Sigma)$, condition number
 - Plot: Double descent curve with curvature diagnostics overlaid
- ☐ Task 2.2: Eigenvalue correlation analysis.
 - Compute: Full spectra of H and Σ at fixed k
 - Plot: Scatter plot $\lambda_i(H)$ vs. $\lambda_i(\Sigma)$
 - Quantify: Pearson/Spearman correlation
- ☐ Task 2.3: Run Experiment 3 (Batch Size Dependence).
 - If peaks correlate with noise: strong support for NESP
 - If not: revisit assumptions

Appendix 1.5.3 Phase 3: Extension and Validation (Weeks 6–8)

- ☐ Task 3.1: Extend to ReLU networks (Experiment 4).
 - Architecture: 2-layer MLP with ReLU, varying width
 - Challenge: Hessian computation (use Lanczos/power iteration)
- ☐ Task 3.2: Escape time numerics (Experiment 2).
 - Protocol: Initialize at sharp minimum, measure escape
- ☒ Statistical analysis: Multiple seeds, confidence intervals
- ☐ Task 3.3: Address top 3 friction points with targeted analysis.
 - Priority: A2 (Fokker-Planck), B2 (Detailed Balance), D1 (Flatness definition)

Appendix 1.5.4 Phase 4: Writing and Refinement (Weeks 9–12)

- ☐ Task 4.1: Write Experimental Results chapter.
 - Figures: Publication-quality plots
 - Analysis: Statistical significance, error bars
- ☐ Task 4.2: Revise theoretical sections based on experimental findings.
 - If experiments confirm: strengthen claims
 - If experiments contradict: revise theory or acknowledge discrepancy
- ☐ Task 4.3: Final polish.
 - Notation consistency check
 - Citation completeness
 - Abstract and introduction refinement

Appendix 1.5.5 Success Criteria

The research will be considered successful if:

1. The curvature-noise coupling (Hypothesis 7.1) is empirically confirmed in the toy model.

2. The double descent peak coincides with maximal $\text{Tr}(\Sigma)/\text{Tr}(H)$ ratio or similar diagnostic.
3. At least one prediction of the NESP framework is validated against equilibrium predictions (showing non-equilibrium deviation).
- Eigenvector alignment: $|v_i(H)^\top v_i(\Sigma)|^2 > \theta$ for threshold θ
- Eigenvalue correlation: Pearson $r(\lambda(H), \lambda(\Sigma)) > 0.5$
- Falsification condition: correlation < 0.2 rejects the hypothesis

Even partial success—confirming some predictions while falsifying others—constitutes scientific progress and publishable results.

Appendix 1.6 Closing Reflection

We have constructed a framework—NESP—that reframes double descent as a dynamical phase transition driven by the anisotropic noise structure of SGD. We have identified the key hypothesis (curvature-noise coupling), derived analytical predictions in a toy model, and designed experiments to test them. The theory makes falsifiable predictions; verification through the experimental program outlined above is the immediate next step.

Appendix 1.7 Critical Self-Review

Following the principle that “obstacles and any problems encountered are insights,” we conducted a systematic review of this manuscript. The findings below represent an honest assessment of where the paper stands and what remains to be done.

Draft Status: This manuscript follows the research workflow: Theory Development → Experiment Design → Experiment Execution → Theory Adjustment. The theoretical framework and experimental design are substantially complete. The experimental phase will be the final step to validate and refine the theory.

Category I: Theoretical Refinements Needed (Before Experiments)

1. “Flatness” Requires Rigorous Definition. Bridge 5 acknowledges that Hessian-based flatness measures are reparameterization-dependent. Yet the NESP selection mechanism relies on “flat minima.” Resolution options: (a) Adopt a reparameterization-invariant measure (PAC-Bayes volume, Fisher information), or (b) restrict to canonical parameterization with explicit justification.
2. Curvature-Noise Coupling Scope. Hypothesis 7.1 ($\Sigma \approx \alpha H$) is analytically confirmed for the linear teacher-student model. Extension to nonlinear networks remains conjectural. Resolution: Experiment 4 will test this extension; until then, claims should be explicitly bounded to linear/random-feature models.
3. Equilibrium Deviation Should Be Quantified. The core claim is that SGD reaches NESS different from Gibbs. A concrete observable measuring this deviation would strengthen the framework. Candidates: entropy production rate, detailed balance violation measure, KL divergence from Boltzmann distribution.
4. Testable Predictions from Hypothesis 7.2.1. Transform the qualitative hypothesis into quantitative predictions:

Category II: Experimental Phase Planning

The experimental agenda (Section 7.10) provides a well-structured validation plan. Prioritized execution order:

1. Experiment 1 (Toy Model Validation): Highest priority—validates core hypothesis in tractable setting. Plot $\text{Tr}(\Sigma)$ vs. test error vs. k/d .
2. Experiment 2 (Eigenvalue Correlation): Direct visualization of curvature-noise coupling via scatter plot $\lambda_i(H)$ vs. $\lambda_i(\Sigma)$.
3. Experiment 3 (Batch Size Dependence): Tests whether noise amplitude modulates double descent peak as predicted.
4. Experiment 4 (ReLU Extension): Tests generalization to nonlinear networks. Even negative results bound the theory’s scope.

Category III: Methodological Refinements

1. Replica Method: Cite conditions under which replica is validated by rigorous methods (cavity method), or treat predictions as heuristics with explicit uncertainty.
2. Translation Protocol: Verify validity conditions in experimental test cases.
3. Escape Time: Derive analytically for the linear model where Fokker-Planck reduces to Ornstein-Uhlenbeck.

Category IV: Content to Strengthen

1. Regularization Analysis: Regularization can suppress/shift the interpolation peak—natural control experiment.
2. Literature Comparison Table: Systematic comparison of key papers on model class, framework, DD type.
3. RSB Detection: Develop into concrete experimental signature of “glassy” phase.

Honest Assessment: Paper Status

Component	Status	Notes
SLT Foundation (Sec. 2–3)	Complete	Standard,
Double Descent Review (Sec. 4)	Adequate	Needs com
Foundational Definitions (Sec. 5)	Complete	Well-docur
Equilibrium Theories (Sec. 6)	Adequate	Replica de
NESP Framework (Sec. 7)	Draft Complete	Core ideas
Toy Model Derivation	Complete	Analytical
Experiment Design	Complete	Well-struct
Experiment Execution	Planned	Final valid

Path Forward: Research Phases

Current Phase: Theory finalization and experimental preparation.

Next Phase: Experimental execution following the 4-experiment agenda.

Final Phase: Theory adjustment based on experimental results.

- ☐ If experiments confirm theory $\rightarrow$ strengthen claims, emphasize predictive power
- ☐ If experiments contradict $\rightarrow$ revise theory, document insights gained
- ☐ Either outcome advances science and is publishable

“Obstacles are insights.” The theoretical obstacles have been documented. They now become experimental hypotheses to test.

References

- [PDF] A Unifeid Bias-Variance Decomposition and its Applications | Semantic Scholar. Accessed July 16, 2024. <https://www.semanticscholar.org/paper/A-Unifeid-Bias-Variance-Decomposition-and-its-Domingos/e1ed9d24db5e8f7ab326aeb797e965a94f5ad6d3>.
- [PDF] A Unifeid Bias-Variance Decomposition and its Applications | Semantic Scholar. Accessed July 16, 2024. <https://www.semanticscholar.org/paper/A-Unifeid-Bias-Variance-Decomposition-and-its-Domingos/e1ed9d24db5e8f7ab326aeb797e965a94f5ad6d3>.
- Achlioptas, Panos. n.d. Stochastic Gradient Descent in Theory and Practice [in en]. Lecture note, Stanford's AI.
- Ackley, D. H., G. E. Hinton, and T. J. Sejnowski. 1985. A learning algorithm for boltzmann machines. *Cognitive Science* 9 (1): 147–169. [https://doi.org/10.1016/0364-0213\(85\)90014-6](https://doi.org/10.1016/0364-0213(85)90014-6). [https://doi.org/10.1016/0364-0213\(85\)90014-6](https://doi.org/10.1016/0364-0213(85)90014-6).
- Adlam, Ben, and Jeffrey Pennington. 2020. Understanding double descent requires a fine-grained bias-variance decomposition. arXiv: 2011.03321 [stat.ML]. <https://arxiv.org/abs/2011.03321>.
- Advani, Madhu S., Subhaneil Lahiri, and Surya Ganguli. 2013. Statistical mechanics of complex neural systems and high dimensional data. ArXiv:1301.7115, *Journal of Statistical Mechanics: Theory and Experiment*, <https://arxiv.org/abs/1301.7115>.
- Advani, Madhu S., and Andrew M. Saxe. 2017a. High-dimensional dynamics of generalization error in neural networks. arXiv: 1710.03667 [stat.ML]. <https://arxiv.org/abs/1710.03667>.
- . 2017b. High-dimensional dynamics of generalization error in neural networks. ArXiv:1710.03667; (published versions/derivatives exist), <https://arxiv.org/abs/1710.03667>.
- al., Molavi et. Model Complexity, Expectations, and Asset Prices | The Review of Economic Studies | Oxford Academic. Accessed July 16, 2024. <https://academic.oup.com/restud/article-abstract/91/4/2462/7222145?redirectedFrom=fulltext&login=false>.
- Allerbo, Oskar. 2025. Changing the kernel during training leads to double descent in kernel regression. arXiv: 2311.01762 [stat.ML]. <https://arxiv.org/abs/2311.01762>.
- Amari, Shun-ichi. 1996. Neural learning in structured parameter spaces - natural riemannian gradient. In *Advances in neural information processing systems* 9 (nips'96), 287–293. Also cited as “Natural Riemannian Gradient”.
- . 1998. Natural gradient works efficiently in learning. *Neural Computation* 10 (2): 251–276. <https://doi.org/10.1162/089976698300017746>.
- Amit, D. J., H. Gutfreund, and H. Sompolinsky. 1985. Spin-glass models of neural networks. *Physical Review A* 32 (2): 1007–1018. <https://doi.org/10.1103/PhysRevA.32.1007>. <https://doi.org/10.1103/PhysRevA.32.1007>.
- Ammar, Mouin Ben, David Brellmann, Arturo Mendoza, Antoine Manzanera, and Gianni Franchi. 2025. Double descent meets out-of-distribution detection: theoretical insights and empirical analysis on the role of model complexity. arXiv: 2411.02184 [stat.ML]. <https://arxiv.org/abs/2411.02184>.
- Arjevani, Yossi, Joan Bruna, Joe Kileel, Elzbieta Polak, and Matthew Trager. 2025. Geometry and optimization of shallow polynomial networks. arXiv: 2501.06074 [cs.LG]. <https://arxiv.org/abs/2501.06074>.
- Ascoli, Stéphane d', Levent Sagun, and Giulio Biroli. 2020. Triple descent and the two kinds of overfitting: where & why do they appear? In *Advances in Neural Information Processing Systems*, 33:3058–3069. Curran Associates, Inc. Accessed August 26, 2024. <https://proceedings.neurips.cc/paper/2020/hash/1fd09c5f59a8ff35d499c0ee25a1d47e-Abstract.html>.
- Bahri, Yasaman, Jonathan Kadmon, Jeffrey Pennington, Sam S. Schoenholz, Jascha Sohl-Dickstein, and Surya Ganguli. 2020. Statistical mechanics of deep learning. *Annual Review of Condensed Matter Physics* 11:501–528. <https://doi.org/10.1146/annurev-conmatphys-031119-050745>. <https://doi.org/10.1146/annurev-conmatphys-031119-050745>.
- Belkin, Mikhail, Daniel Hsu, Siyuan Ma, and Soumik Mandal. 2019a. Reconciling modern machine learning practice and the bias-variance trade-off. ArXiv:1812.11118 [cs, stat], *Proc. Natl. Acad. Sci. U.S.A.* 116, no. 32 (August): 15849–15854. ISSN: 0027-8424, 1091-6490, accessed July 16, 2024. <https://doi.org/10.1073/pnas.1903070116>. <http://arxiv.org/abs/1812.11118>.
- . 2019b. Reconciling modern machine learning practice and the bias-variance trade-off [in en]. ArXiv:1812.11118 [cs, stat], *Proc. Natl. Acad. Sci. U.S.A.* 116, no. 32 (August): 15849–15854. ISSN: 0027-8424, 1091-6490, accessed July 16, 2024. <https://doi.org/10.1073/pnas.1903070116>. <http://arxiv.org/abs/1812.11118>.
- Belkin, Mikhail, Daniel Hsu, and Ji Xu. 2020. Two models of double descent for weak features. *SIAM Journal on Mathematics of Data Science* 2, no. 4 (January): 1167–1180. ISSN: 2577-0187. <https://doi.org/10.1137/20m1336072>. <http://dx.doi.org/10.1137/20M1336072>.
- Belkin, Mikhail, Siyuan Ma, and Soumik Mandal. 2018. To understand deep learning we need to understand kernel learning. arXiv: 1802.01396 [stat.ML]. <https://arxiv.org/abs/1802.01396>.
- Bias-variance tradeoffs in program analysis | Proceedings of the 41st ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages. Accessed July 16, 2024. <https://dl.acm.org/doi/10.1145/2535838.2535853>.
- Bousquet, Olivier, Steve Hanneke, Shay Moran, Ramon van Handel, and Amir Yehudayoff. 2020. A theory of universal learning. arXiv: 2011.04483 [cs.LG]. <https://arxiv.org/abs/2011.04483>.
- Brellmann, David, Eloise Berthier, David Filliat, and Goran Frehse. 2024. On double descent in reinforcement learning with LSTD and random features. In *The twelfth international conference on learning representations*. <https://openreview.net/forum?id=9RIbNmX984>.
- Brown, Gavin, and Riccardo Ali. 2024. Bias/variance is not the same as approximation/estimation. *Transactions on Machine Learning Research*, ISSN: 2835-8856. <https://openreview.net/forum?id=4TnFbv16hK>.
- Buschjäger, Sebastian, Lukas Pfahler, and Katharina Morik. 2020a. Generalized Negative Correlation Learning for Deep Ensembling. ArXiv:2011.02952 [cs, stat], December. Accessed July 16, 2024. <https://doi.org/10.48550/arXiv.2011.02952>. <http://arxiv.org/abs/2011.02952>.
- . 2020b. Generalized Negative Correlation Learning for Deep Ensembling [in en]. ArXiv:2011.02952 [cs, stat], December. Accessed July 16, 2024. <http://arxiv.org/abs/2011.02952>.
- Cauchy, Augustin-Louis. 1847. General method for the resolution of simultaneous systems of equations. *Comp. Rend. Sci. Paris* 25:536–538.

- Chaudhari, Pratik, and Stefano Soatto. 2018. Stochastic gradient descent performs variational inference, converges to limit cycles for deep networks. Analysis of SGD as variational inference and its connection to Fokker-Planck dynamics, arXiv preprint arXiv:1710.11029, <https://arxiv.org/abs/1710.11029>.
- Cherkassky, Vladimir, and Eng Hock Lee. 2024. To understand double descent, we need to understand vc theory. *Neural Networks* 169:242–256. <https://doi.org/10.1016/j.neunet.2023.10.014>. <https://doi.org/10.1016/j.neunet.2023.10.014>.
- Cohen, Jeremy M., Simran Kaur, Yuanzhi Li, J. Zico Kolter, and Ameet Talwalkar. 2021. Gradient descent on neural networks typically occurs at the edge of stability. Introduces the concept of Edge of Stability in neural network training dynamics, arXiv preprint arXiv:2103.00065, <https://arxiv.org/abs/2103.00065>.
- Curth, Alicia, Alan Jeffares, and Mihaela van der Schaar. 2023. A u-turn on double descent: rethinking parameter counting in statistical learning. arXiv: 2310.18988 [stat.ML]. <https://arxiv.org/abs/2310.18988>.
- Davies, Xander, Lauro Langosco, and David Krueger. 2023. Unifying Grokking and Double Descent [in en]. ArXiv:2303.06173 [cs], March. Accessed August 26, 2024. <http://arxiv.org/abs/2303.06173>.
- Domingos, Pedro. 2000a. A unified bias-variance decomposition and its applications. In *Proceedings of the 17th international conference on machine learning (icml-2000)*, 231–238. Morgan Kaufmann. <https://homes.cs.washington.edu/~pedrod/papers/mlc00a.pdf>.
- . 2000b. A unified bias-variance decomposition for zero-one and squared loss. In *Proceedings of the 17th national conference on artificial intelligence (aaai-2000)*, 564–569. AAAI Press. <https://homes.cs.washington.edu/~pedrod/papers/aaai00.pdf>.
- Domingos, Pedro M. 2000a. A Unifeid Bias-Variance Decomposition and its Applications. In *Semantic scholar*. June. Accessed July 16, 2024. <https://www.semanticscholar.org/paper/A-Unifeid-Bias-Variance-Decomposition-and-its-Domingos/e1ed9d24db5e8f7ab326aeb797e965a94f5ad6d3>.
- . 2000b. A unified bias-variance decomposition for zero-one and squared loss. In *Aaai/iaai*. <https://api.semanticscholar.org/CorpusID:2063488>.
- E. L. Lehmann, George Casella. 1998. *Theory of Point Estimation* [in en]. Springer Texts in Statistics. New York: Springer-Verlag. ISBN: 978-0-387-98502-2, accessed May 25, 2025. <https://doi.org/10.1007/b98854>. <http://link.springer.com/10.1007/b98854>.
- Engel, A., and C. Van den Broeck. 2001. *Statistical mechanics of learning*. Cambridge: Cambridge University Press. ISBN: 978-0521773072.
- Ferraro, Gino Del, Chuang Wang, Dani Martí, and Marc Mézard. 2014. Cavity method: message passing from a physics perspective, arXiv: 1409.3048 [cond-mat.dis-nn]. <https://arxiv.org/abs/1409.3048>.
- Fletcher, R., and C. M. Reeves. 1964. Function minimization by conjugate gradients. *The Computer Journal* 7 (2): 149–154. <https://doi.org/10.1093/comjnl/7.2.149>.
- Gardner, E., and B. Derrida. 1988. Optimal storage properties of neural network models. *Journal of Physics A: Mathematical and General* 21 (1): 271–284. <https://doi.org/10.1088/0305-4470/21/1/031>. <https://doi.org/10.1088/0305-4470/21/1/031>.
- Geiger, Mario, Stefano Spigler, Stéphane d’Ascoli, Levent Sagun, Marco Baity-Jesi, Giulio Biroli, and Matthieu Wyart. 2019. Jamming transition as a paradigm to understand the loss landscape of deep neural networks. *Statistical physics perspective on interpolation threshold and loss landscapes*, *Physical Review E* 100 (1): 012115. <https://arxiv.org/abs/1809.09349>.
- Geman, Stuart, Elie Bienenstock, and Rene Doursat. 1992. Neural networks and the bias/variance dilemma. *Neural Computation* 4 (1): 1–58. <https://doi.org/10.1162/neco.1992.4.1.1>.
- Gerace, Federica, Bruno Loureiro, Florent Krzakala, Marc Mézard, and Lenka Zdeborová. 2020. Generalisation error in learning with random features and the hidden manifold model. *Replica method analysis of generalization in random feature models*, *Proceedings of the 37th International Conference on Machine Learning (ICML)* 119:3452–3462. <https://arxiv.org/abs/2002.09339>.
- Goodfellow, Ian, Yoshua Bengio, Aaron Courville, and Yoshua Bengio. 2016. *Deep learning*. Vol. 1. MIT Press.
- Gunasekar, Suriya, Blake Woodworth, and Nathan Srebro. 2021. Mirrorless mirror descent: a natural derivation of mirror descent. In *Proceedings of the 24th international conference on artificial intelligence and statistics (aistats 2021)*, 130:2305–2313. PMLR.
- Hajek, Bruce, and Maxim Raginsky. 2021. *Statistical learning theory*. Vol. 1. <https://maxim.ece.illinois.edu/teaching/SLT/>.
- Harzli, Ouns El, Bernardo Cuenca Grau, Guillermo Valle-Perez, and Ard A. Louis. 2023. Double-descent curves in neural networks: a new perspective using gaussian processes. arXiv: 2102.07238 [stat.ML]. <https://arxiv.org/abs/2102.07238>.
- Hastie, Trevor, Andrea Montanari, Saharon Rosset, and Ryan J Tibshirani. 2019. Surprises in high-dimensional ridgeless least squares interpolation. arXiv preprint arXiv:1903.08560.
- Heckel, Rebekka, and Murat Yilmaz. 2020. Early stopping in deep networks: double descent and how to eliminate it. arXiv preprint. Discusses epoch-wise double descent and mitigation strategies. <https://arxiv.org/abs/2007.10099>.
- Hellström, Thomas, Virginia Dignum, and Suna Bensch. 2020. Bias in Machine Learning – What is it Good for? ArXiv:2004.00686 [cs], September. Accessed July 16, 2024. <https://doi.org/10.48550/arXiv.2004.00686>. <http://arxiv.org/abs/2004.00686>.
- Hestenes, Magnus R., and Eduard Stiefel. 1952. Methods of conjugate gradients for solving linear systems. *Journal of Research of the National Bureau of Standards* 49 (6): 409–436. <https://doi.org/10.6028/jres.049.044>.
- Hinton, G. E. 2012. A practical guide to training restricted boltzmann machines. In *Neural networks: tricks of the trade* (2nd ed.) Edited by G. Montavon, G. B. Orr, and K.-R. Müller, 599–619. Springer. https://doi.org/10.1007/978-3-642-35289-8_32. https://doi.org/10.1007/978-3-642-35289-8_32.
- Hopfield, J. J. 1982. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences* 79 (8): 2554–2558. <https://doi.org/10.1073/pnas.79.8.2554>. <https://www.pnas.org/doi/10.1073/pnas.79.8.2554>.
- Hu, Erdong. 2021. An overview of double descent and overparameterization. independent, no. 0.
- Hu, Xia, Lingyang Chu, Jian Pei, Weiqing Liu, and Jiang Bian. 2021. Model Complexity of Deep Learning: A Survey. ArXiv:2103.05127 [cs], August. Accessed July 18, 2024. <https://doi.org/10.48550/arXiv.2103.05127>. <http://arxiv.org/abs/2103.05127>.

- Huang, Baihe, Shanda Li, Tianhao Wu, Yiming Yang, Ameet Talwalkar, Kannan Ramchandran, Michael I. Jordan, and Jiantao Jiao. 2025. Sample complexity and representation ability of test-time scaling paradigms. *arXiv*: 2506.05295 [cs.LG]. <https://arxiv.org/abs/2506.05295>.
- Jacot, Arthur, Francois Gabriel, and Clement Hongler. 2018. Neural tangent kernel: convergence and generalization in neural networks. In *Advances in neural information processing systems* (neurips). Kernel view of wide-network behavior.
- James, Gareth, Trevor Hastie, Robert Tibshirani, and Jerome H. Friedman. 2009. *The elements of statistical learning: data mining, inference, and prediction*. 2nd. Springer Series in Statistics. Springer.
- Janik, Romuald A., and Przemek Witaszczyk. 2021. Complexity for deep neural networks and other characteristics of deep feature representations. *arXiv*: 2006.04791 [cs.LG]. <https://arxiv.org/abs/2006.04791>.
- Kearns, Michael J., and Umesh V. Vazirani. 1994. *An introduction to computational learning theory*. Cambridge, MA, USA: MIT Press. ISBN: 0262111934.
- Khan, Mohammad Emtiyaz, and Håvard Rue. 2024. The Bayesian Learning Rule. *ArXiv*:2107.04562 [cs, stat] version: 4, June. Accessed July 16, 2024. <https://doi.org/10.48550/arXiv.2107.04562>. <http://arxiv.org/abs/2107.04562>.
- Lafon, Marc, and Alexandre Thomas. 2024. Understanding the Double Descent Phenomenon in Deep Learning [in en]. *ArXiv*:2403.10459 [cs, stat], March. Accessed August 26, 2024. <http://arxiv.org/abs/2403.10459>.
- Lee, Eng Hock, and Vladimir Cherkassky. 2022. Vc theoretical explanation of double descent. *arXiv*: 2205.15549 [stat.ML]. <https://arxiv.org/abs/2205.15549>.
- Li, Qianyi, and Haim Sompolsky. 2021. Statistical mechanics of deep linear neural networks: the backpropagating kernel renormalization. *Physical Review X* 11 (3): 031059. <https://doi.org/10.1103/PhysRevX.11.031059>. <https://doi.org/10.1103/PhysRevX.11.031059>.
- Li, Xinyue, and Rishi Sonthalia. 2024. Least squares regression can exhibit under-parameterized double descent. In *Advances in neural information processing systems*, edited by A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, 37:25510–25560. Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2024/file/2d43f7a61b57f83619f82c971e4bddc0-Paper-Conference.pdf.
- Liu, Chris Yuhao, and Jeffrey Flanigan. 2023. Understanding the role of optimization in double descent. *arXiv*: 2312.03951 [cs.LG]. <https://arxiv.org/abs/2312.03951>.
- Liu, Fanghui, Zhenyu Liao, and Johan A. K. Suykens. 2021. Kernel regression in high dimensions: refined analysis beyond double descent. *arXiv*: 2010.02681 [stat.ML]. <https://arxiv.org/abs/2010.02681>.
- Luo, Jing, Huiyuan Wang, and Weiran Huang. 2024. Investigating the impact of model complexity in large language models. *arXiv*: 2410.00699 [cs.LG]. <https://arxiv.org/abs/2410.00699>.
- Luo, Yifan, and Bin Dong. 2023. Double descent of discrepancy: a task-, data-, and model-agnostic phenomenon. *arXiv*: 2305.15907 [cs.LG]. <https://arxiv.org/abs/2305.15907>.
- Luong, Duy V. N., Panos Pappas, Daniel Rueckert, and Berc Rustem. 2016. A weighted mirror descent algorithm for nonsmooth convex optimization problem. *Journal of Optimization Theory and Applications* 170 (3): 900–915. <https://doi.org/10.1007/s10957-016-0963-5>.
- Manin, Yuri, and Matilde Marcolli. 2024. Homotopy theoretic and categorical models of neural information networks. *Compositionality Volume 6* (2024) (September). ISSN: 2631-4444. <https://doi.org/10.46298/compositionality-6-4>. <http://dx.doi.org/10.46298/compositionality-6-4>.
- Mehrabi, Ninareh, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. 2022. A Survey on Bias and Fairness in Machine Learning. *ArXiv*:1908.09635 [cs], January. Accessed July 16, 2024. <https://doi.org/10.48550/arXiv.1908.09635>. <http://arxiv.org/abs/1908.09635>.
- Mehta, Pankaj, and David J. Schwab. 2014. An exact mapping between the variational renormalization group and deep learning. *ArXiv*:1410.3831, <https://arxiv.org/abs/1410.3831>.
- Mei, Song, and Andrea Montanari. 2019a. The generalization error of random features regression: precise asymptotics and double descent curve. *arXiv preprint*, <https://arxiv.org/abs/1908.05355>.
- . 2019b. The generalization error of random features regression: precise asymptotics and double descent curve. *arXiv preprint arXiv:1908.05355*.
- . 2020. The generalization error of random features regression: precise asymptotics and double descent curve. *arXiv*: 1908.05355 [math.ST]. <https://arxiv.org/abs/1908.05355>.
- Mezard, M., G. Parisi, and M. A. Virasoro. 1987. *Spin glass theory and beyond*. Singapore: World Scientific. ISBN: 978-9971501533.
- Mezard, Marc, and Andrea Montanari. 2009. *Information, physics, and computation*. Oxford: Oxford University Press. ISBN: 978-0199233210.
- Mohri, Mehryar, Afshin Rostamizadeh, and Ameet Talwalkar. 2012. *Foundations of machine learning*. The MIT Press. ISBN: 026201825X.
- Molavi, Pooya, Alireza Tahbaz-Salehi, and Andrea Vedolin. 2024. Model Complexity, Expectations, and Asset Prices. *The Review of Economic Studies* 91, no. 4 (July): 2462–2507. ISSN: 0034-6527, accessed July 16, 2024. <https://doi.org/10.1093/restud/rdad073>. <https://doi.org/10.1093/restud/rdad073>.
- Nakkiran, Preetum. 2019a. More data can hurt for linear regression: sample-wise double descent. *arXiv*:1912.07242. *ArXiv preprint*. <https://arxiv.org/abs/1912.07242>.
- . 2019b. More data can hurt for linear regression: sample-wise double descent. *arXiv*: 1912.07242 [stat.ML]. <https://arxiv.org/abs/1912.07242>.
- Nakkiran, Preetum, Gal Kaplun, Yamini Bansal, Tristan Yang, Boaz Barak, and Ilya Sutskever. 2019. Deep Double Descent: Where Bigger Models and More Data Hurt. *ArXiv*:1912.02292 [cs, stat], December. Accessed July 16, 2024. <https://doi.org/10.48550/arXiv.1912.02292>. <http://arxiv.org/abs/1912.02292>.
- Nakkiran, Preetum, Prayaag Venkat, Sham Kakade, and Tengyu Ma. 2020. Optimal regularization can mitigate double descent (openreview entry). OpenReview forum submission. <https://openreview.net/forum?id=7R7fAoUygoa>.
- . 2021. Optimal regularization can mitigate double descent. *arXiv*: 2003.01897 [cs.LG]. <https://arxiv.org/abs/2003.01897>.
- Neal, Brady. 2019. On the bias-variance tradeoff: textbooks need an update. *arXiv*: 1912.08286 [cs.LG]. <https://arxiv.org/abs/1912.08286>.
- Neal, Brady, Sarthak Mittal, Aristide Baratin, Vinayak Tania, Matthew Scicluna, Simon Lacoste-Julien, and Ioannis Mitliagkas. 2018. A modern take on the bias-variance tradeoff in neural networks. *arXiv preprint arXiv:1810.08591*.

- Nemirovski, Arkadii S., and David B. Yudin. 1983. Problem complexity and method efficiency in optimization. Wiley-Interscience Series in Discrete Mathematics. Chichester, UK: John Wiley / Sons. ISBN: 9780471103455.
- Olmin, Amanda, and Fredrik Lindsten. 2024. Towards understanding epoch-wise double descent in two-layer linear neural networks. arXiv: 2407.09845 [stat.ML]. <https://arxiv.org/abs/2407.09845>.
- Oono, Kenta, and Taiji Suzuki. 2020. Graph neural networks exponentially lose expressive power for node classification. In International conference on learning representations. <https://openreview.net/forum?id=S1ldO2EFPr>.
- Paninski, Liam. 2005. Statistics 4107: Intro to Math Stat (fall 2005). Accessed May 25, 2025. <https://sites.stat.columbia.edu/liam/teaching/4107-fall05/>.
- Pezeshki, Mohammad, Amartya Mitra, Yoshua Bengio, and Guillaume Lajoie. 2021. Multi-scale feature learning dynamics: insights for double descent. arXiv: 2112.03215 [cs.LG]. <https://arxiv.org/abs/2112.03215>.
- Pfau, David. 2013. A generalized bias-variance decomposition for bregman divergences. Technical report.
- Quétu, Victor, and Enzo Tartaglione. 2023a. Can we avoid Double Descent in Deep Neural Networks? [In en]. In 2023 IEEE International Conference on Image Processing (ICIP), 1625–1629. ArXiv:2302.13259 [cs]. October. Accessed August 26, 2024. <https://doi.org/10.1109/ICIP49359.2023.10222624>. <http://arxiv.org/abs/2302.13259>.
- . 2023b. Can we avoid Double Descent in Deep Neural Networks? [In en]. In 2023 IEEE International Conference on Image Processing (ICIP), 1625–1629. ArXiv:2302.13259 [cs]. October. Accessed August 26, 2024. <https://doi.org/10.1109/ICIP49359.2023.10222624>. <http://arxiv.org/abs/2302.13259>.
- Raskutti, Garvesh, and Sayan Mukherjee. 2013. The information geometry of mirror descent. ArXiv:1310.7780, arXiv Preprint.
- Ratray, M., D. Saad, and Shun-ichi Amari. 1998. Natural gradient descent for on-line learning. Physical Review Letters 81:5461–5464. <https://doi.org/10.1103/PhysRevLett.81.5461>.
- Robbins, Herbert, and Sutton Monro. 1951. A stochastic approximation method. Annals of Mathematical Statistics 22 (3): 400–407. <https://doi.org/10.1214/aoms/117729586>.
- Roberts, Daniel A., Sho Yaida, and Boris Hanin. 2021. The principles of deep learning theory. Technical report. ArXiv:2106.10165; expanded into a Cambridge Univ. Press book. <https://arxiv.org/abs/2106.10165>.
- Rosenblatt, Frank. 1958. The perceptron: a probabilistic model for information storage and organization in the brain. Psychological review 65 6:386–408. <https://api.semanticscholar.org/CorpusID:12781225>.
- Ruder, Sebastian. 2017. An overview of gradient descent optimization algorithms. ArXiv:1609.04747 [cs], June. Accessed June 28, 2025. <https://doi.org/10.48550/arXiv.1609.04747>. <http://arxiv.org/abs/1609.04747>.
- Russo, Daniel, and James Zou. 2016. Controlling bias in adaptive data analysis using information theory. In Proceedings of aistats. Introduced info-theoretic generalization bounds.
- Schaeffer, Rylan, Mikail Khona, Zachary Robertson, Akhilan Boopathy, Kateryna Pistunova, Jason W. Rocks, Ila Rani Fiete, and Oluwasanmi Koyejo. 2023. Double Descent Demystified: Identifying, Interpreting & Ablating the Sources of a Deep Learning Puzzle. ArXiv:2303.14151 [cs, stat], March. Accessed July 16, 2024. <https://doi.org/10.48550/arXiv.2303.14151>. <http://arxiv.org/abs/2303.14151>.
- Seung, H. S., H. Sompolinsky, and N. Tishby. 1992. Statistical mechanics of learning from examples. Physical Review A 45 (8): 6056–6091. <https://doi.org/10.1103/PhysRevA.45.6056>. <https://doi.org/10.1103/PhysRevA.45.6056>.
- Shalev-Shwartz, Shai, and Shai Ben-David. 2014. Understanding machine learning: from theory to algorithms. USA: Cambridge University Press. ISBN: 1107057132.
- Shalev-Shwartz, Shai, Ohad Shamir, Nathan Srebro, and Karthik Sridharan. 2010. Learnability, stability and uniform convergence. Journal of Machine Learning Research 11 (90): 2635–2670. <http://jmlr.org/papers/v11/shalev-shwartz10a.html>.
- Sharma, Rahul, and Alex Aiken. 2014. Bias-variance tradeoffs in program analysis. In Proceedings of the 41st ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages, 127–137. POPL '14. New York, NY, USA: Association for Computing Machinery. ISBN: 978-1-4503-2544-8, accessed July 15, 2024. <https://doi.org/10.1145/2535838.2535853>. <https://doi.org/10.1145/2535838.2535853>.
- Shi, Cheng, Liming Pan, Hong Hu, and Ivan Dokmanić. 2024. Homophily modulates double descent generalization in graph convolution networks. arXiv: 2212.13069 [cs.LG]. <https://arxiv.org/abs/2212.13069>.
- Sohl-Dickstein, Jascha. 2012. The natural gradient by analogy to signal whitening, and recipes and tricks for its use. ArXiv:1205.1828, arXiv Preprint.
- Spieß, Jann, Guido Imbens, and Amar Venugopal. 2023. Double and single descent in causal inference with an application to high-dimensional synthetic control. arXiv: 2305.00700 [econ.EM]. <https://arxiv.org/abs/2305.00700>.
- Srivastava, Nitish, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. 2014. Dropout: a simple way to prevent neural networks from overfitting. Journal of Machine Learning Research 15 (56): 1929–1958. <https://jmlr.org/papers/volume15/srivastava14a/srivastava14a.pdf>.
- Sterkenburg, Tom F. 2024. Statistical learning theory and occam's razor: the core argument. Minds and Machines 35, no. 1 (November). ISSN: 1572-8641. <https://doi.org/10.1007/s11023-024-09703-y>. <http://dx.doi.org/10.1007/s11023-024-09703-y>.
- Sugiyama, Masashi. 2015. Introduction to statistical machine learning. San Francisco, CA, USA: Morgan Kaufmann Publishers Inc. ISBN: 9780128023501.
- Transtrum, Mark K., Gus L. W. Hart, Tyler J. Jarvis, and Jared P. Whitehead. 2025. Ega! double descent is explained by generalized aliasing decomposition. arXiv: 2408.08294 [math.ST]. <https://arxiv.org/abs/2408.08294>.
- Truong, Lan V. 2025. On rademacher complexity-based generalization bounds for deep learning. arXiv: 2208.04284 [stat.ML]. <https://arxiv.org/abs/2208.04284>.
- Valiant, L. G. 1984. A theory of the learnable. Commun. ACM (New York, NY, USA) 27, no. 11 (November): 1134–1142. ISSN: 0001-0782. <https://doi.org/10.1145/1968.1972>. <https://doi.org/10.1145/1968.1972>.
- Vapnik, Vladimir. 1999. The nature of statistical learning theory. Springer: New York.
- Widrow, Bernard, and Marcian E. Hoff. 1960. Adaptive switching circuits. In 1960 ire wescon convention record, part 4, 96–104. New York: Institute of Radio Engineers.
- Xiao, Lin, and Tong Zhang. 2014. A proximal stochastic gradient method with progressive variance reduction. SIAM Journal on Optimization 24 (4): 2057–2075. <https://doi.org/10.1137/140961791>.

- Xu, An, and Maxim Raginsky. 2017. Information-theoretic analysis of generalization capability of learning algorithms. In *Neurips*. Mutual information bounds.
- Yang, H. H., and Shun-ichi Amari. 1998. Complexity issues in natural gradient descent method for training multilayer perceptrons. *Neural Computation* 10 (8): 2137–2157. <https://doi.org/10.1162/089976698300017007>.
- Yang, Tian-Le, and Joe Suzuki. 2024. Dropout drops double descent. *Japanese Journal of Statistics and Data Science* 7, no. 2 (March): 615–632. ISSN: 2520-8764. <https://doi.org/10.1007/s42081-024-00242-5>. <http://dx.doi.org/10.1007/s42081-024-00242-5>.
- Yang, Zitong, Yaodong Yu, Chong You, Jacob Steinhardt, and Yi Ma. 2020. Rethinking Bias-Variance Trade-off for Generalization of Neural Networks [in en]. *ArXiv:2002.11328* [cs, stat], December. Accessed July 16, 2024. <http://arxiv.org/abs/2002.11328>.
- Zhang, Jiawei. 2019. Gradient Descent based Optimization Algorithms for Deep Learning Models Training. *ArXiv:1903.03614* [cs], March. Accessed June 28, 2025. <https://doi.org/10.48550/arXiv.1903.03614>. <http://arxiv.org/abs/1903.03614>.
- Zhang, Ya Shi. 2024. Manipulating sparse double descent. *arXiv:2401.10686* [cs.LG]. <https://arxiv.org/abs/2401.10686>.